

**TECH-VENTURE: SPATIAL FEATURE TRANSFORM NEURAL NETWORK
FOR REGION AWARE STARTUP SUCCESS PREDICTION**

A Thesis Proposal

Presented to the Faculty of the

Department of Computer, Information Sciences and Mathematics

University of San Carlos

In Partial Fulfillment

of the Requirements for the Degree

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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July 2026

ABSTRACT

Startups are innovation-driven, early-stage ventures that pursue rapid growth through scalable business models and external investment, distinguishing them from conventional small and medium-sized enterprises that primarily emphasize stable operations. In this study, startup success refers to achieving favorable venture outcomes, such as acquisition or initial public offering, reflecting sustained growth and market viability. Accurately predicting these outcomes across distinct global startup ecosystems, characterized by different economic, institutional, and investment conditions, remains challenging because regional differences influence how startup characteristics relate to venture success. Existing tabular machine learning models treat geographic location as a static input or ignore regional context, favoring data-rich ecosystems. This study proposes the Spatial Feature Transform Neural Network (SFTNN), a region-aware deep learning model that incorporates geographic context into its internal feature learning process using learned regional embeddings and Feature-wise Linear Modulation (FiLM). Using an age-conditioned funding-round momentum heuristic, a balanced binary dataset of 38,090 Crunchbase-derived and regional startup records across seven global regions was constructed. SFTNN was evaluated against MLP, XGBoost, and TabNet across Accuracy, Precision, Recall, F1-Score, AUC-ROC, and PR-AUC. Interpretability was established by directly inspecting the learned scaling (γ) and shifting (β) modulation vectors and performing counterfactual regional embedding ablation. Results demonstrate that conditioning shared latent representations on regional context improves predictive performance in underrepresented markets while maintaining a transparent, architecturally grounded interpretability mechanism.

Keywords: Startup Success Prediction, Spatial Feature Transform Neural Network (SFTNN), Region-Aware Deep Learning, Feature-wise Linear Modulation (FiLM), Spatial Non-Stationarity, Model Interpretability

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CHAPTER 1

INTRODUCTION

Machine learning has become an essential tool for data-driven decision-making across domains such as healthcare, finance, fraud detection, and venture capital. In startup ecosystems, the growing availability of large-scale startup datasets has enabled the development of predictive models that identify patterns associated with startup success. However, accurately predicting startup outcomes remains challenging because entrepreneurial ecosystems differ substantially across geographic regions.

To address this challenge, this study proposes the Spatial Feature Transform Neural Network (SFTNN), a region-aware deep learning model that improves startup success prediction by incorporating geographic context directly into the model architecture. Unlike conventional approaches that treat geographic region as a static categorical input, SFTNN allows regional context to influence how startup characteristics are interpreted during prediction, enabling the model to better capture differences across startup ecosystems. This chapter presents the study's background, research gap, statement of the problem, objectives, significance, and scope and limitations.

1.1 Background of the Study

Startups have become an important driver of innovation, technological advancement, and economic growth. Unlike traditional Small and Medium Enterprises (SMEs), which primarily focus on stable operations and sustainable profitability within established markets, startups are innovation-driven ventures that operate under high uncertainty while pursuing rapid growth through scalable business models and external investment (Abdelghani, 2024). Established corporations, on the other hand, focus on improving already validated business models rather than discovering new markets. Because of these characteristics, startup success is commonly associated with achieving sustained growth that leads to favorable outcomes such as acquisition, an Initial Public Offering (IPO),

or other indicators of long-term venture viability (Ningrum et al., 2024). Accurately identifying startups with high growth potential is therefore important for investors, entrepreneurs, startup incubators, and policymakers responsible for supporting entrepreneurial development.

The increasing availability of startup registries, particularly Crunchbase, has enabled researchers to apply machine learning techniques to startup success prediction (Ferrati & Muffatto, 2020). These models analyze structured startup information, including funding history, investment activity, organizational characteristics, and industry sectors, to identify patterns associated with successful ventures. As startup ecosystems continue to expand globally, startup success prediction has evolved alongside advances in tabular machine learning, improvements in startup data availability, and the emergence of techniques that enable models to incorporate contextual information during learning.

Machine learning approaches for startup success prediction have progressed through several major algorithmic paradigms. Early studies commonly relied on standard feedforward Multi-Layer Perceptrons (MLPs), which process startup characteristics through uniformly connected neural layers but often struggle with noisy, high-dimensional tabular data. Tree-based ensemble methods later became the dominant approach because of their ability to model complex, non-linear relationships while remaining robust to noisy and correlated features. Among these methods, XGBoost became a widely adopted baseline because of its regularized gradient-boosting framework (Chen & Guestrin, 2016). More recently, attention-based architectures such as TabNet introduced sequential feature selection while remaining trainable end-to-end through backpropagation (Arik & Pfister, 2019). These three approaches serve as the baseline models evaluated against the proposed SFTNN architecture.

Despite these advances, existing startup prediction models generally assume that the relationship between startup characteristics and success is consistent across all geographic regions. In reality, entrepreneurial ecosystems

differ in venture capital availability, regulatory environments, market maturity, technological infrastructure, and entrepreneurial culture. Consequently, characteristics associated with startup success in one region may not have the same significance in another. This challenge is particularly evident in publicly available startup registries such as Crunchbase, which remain heavily concentrated in data-rich ecosystems while comparatively underrepresenting emerging startup regions, including Southeast Asia, the primary regional focus of this study (Ferrati & Muffatto, 2020). Geographic context has therefore been recognized as an important factor influencing entrepreneurial outcomes beyond firm-level characteristics (Vedula & Kim, 2019).

Another limitation concerns how regional information is represented during prediction. Conventional machine learning models typically encode geographic regions using one-hot encoding, treating each region as an independent category. While computationally simple, this representation prevents models from learning shared relationships and localized variation across startup ecosystems. This assumption conflicts with the principle of spatial non-stationarity, which states that relationships between variables vary across geographic locations and therefore cannot always be represented by a single global model (Brunsdon et al., 1996).

A second challenge concerns the prediction target itself. Most startup records remain in an unresolved operating state rather than reaching a confirmed exit event, creating a form of right-censoring in which the final outcome is not observed within the study period (Mashhadi et al., 2026). Conventional calendar-based labeling methods discard these records, reducing the available training data and worsening class imbalance. To address this limitation, this study introduces an age-conditioned funding-round momentum heuristic that constructs balanced target labels while retaining valuable startup observations.

To overcome these limitations, this study adapts Feature-wise Linear Modulation (FiLM), a conditioning framework that dynamically scales and shifts a

neural network's internal feature representations using external information (Perez et al., 2018). A related architecture, Spatial Feature Transform (SFT), applies the same principle to image super-resolution by conditioning feature transformations on semantic information (Wang et al., 2018). Building on these concepts, the proposed Spatial Feature Transform Neural Network (SFTNN) replaces semantic information with learned regional embeddings that generate region-specific scaling (γ) and shifting (β) parameters. Rather than treating geographic region as an additional input feature, the model uses regional context to directly influence how startup characteristics are represented throughout the network. This design enables the model to capture both globally shared and region-specific patterns while preserving interpretability through direct inspection of the learned regional modulation parameters.

1.2 Problem Statement

Despite recent advances in machine learning for startup success prediction, existing models remain limited in their ability to generalize across geographically diverse entrepreneurial ecosystems. Most approaches assume that the relationship between startup characteristics and success is consistent across regions. In reality, startup ecosystems differ in venture capital availability, government support, market maturity, regulations, technological infrastructure, and entrepreneurial culture, causing indicators of startup success to vary across geographic contexts. Reinders (2024) emphasized that geographic context plays a significant role in entrepreneurial outcomes. However, most existing models either treat geographic region as a simple categorical feature or rely on complex relational data that are often unavailable in publicly accessible startup datasets. Consequently, they are unable to effectively incorporate regional context into the feature learning process.

These limitations reduce the reliability of startup success prediction. Public startup datasets are heavily concentrated in well-developed entrepreneurial ecosystems, causing models to learn patterns that favor data-rich regions while

underrepresenting emerging markets. As a result, prediction accuracy may decline and geographic bias may increase for startups operating in less-represented ecosystems. Furthermore, many startup records remain in an unresolved operating state rather than reaching a confirmed exit. Using only confirmed outcomes as training labels discards a substantial portion of the available data and creates severe class imbalance, further limiting model performance.

Although previous studies have explored startup success prediction using conventional neural networks, ensemble learning methods, graph-based models, and domain adaptation techniques (Vasquez et al., 2023), limited research has investigated architectures that directly integrate regional entrepreneurial ecosystem information into internal feature learning. Existing approaches generally include geographic region as an additional input feature rather than allowing regional context to influence how startup characteristics are represented throughout the network. This represents a significant gap in developing region-aware machine learning models for startup success prediction.

To address this gap, this study proposes the Spatial Feature Transform Neural Network (SFTNN), which incorporates regional entrepreneurial ecosystem information through learned regional embeddings and Feature-wise Linear Modulation (FiLM) (Perez et al., 2018). Instead of treating geographic regions as a static input, SFTNN generates learned scaling (γ) and shifting (β) parameters that dynamically adapt internal feature representations to regional context. The study also analyzes these learned modulation parameters to provide an interpretable understanding of how regional context influences startup success prediction without relying on post-hoc explanation methods.

1.3 Objectives

This study investigates how regional entrepreneurial ecosystem information can be integrated into a neural network architecture to improve startup success prediction. Specifically, it examines regional differences in

startup data, develops and evaluates a region-aware prediction model, and analyzes how regional context influences the model's learned feature representations.

1.3.1 General Objective

Generally, this study aims to:

Develop a Spatial Feature Transform Neural Network (SFTNN) that incorporates regional entrepreneurial ecosystem information through learned regional embeddings and affine feature modulation to improve startup success prediction across geographic regions while maintaining model interpretability through analysis of the learned regional modulation parameters.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. **Analyze** how startup characteristics, including total funding raised, number of funding rounds, funding interval, investor count, startup category, and binary success outcomes, vary across regional entrepreneurial ecosystem groups within the selected Crunchbase dataset.
2. **Design and implement** a Spatial Feature Transform Neural Network (SFTNN) that converts regional information into learned embeddings and generates scaling (gamma) and shifting (beta) parameters to adapt internal feature representations to regional entrepreneurial contexts.
3. **Evaluate** the proposed SFTNN against a standard Multi-Layer Perceptron (MLP), XGBoost, and TabNet using Accuracy, Precision, Recall, F1-Score, AUC-ROC, PR-AUC, and per-region confusion matrices to assess the effectiveness of region-aware feature modulation.
4. Interpret the learned regional modulation parameters (gamma and beta) by visualizing scaling and shifting vectors, apply **Integrated**

Gradients to measure feature importance across regional contexts, and perform counterfactual ablation using a global mean regional embedding to quantify the contribution of regional conditioning.

1.4 Significance of the Study

This study contributes to both machine learning and startup analytics by introducing a region-aware deep learning architecture that models geographic variation in startup ecosystems. By incorporating regional context directly into feature learning, the proposed approach provides a practical framework for more balanced and context-aware startup success prediction. The findings of this study may benefit the following stakeholders:

Field of Computer Science and Data Science

This study contributes an alternative approach for representing geographic context in deep tabular learning. Unlike traditional neural networks that treat geographic regions as a static categorical feature, the proposed architecture uses Feature-wise Linear Modulation (FiLM) to learn region-specific scaling (gamma) and shifting (beta) parameters that adapt internal feature representations. The study also improves model interpretability by allowing these learned regional adjustments to be examined directly, reducing reliance on post-hoc explanation methods such as SHAP or LIME.

Industry Practitioners and Venture Capitalists

This study provides a framework for startup evaluation that accounts for regional differences rather than relying on a single global standard. By adapting predictions to local entrepreneurial ecosystems, the proposed model may reduce geographic bias and improve the identification of promising startups in underrepresented and emerging markets, supporting more informed investment and funding decisions.

Future Researchers

This study establishes a foundation for future research on region-aware machine learning for startup success prediction. The proposed framework can be extended to country-level modeling as more comprehensive startup datasets become available. Beyond startup analytics, the architecture may also be adapted to other tabular prediction problems affected by geographic or demographic variation, such as credit risk assessment and healthcare prediction.

1.5 Research Questions

This study is guided by one general research question and five specific research questions.

1.5.1 General Research Question

How does the proposed Spatial Feature Transform Neural Network (SFTNN), which conditions feature learning on a startup's geographic region, improve the accuracy, regional fairness, and interpretability of startup success prediction compared with conventional region-blind and one-hot region models?

1.5.2 Specific Research Questions

1. How can Feature-wise Linear Modulation (FiLM) be used to reduce spatial feature-interaction bias by dynamically scaling (γ) and shifting (β) startup feature representations?
2. How can an age-conditioned funding-round momentum heuristic generate balanced target labels for right-censored startup records compared with conventional calendar-based labeling?
3. To what extent does conditioning feature learning on regional embeddings improve F1-score, AUC-ROC, PR-AUC, and classification performance in underrepresented regions compared with a standard Multi-Layer Perceptron (MLP), XGBoost, and TabNet?

4. How can the learned scaling (gamma) and shifting (beta) parameters improve the interpretability of regional adaptation in startup success prediction?
5. Does replacing learned regional embeddings with a global mean embedding produce a measurable decline in predictive performance, particularly in underrepresented regions?

1.6 Scope and Limitations

This section defines the boundaries of the study by identifying the aspects included within the research scope, the deliberate exclusions made by the researchers (delimitations), and the practical constraints encountered during the conduct of the study (limitations).

1.6.1 Scope

This study focuses on the development, implementation, and evaluation of the proposed Spatial Feature Transform Neural Network (SFTNN) for region-aware startup success prediction. Specifically, the study covers the following:

- **Dataset.** A consolidated dataset of 38,090 startup records from a Crunchbase-derived global dataset and publicly available regional startup registries representing seven geographic regions, with Southeast Asia as the primary regional focus.
- **Prediction Task.** Binary startup success prediction using observed outcomes (IPO, acquisition, or closure) together with a heuristic distant-supervision labeling strategy for startups with unresolved operating status.
- **Input Features.** Structured tabular features, primarily funding total, capital velocity, funding rounds, industry, and region, supported with last funding amount, year since last funded, and funding frequency. Unstructured data, including founder information, news articles, social media activity, and product descriptions, are excluded.

- **Model Comparison.** Comparative evaluation of SFTNN against Multi-Layer Perceptron (MLP), XGBoost, and TabNet under a common preprocessing and evaluation pipeline.
- **Implementation.** Development using Python-based machine learning frameworks with a FastAPI-Streamlit proof-of-concept application for real-time inference.
- **Performance Evaluation.** Evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC at both the overall and regional levels, together with regional modulation analysis, Integrated Gradients feature attribution, and region-ablation experiments.

1.6.2 Delimitations

The study intentionally excludes the following:

- The study predicts startup success but does not establish causal relationships between regional characteristics and startup outcomes.
- Proprietary startup databases and commercial data sources, including Crunchbase Pro and Tracxn, are not used. Only publicly accessible datasets are included.
- The study relies exclusively on historical datasets and does not incorporate live or continuously updated startup information.
- Hyperparameter optimization is limited to a bounded Optuna search rather than exhaustive optimization due to available computational resources and project duration.
- The comparison is limited to SFTNN, MLP, XGBoost, and TabNet. Other architectures, such as Graph Neural Networks and transformer-based tabular models, are beyond the scope of this study.
- The deployed FastAPI-Streamlit application serves only as a proof-of-concept demonstration and is not evaluated for production-level performance, scalability, or security.

- Geographic conditioning is performed at the regional level only. Country-level, province-level, and city-level variations are not modeled.

1.6.3 Limitations

Despite careful design and implementation, the study is subject to several limitations.

- **Unmodeled startup characteristics.** The study relies on structured tabular data and does not include qualitative factors such as founder experience, team composition, product-market fit indicators (e.g., customer acquisition cost and burn rate), or unstructured information such as pitch decks and media sentiment because these are not consistently available in registry-derived datasets.
- **Label uncertainty.** Some training labels are generated using a heuristic distant-supervision strategy for startups with incomplete outcome information, making model performance dependent on the quality of the adopted labeling approach.
- **Regional data imbalance.** Startup records remain concentrated in North America and Europe, with comparatively fewer observations from regions such as Southeast Asia, which may affect model learning despite the proposed region-aware architecture.
- **Computational and time constraints.** Limited computing resources and the project's academic timeline restricted the extent of hyperparameter optimization and the evaluation of additional model architectures and interpretability methods.

CHAPTER 2

REVIEW OF RELATED LITERATURE

The literature on startup success prediction has evolved from traditional financial distress models to advanced machine learning approaches capable of learning complex patterns from entrepreneurial data. Growing evidence also shows that startup outcomes are influenced by both firm-level characteristics and regional ecosystem conditions. This chapter reviews the literature on startup forecasting, entrepreneurial ecosystem heterogeneity, context-aware learning, domain adaptation, region-aware machine learning, and explainable artificial intelligence. Together, these studies establish the theoretical and empirical foundations of the proposed Region-Aware Neural Network with Spatial Feature Transform Neural Network (SFTNN).

2.1 Startup Success Prediction and Tabular Machine Learning

2.1.1 Historical Foundations of Corporate Risk Forecasting

Research on startup success prediction originates from the broader field of corporate risk forecasting, which historically focused on identifying financially distressed firms through statistical modeling. Early approaches relied heavily on accounting-based indicators and linear classification techniques, with the Altman Z-score emerging as one of the most influential frameworks for evaluating corporate viability. By combining multiple financial ratios into a single predictive index, the model provided a standardized method for assessing bankruptcy risk and became widely adopted across industries.

Despite its historical significance, subsequent research has identified important limitations in applying traditional financial distress models to contemporary entrepreneurial environments. Dai and Cai (2026) demonstrate that the predictive effectiveness of the Altman Z-score

declines substantially in rapidly evolving and non-Western economic settings, where financial conditions change more dynamically than the assumptions underlying classical statistical models. These limitations become even more pronounced in startup ecosystems, where firms frequently lack the extensive financial histories required by traditional accounting-based approaches. Early-stage ventures often operate under conditions of uncertainty, rapid growth, and incomplete financial reporting, making static ratio-based methods insufficient for capturing the complex factors associated with startup success and failure.

As entrepreneurial ecosystems became increasingly data-rich, researchers shifted toward computational approaches capable of modeling nonlinear relationships among diverse organizational characteristics. Gabrielli et al. (2026) argue that modern forecasting environments benefit from machine learning architectures that can process large-scale, multidimensional datasets and uncover complex interactions that traditional statistical methods cannot adequately represent. This transition reflects a broader movement from deterministic financial evaluation toward data-driven prediction, laying the foundation for contemporary machine learning approaches to startup forecasting.

2.1.2 Machine Learning Approaches to Startup Prediction

The availability of entrepreneurial databases such as Crunchbase has accelerated the adoption of machine learning techniques for startup prediction. Rather than relying solely on financial statements, contemporary models incorporate a wide range of startup characteristics, including funding activity, investment history, organizational structure, founder attributes, and growth-related indicators. These variables are typically represented in tabular form, making tabular machine learning one of the dominant paradigms in startup forecasting research.

Among the most widely used architectures are Multi-Layer Perceptrons (MLPs) and other supervised learning models that learn predictive relationships from structured startup metadata. Razaghzadeh Bidgoli et al. (2024) show that neural network-based approaches can successfully identify patterns associated with startup outcomes when trained on sufficiently representative datasets. Their findings align with a broader body of research demonstrating that machine learning models consistently outperform traditional statistical approaches in environments characterized by nonlinear interactions and high-dimensional feature spaces.

A critical aspect of startup prediction research involves defining what constitutes entrepreneurial success. Recent studies increasingly operationalize success through objective liquidity events, particularly Initial Public Offerings (IPOs) and Mergers and Acquisitions (M&As), because these outcomes provide measurable indicators of venture performance and market validation (Wei et al., 2025). Using standardized success definitions improves comparability across studies while reducing ambiguity in model evaluation.

Beyond static firm characteristics, temporal indicators have also emerged as important predictors of startup outcomes. Te et al. (2023) demonstrate that variables such as funding velocity, fundraising frequency, and the duration between investment rounds provide meaningful signals regarding organizational growth trajectories. These findings suggest that startup success is shaped not only by the presence of resources but also by the pace at which ventures acquire and deploy them. Collectively, the literature indicates that machine learning approaches are increasingly capable of capturing these complex relationships, making them valuable tools for entrepreneurial forecasting.

However, despite improvements in predictive performance, startup prediction models continue to face significant challenges related to data quality, representation, and class distribution. These limitations affect both model reliability and generalizability, particularly when predictions are applied across diverse entrepreneurial environments.

2.1.3 Data Quality, Scarcity, and Structural Imbalance

Despite the predictive capabilities of contemporary machine learning models, startup forecasting remains constrained by the quality and availability of entrepreneurial data. Early-stage startups frequently operate under conditions of uncertainty, limited reporting, and incomplete information, creating challenges for data-driven decision-making processes (Eranki et al., 2023). Compared with mature organizations, early-stage ventures often possess shorter operational histories, fewer observable performance indicators, and less standardized reporting practices, making reliable data collection inherently difficult.

These challenges are amplified when entrepreneurial data are used for machine learning. Startup repositories commonly contain missing values, incomplete records, and uneven coverage across firms, industries, and regions. In addition, successful ventures tend to receive greater visibility and documentation than failed startups, contributing to imbalanced datasets that may not accurately reflect the full distribution of entrepreneurial outcomes. Such conditions complicate model training and increase the risk of biased predictions.

From a machine learning perspective, data scarcity and class imbalance can significantly affect model performance. Rodriguez-Belenguer et al. (2026) demonstrate that predictive systems trained on limited or highly imbalanced data often struggle to learn meaningful patterns associated with underrepresented classes. As optimization procedures naturally prioritize majority patterns during

training, weak but informative signals may be overlooked. Similarly, Hakami (2024) argues that challenges related to data scarcity, imbalance, and feature selection frequently require architectural and methodological interventions rather than simple preprocessing techniques.

These limitations suggest that improving startup prediction requires more than selecting increasingly sophisticated algorithms. Predictive frameworks must also account for the structural characteristics of entrepreneurial data and preserve informative signals despite incomplete, imbalanced, and heterogeneous observations. Furthermore, because data availability and reporting practices differ across entrepreneurial ecosystems, these challenges become even more pronounced when prediction models are applied across multiple geographic regions. This raises an important question regarding whether a single globally optimized model can adequately capture startup dynamics in diverse economic environments.

2.2 Entrepreneurial Ecosystems and Regional Heterogeneity

2.2.1 Macroeconomic and Institutional Influences on Startup Outcomes

Research on startup success increasingly recognizes that entrepreneurial outcomes are shaped not only by firm-level characteristics but also by the broader environments in which ventures operate. Entrepreneurial ecosystems consist of interconnected economic, institutional, and social conditions that influence how startups access resources, attract investment, develop capabilities, and sustain growth. Consequently, startups with similar organizational profiles may experience substantially different outcomes depending on the regions in which they operate.

Regional ecosystem quality has emerged as one of the strongest predictors of entrepreneurial persistence and survival. Vedula and Kim (2019) found that ventures operating within high-quality entrepreneurial ecosystems exhibit greater survival rates than those located in weaker ecosystems, suggesting that regional conditions directly influence firm outcomes. This finding is consistent with broader evidence demonstrating that startup success is facilitated by ecosystem factors such as infrastructure accessibility, institutional support, and market maturity (Eggert et al., 2022). Similarly, regional culture, policy environments, and local demand conditions have been shown to influence organizational survival, particularly among mission-driven and sustainability-oriented ventures (Vedula et al., 2022).

These findings suggest that startup success emerges from the interaction between organizational capabilities and environmental opportunities. While internal resources remain important, entrepreneurial ecosystems determine how effectively those resources can be mobilized and transformed into growth. Consequently, startup characteristics should not be interpreted independently of the regional contexts in which they are observed.

Macroeconomic conditions further contribute to regional variation in entrepreneurial outcomes. Venture capital availability, regulatory stability, economic growth, and market accessibility influence both startup development and investment decision-making. Comparative analyses of venture capital markets indicate substantial differences in investment behavior across global regions, reflecting variations in institutional structures, cultural dynamics, and regional market maturity (Mortada & Nwachukwu, 2025). Likewise, cross-border venture investments are affected by ecosystem-specific conditions that influence venture selection, monitoring, and exit opportunities (Dai et al., 2012). Recent work by Harrison (2025) similarly emphasizes that national innovation systems and

local institutional arrangements shape the flow of entrepreneurial capital, creating distinct developmental pathways across regions.

Collectively, these studies indicate that entrepreneurial ecosystems cannot be viewed as passive backdrops to startup activity. Rather, they actively influence how entrepreneurial signals are generated, interpreted, and rewarded. As a result, predictive models that rely exclusively on firm-level variables may overlook important environmental factors that shape startup trajectories.

2.2.2 Institutional Voids and Emerging Market Dynamics

The influence of regional context becomes particularly evident in emerging economies, where entrepreneurial activity often occurs under conditions characterized by institutional constraints and limited resource availability. The concept of institutional voids provides a useful framework for understanding these differences. Mair and Marti (2009) define institutional voids as the absence or underdevelopment of market-supporting institutions, including regulatory mechanisms, specialized intermediaries, and contract enforcement systems that facilitate economic exchange.

Subsequent research demonstrates that these institutional conditions significantly affect entrepreneurial behavior. Yildirim et al. (2022) argue that startups operating within emerging ecosystems frequently develop alternative organizational practices to compensate for gaps in institutional support. Rather than relying on established entrepreneurial infrastructures, firms often allocate resources toward overcoming operational barriers that would otherwise be handled by external institutions.

The consequences of these institutional differences extend beyond operational efficiency and directly influence startup growth trajectories.

Venture capital studies suggest that entrepreneurial ecosystems in emerging markets frequently exhibit different investment dynamics from those observed in mature ecosystems, with financing mechanisms often serving both developmental and survival functions (Kato, 2025). Similarly, entrepreneurial organizations adapt their strategies to local ecosystem realities, creating patterns of growth and investment behavior that may differ substantially from those found in developed economies (Sun et al., 2019).

Another challenge arises from the unequal integration of entrepreneurial ecosystems into global investment networks. Zhou (2024) describes this phenomenon as a liability of outsidership, wherein organizations operating outside dominant economic centers experience disadvantages in attracting resources, visibility, and strategic partnerships. Consequently, startups in peripheral or developing ecosystems may appear less competitive when evaluated using metrics derived primarily from mature entrepreneurial environments.

Recent evidence further suggests that macro-environmental factors such as inflation, corruption, institutional instability, and currency volatility exert significant influence on entrepreneurial outcomes in developing economies (Ali et al., 2023). These factors introduce additional complexity into startup prediction because they affect the interpretation of commonly used venture indicators. Funding frequency, investor participation, and growth rates may therefore reflect fundamentally different underlying conditions depending on the economic environment in which they occur.

Taken together, the literature indicates that entrepreneurial behavior is strongly conditioned by regional institutional realities. What appears to be weak performance in one context may represent effective adaptation in another. Therefore, regional variation should not be treated as statistical

noise but rather as a meaningful source of information regarding startup outcomes.

2.2.3 Limitations of Geographically Invariant Prediction Models

Despite extensive evidence demonstrating the importance of entrepreneurial ecosystems, many startup prediction models continue to adopt geographically invariant assumptions. Most machine learning frameworks are designed to identify global relationships across pooled datasets, implicitly assuming that predictive features carry similar meanings regardless of regional context. While this assumption simplifies model development and increases data availability during training, it conflicts with growing evidence from entrepreneurship research suggesting that startup success determinants vary across ecosystems. Rather than representing universal entrepreneurial principles, many commonly used indicators appear to be context dependent, reflecting differences in institutional quality, investment behavior, market maturity, and ecosystem support structures. Consequently, models optimized on aggregated global datasets may learn dominant regional patterns while overlooking mechanisms that operate in less represented entrepreneurial environments.

Research increasingly challenges the assumption of geographic invariance. Argaw and Liu (2024) conclude that startup success determinants differ substantially between developed and emerging economies, suggesting that entrepreneurial signals cannot be interpreted uniformly across regions. This finding is reinforced by Ren et al. (2025), who demonstrate that geographic proximity to entrepreneurial support structures alters investor perceptions of startup quality and risk. Taken together, these studies imply that identical startup characteristics may produce different outcomes depending on the ecosystem in which they are observed. Consequently, geographic variation influences not only

resource availability but also the predictive meaning of commonly used startup indicators.

Beyond entrepreneurship research, studies examining entrepreneurial ecosystems from a spatial perspective reveal substantial geographic heterogeneity. Perugini (2023) identifies distinct spatial and temporal ecosystem patterns that become obscured when regions are aggregated into broad analytical categories. Likewise, Schäfer et al. (2024) argue that neglecting spatial structure limits understanding of entrepreneurial ecosystem dynamics because ecosystem interactions frequently extend beyond local administrative boundaries.

Recent ecosystem research further reinforces this view. Pigola et al. (2025) conceptualize entrepreneurial ecosystems as spatial mosaics composed of interconnected yet distinct regional environments, while Vicentin et al. (2026) demonstrate that global investment patterns exhibit identifiable geographic clusters that evolve differently over time. At a more operational level, Pilc and Taušl Procházková (2025) show that access to incubators, accelerators, mentorship programs, and institutional support structures varies substantially across regions and contributes to differences in startup development outcomes.

The implications of these findings are significant for startup prediction. If entrepreneurial ecosystems differ in their institutional structures, investment patterns, and developmental trajectories, then predictive variables are unlikely to exhibit identical relationships with outcomes across all regions. Models trained under assumptions of geographic homogeneity may therefore disproportionately learn patterns associated with dominant ecosystems while underrepresenting the realities of smaller or emerging entrepreneurial hubs.

Consequently, a growing gap exists between entrepreneurial ecosystem theory and prevailing machine learning practices. While

entrepreneurship research increasingly recognizes regional heterogeneity as a fundamental characteristic of startup development, many predictive frameworks continue to represent geographic information as static categorical variables or omit it entirely. Addressing this gap requires learning approaches capable of adapting internal representations according to environmental context rather than assuming universal relationships across heterogeneous ecosystems. This challenge aligns closely with developments in context-aware learning and domain adaptation, which provide theoretical foundations for modeling variation across distinct environments.

2.3 Region-Aware Neural Networks and Context-Aware Learning

2.3.1 Context-Aware Learning and Regional Heterogeneity

Recent advances in machine learning increasingly challenge the assumption that a single global model can adequately represent heterogeneous environments. In context-aware learning, contextual variables are not merely additional inputs but environmental factors that alter how predictive signals are interpreted. Unger and Tuzhilin (2022) show that modeling context as hierarchical latent representations, rather than static features, improves performance in recommendation and predictive systems.

This perspective is particularly relevant to startup forecasting because entrepreneurship research consistently demonstrates that regional ecosystems influence how entrepreneurial resources are translated into venture outcomes. Vedula and Kim (2019) found that ecosystem quality significantly affects startup survival, while Eggert et al. (2022) show that infrastructure accessibility and institutional support shape digital startup success. Similarly, Vedula et al. (2022) identify regional culture, policy environments, and local market demand as important determinants of green startup survival. Although these studies

examine different entrepreneurial contexts, they collectively suggest that regional environments alter the effectiveness of entrepreneurial strategies and resources. Nevertheless, Kim et al. (2023) observe that most startup prediction models continue to represent geographic location as a static categorical attribute rather than a contextual factor capable of modifying feature interpretation. This disconnect highlights a growing gap between entrepreneurial ecosystem theory and contemporary machine learning practice.

This gap is compounded by a tendency to treat geography as homogeneous. Schäfer et al. (2024) emphasize that ignoring spatial structure leads to incomplete theoretical understanding of ecosystem dynamics. In the same way, Perugini (2023) reveals heterogeneous regional patterns that coarse spatial aggregation methods obscure, and macro-environmental constraints such as inflation, corruption, and currency instability remain underrepresented in predictive models despite significantly affecting startup outcomes in developing economies (Ali et al., 2023). However, Taghiyarrenani et al. (2023) show that conditioning models on domain information improves learning under distribution shift in low-resource settings. Nevertheless, there remains a gap in understanding how this principle applies to startup prediction, as existing frameworks rarely treat geographic regions as explicit domains requiring internal representation adaptation.

Spatial machine learning studies reinforce that explicitly modeling spatial variation improves accuracy over non-spatial baselines. Boegl and Kounadi (2024) demonstrate this with regionalized approaches such as Regionalization Random Forest. Similarly, entrepreneurial ecosystems are spatially heterogeneous systems with distinct regional dynamics that unsupervised learning can help uncover (Pigola et al., 2025; Vicentin et al., 2026), and Pilc and Taušl Procházková (2025) find that institutional support structures such as incubators, accelerators, and mentorship vary

significantly across regions in ways that shape startup growth trajectories. On the other hand, most startup prediction models still rely on standard architectures without region-aware modulation mechanisms, leaving a gap between theoretical spatial understanding and model design.

Collectively, these findings provide a strong theoretical rationale for investigating region-aware learning approaches within startup forecasting. Existing research demonstrates that entrepreneurial ecosystems influence startup development and that spatially informed machine learning methods can improve predictive performance in heterogeneous environments. However, the extent to which contextual modulation mechanisms can enhance startup success prediction remains underexplored. This gap motivates the investigation of architectures capable of adapting feature representations according to regional ecosystem conditions while maintaining a unified predictive framework.

2.3.2 Limitations of Existing Geographic Representation Approaches

Despite growing recognition of regional heterogeneity, existing machine learning approaches continue to exhibit limitations in how geographic information is incorporated into predictive models. One common strategy involves feature gating mechanisms that selectively emphasize or suppress variables during model training. Wei et al. (2025) demonstrate that feature gating architectures can improve predictive performance by reducing the influence of noisy or unstable inputs. However, because these mechanisms operate globally, they may inadvertently suppress features that remain highly informative within specific entrepreneurial ecosystems.

This limitation is particularly relevant in startup forecasting, where variables such as funding frequency, investor participation, and fundraising intervals may carry different meanings across regional environments. Consequently, globally applied gating mechanisms risk

removing ecosystem-specific information that reflects meaningful contextual variation rather than noise. Similar concerns have been identified in studies examining the propagation of missing values and structural noise in tabular learning systems, where weak but informative signals may be overlooked during optimization processes (Imran et al., 2026).

Another common strategy involves representing geographic information through high-cardinality categorical encodings. In their pioneering work, Guo and Berkhahn (2016) demonstrated that neural entity embeddings preserve multidimensional relationships more effectively than traditional one-hot encoding approaches. Nevertheless, these representations remain static once learned and often struggle to generalize when observations are unevenly distributed across numerous geographic categories. Avanzi et al. (2024) report that high-cardinality categorical variables can increase susceptibility to overfitting, particularly in data-scarce environments.

More sophisticated approaches utilize graph-based representations, federated learning frameworks, and distributed knowledge networks to capture relationships across geographic environments (Pan et al., 2026; Ma et al., 2025). Although these approaches provide powerful mechanisms for integrating decentralized information, they often require extensive relational infrastructure and highly connected datasets that may not be readily available within entrepreneurial ecosystems, particularly in emerging regions.

Taken together, these limitations suggest that existing approaches either treat geography as static information, suppress ecosystem-specific variation, or introduce substantial computational complexity. Consequently, there remains a need for architectures capable of dynamically adapting

feature representations according to regional context while maintaining scalability and interpretability.

Although alternative approaches such as graph-based learning, federated architectures, and distributed knowledge networks provide mechanisms for modeling geographic variation, they often prioritize relational complexity over interpretability and may require extensive ecosystem connectivity information (Pan et al., 2026; Ma et al., 2025). In contrast, startup datasets are frequently tabular, incomplete, and unevenly distributed across regions, making highly structured relational approaches difficult to deploy consistently. This creates a methodological tradeoff between representational richness and practical applicability. Consequently, there remains a need for approaches that can adapt to regional variation while remaining compatible with conventional tabular startup datasets.

2.3.3 Theoretical Foundations of Domain Adaptation

The challenge of learning across heterogeneous entrepreneurial ecosystems aligns closely with the broader field of domain adaptation. Domain adaptation seeks to improve predictive performance when data distributions differ across environments by adjusting internal representations to account for distributional variation. In their comprehensive survey, Wilson and Cook (2020) identify representation alignment as one of the most effective strategies for addressing domain shifts while preserving predictive performance.

Rather than forcing a model to learn a single globally invariant relationship, domain adaptation frameworks permit environment-specific adjustments that preserve localized information (Stojanov et al., 2021). Ge et al. (2023) further argue that conditioning mechanisms enable models to align divergent distributions without eliminating meaningful contextual differences. These principles are particularly relevant to startup

forecasting, where entrepreneurial ecosystems exhibit substantial variation in institutional structures, investment patterns, and developmental trajectories.

Within this perspective, geographic regions can be conceptualized as distinct domains that influence the interpretation of entrepreneurial signals. Features associated with startup growth may therefore exhibit different relationships with outcomes across regions, requiring learning approaches capable of adapting internal representations accordingly. Domain adaptation thus provides the theoretical basis for architectures that explicitly incorporate regional variation into the predictive process.

2.3.4 Feature-Wise Linear Modulation and Spatial Conditioning

A particularly influential development in contextual adaptation is Feature-wise Linear Modulation (FiLM), originally introduced by Perez et al. (2018) for visual reasoning tasks. FiLM enables contextual information to modify neural representations through feature-level scaling and shifting operations. Given an input representation x , the transformed output y is computed as:

$$y = \gamma x + \beta$$

where γ represents a learned scaling parameter and β represents a learned shifting parameter.

Through these operations, contextual information dynamically influences feature representations without requiring separate models for different environments. Recent studies suggest that modulation-based approaches extend beyond computer vision and can be effectively applied to structured tabular data (Cai & Ye, 2025). Furthermore, adaptive modulation mechanisms have demonstrated strong performance under data-scarce conditions by enabling efficient contextual adaptation from limited observations (Feng et al., 2021).

These findings indicate that feature modulation provides a promising mechanism for incorporating environmental context into predictive systems. Rather than treating contextual information as an auxiliary feature, modulation frameworks allow context to directly influence how predictive signals are processed within the model.

2.3.5 Modulation-Based Approaches for Contextual Adaptation

Building upon principles from domain adaptation and feature-wise modulation, Spatial Feature Modulation (SFM) provides a mechanism for incorporating regional context directly into startup prediction models. Rather than treating geographic information as a static categorical attribute, SFM allows regional information to dynamically modify internal feature representations through learned scaling and shifting operations.

Under this framework, startup characteristics are first processed through shared neural layers that capture general entrepreneurial patterns. Regional embeddings subsequently generate modulation parameters that adjust intermediate representations according to ecosystem context. This process enables the model to preserve global predictive relationships while simultaneously accounting for localized ecosystem dynamics.

Empirical evidence from spatial machine learning supports this approach. Regionalized learning frameworks have demonstrated improved predictive performance compared with geographically invariant baselines (Boegl & Kounadi, 2024). Entrepreneurial ecosystem research likewise indicates that startup development occurs within spatially heterogeneous environments characterized by distinct support structures, investment networks, and developmental trajectories (Pigola et al., 2025; Vicentin et al., 2026; Pilc & Taušl Procházková, 2025).

Consequently, Spatial Feature Modulation represents a theoretically grounded approach for addressing the limitations of conventional startup prediction models. By conditioning feature representations on regional context, SFM enables the simultaneous modeling of global entrepreneurial regularities and ecosystem-specific variation within a unified predictive framework.

2.4 Explainable Artificial Intelligence in Startup Forecasting

2.4.1 Explainability and Model Transparency

As machine learning systems become increasingly integrated into entrepreneurial finance and decision-support environments, model transparency has emerged as an important consideration. In their definitive taxonomy of Explainable AI, Arrieta et al. (2020) highlight that deep learning architectures frequently operate as black-box systems that generate predictions without clearly revealing the factors responsible for decision outcomes. This limitation may reduce stakeholder trust and complicate the practical adoption of predictive systems.

Explainable Artificial Intelligence (XAI) addresses this challenge by providing methods for understanding model behavior and feature contributions. Techniques such as SHAP and LIME identify the relative importance of individual variables and help explain how specific predictions are generated (Ribeiro et al., 2016; Lundberg & Lee, 2017). These methods contribute to transparency, accountability, and informed decision-making.

Region-aware architectures provide an additional dimension of interpretability because contextual modulation parameters can themselves be examined. Within modulation-based frameworks, scaling and shifting parameters reflect how the model adapts feature representations across different environments. Analyzing these parameters may provide insights

into how entrepreneurial ecosystems influence prediction outcomes and whether regional adaptations align with theoretical expectations.

Accordingly, integrating explainability mechanisms within region-aware startup prediction frameworks contributes not only to predictive performance but also to transparency and practical applicability. Such capabilities are particularly valuable in entrepreneurial forecasting, where understanding the rationale behind model predictions is often as important as the predictions themselves.

Overall, the literature shows that startup success is shaped by the interaction of organizational characteristics, funding dynamics, investment networks, and entrepreneurial ecosystem conditions. Although machine learning has significantly improved startup success prediction, existing models remain challenged by data quality, class imbalance, regional heterogeneity, and limited interpretability. Studies on entrepreneurial ecosystems consistently demonstrate that geographic context influences entrepreneurial outcomes, yet most predictive models either assume globally consistent relationships or treat geographic information as a static input. Recent advances in context-aware learning, domain adaptation, and feature-wise modulation show that contextual information can be integrated directly into internal model representations. However, limited research has explored applying these approaches to startup success prediction to address geographic variation. Consequently, a gap remains in developing prediction models that capture both globally shared patterns and region-specific entrepreneurial characteristics.

CHAPTER 3

TECHNICAL BACKGROUND

The previous chapter presented the review of related literature and studies relevant to startup success prediction, entrepreneurial ecosystems, region-aware machine learning, and explainable artificial intelligence. Building on these studies, this chapter discusses the technical concepts, theories, tools, and frameworks that form the foundation of the proposed Spatial Feature Transform Neural Network (SFTNN). Rather than reviewing previous research, this chapter explains the technical knowledge required to understand how the proposed model is designed, implemented, and evaluated. It provides the necessary background for the methodology presented in the succeeding chapter.

3.1 Introduction to the Domain

Startup success prediction is a specialized application of machine learning that focuses on determining whether a startup is likely to achieve a successful business outcome based on its observable characteristics. Instead of relying solely on expert judgment or traditional financial analysis, machine learning models learn patterns from historical startup data to estimate the likelihood of future success. These predictions assist investors, entrepreneurs, venture capital firms, and policymakers in making more informed decisions regarding investments, resource allocation, and business strategies (Wei et al., 2025).

As startup ecosystems continue to grow worldwide, large datasets containing information on funding history, company profiles, investors, industries, and business outcomes have become increasingly available through platforms such as Crunchbase and Kaggle. The availability of these datasets has encouraged researchers to develop machine learning models capable of identifying complex relationships among startup characteristics that may not be easily recognized through conventional statistical approaches (Razaghzadeh Bidgoli et al., 2024). Unlike traditional models that often assume linear

relationships between variables, machine learning algorithms can capture nonlinear interactions and discover hidden patterns within large amounts of structured data.

This study belongs to the field of tabular machine learning, where data are organized into rows and columns representing different observations and features. In the context of startup prediction, each row corresponds to an individual startup, while each column contains information such as funding rounds, total investment received, company age, industry sector, location, and other measurable characteristics. Because startup information is naturally stored in this structured format, tabular machine learning provides an effective approach for developing predictive models.

Although existing machine learning models have shown promising results in startup success prediction, many assume that the relationship between startup characteristics and business success is the same across all geographic regions. In reality, startups operate within different entrepreneurial ecosystems, each having unique economic conditions, investment environments, government policies, and levels of technological development. As a result, startup features that strongly indicate success in one region may have a different level of importance in another (Argaw & Liu, 2024).

To address this challenge, this study proposes the Spatial Feature Transform Neural Network (SFTNN), a region-aware neural network that incorporates geographic context into the prediction process. Instead of treating the startup's region as another ordinary input feature, the proposed model allows regional information to influence how other startup characteristics are interpreted during prediction. This enables the model to adapt its internal feature representation according to the entrepreneurial ecosystem in which a startup operates, improving its ability to generalize across diverse geographic regions.

Understanding the proposed approach requires familiarity with several technical concepts, including tabular machine learning, binary classification, neural

networks, feature modulation, domain adaptation, regional embeddings, and model interpretability. These concepts are discussed in the succeeding sections as the technical foundation of the proposed SFTNN.

3.2 Core Concepts and Definitions

This section presents the fundamental concepts that serve as the basis of the proposed Spatial Feature Transform Neural Network (SFTNN). Understanding these concepts is essential for explaining how the model processes startup data, incorporates regional information, and generates interpretable predictions. The concepts are introduced from the most fundamental principles of machine learning to the techniques that are directly applied in this study.

Tabular Machine Learning

Tabular machine learning is a branch of machine learning that focuses on structured data organized into rows and columns, similar to a spreadsheet or relational database. Each row represents an individual observation, while each column corresponds to a specific feature or attribute of that observation. In this study, each row represents a startup, and each column contains measurable characteristics such as the number of funding rounds, total funding raised, company age, industry sector, investor information, and geographic region.

Since startup databases naturally store information in tabular form, this type of machine learning has become one of the most widely used approaches for startup success prediction. Unlike image or text data, tabular datasets contain heterogeneous numerical and categorical features that require specialized preprocessing and modeling techniques to capture meaningful relationships among variables (Razaghzadeh Bidgoli et al., 2024).

Binary Classification

Binary classification is a supervised machine learning task in which a model predicts one of two possible outcomes for each observation. The objective

is to learn patterns from labeled training data so that the model can correctly classify new, unseen observations into one of the predefined classes.

In this study, the two classes are successful and unsuccessful startups. A startup is considered successful if it reaches a major liquidity event, specifically an Initial Public Offering (IPO) or a Merger and Acquisition (M&A). These outcomes are widely recognized as objective indicators of startup success because they represent measurable business achievements rather than subjective assessments (Wei et al., 2025). After processing the input features, the model produces a probability value between 0 and 1 that represents the likelihood of a startup belonging to the successful class.

Class Imbalance

Class imbalance occurs when one class in a dataset contains significantly more observations than the other. This problem is common in startup datasets because only a small percentage of startups eventually achieve an IPO or acquisition, while the majority either remain inactive, cease operations, or fail to reach these milestones.

When trained on imbalanced data, machine learning models tend to favor the majority class because doing so minimizes the overall prediction error. As a result, the model may perform well on the training data while failing to accurately identify successful startups, which are the minority class of greater interest. Previous studies have shown that addressing class imbalance requires appropriate data preprocessing and model design to improve the model's ability to learn meaningful patterns from underrepresented observations (Rodriguez-Belenguer et al., 2026).

Geographic Distribution Shift

Geographic distribution shift refers to differences in the statistical relationship between input features and target outcomes across different

geographic regions. In startup prediction, this means that the factors influencing startup success are not identical across all entrepreneurial ecosystems.

For example, a funding amount that is considered average in Silicon Valley may represent a substantial investment in an emerging economy. Likewise, investor participation, government support, and market maturity vary considerably from one region to another. Because of these differences, models trained using globally pooled data may learn patterns that primarily represent dominant regions while performing poorly in less-represented ecosystems (Wilson & Cook, 2020).

This challenge is one of the primary motivations for the proposed SFTNN, which incorporates regional information directly into the feature learning process instead of assuming that all startups follow the same predictive patterns.

Entrepreneurial Ecosystem

An entrepreneurial ecosystem refers to the collection of economic, institutional, and social factors that influence the creation, growth, and success of startups within a particular geographic region. These factors include the availability of venture capital, government policies, business infrastructure, educational institutions, innovation networks, accelerators, incubators, and the overall entrepreneurial culture.

The quality of an entrepreneurial ecosystem significantly affects startup performance. Even startups with similar business characteristics may experience different outcomes because they operate under different environmental conditions. Studies have consistently shown that regional ecosystem quality plays a major role in startup survival and long-term growth (Vedula & Kim, 2019; Eggert et al., 2022). For this reason, regional context is treated as an important source of information in the proposed prediction model.

Institutional Voids

Institutional voids refer to the absence or weakness of formal institutions that support business operations within a region. These institutions include financial systems, legal frameworks, regulatory agencies, investor networks, and other organizations that facilitate entrepreneurial activities.

Startups operating in regions with strong institutional support generally have easier access to funding, legal protection, and business resources. In contrast, startups in emerging markets often encounter institutional gaps that require them to adopt different operational strategies to overcome resource limitations (Mair & Marti, 2009; Yildirim et al., 2022). Because these differences influence startup behavior and performance, they contribute to regional variations that machine learning models must consider during prediction.

Feature-wise Linear Modulation (FiLM)

Feature-wise Linear Modulation (FiLM) is a neural network conditioning technique that allows external contextual information to modify how internal feature representations are processed. Instead of applying the same transformation to every input, FiLM generates parameters that adjust feature values according to additional contextual information.

The FiLM operation is defined as

$$y = \gamma \odot x + \beta$$

where γ is a learned element-wise scaling vector, β is a learned element-wise shifting vector, and $\odot$ denotes element-wise multiplication. Both γ and β are generated by a conditioning network from a contextual input — in this study, a regional ecosystem embedding.

Spatial Feature Modulation (SFM)

Spatial Feature Modulation (SFM) extends the concept of FiLM to address regional variation in startup prediction. Instead of using visual or textual

information as contextual input, SFM uses geographic region to influence how startup features are interpreted during model inference.

Under this approach, the model does not simply append region as another feature in the input vector. Instead, regional information modifies the internal feature representations learned by the neural network. Consequently, identical startup characteristics can be interpreted differently depending on the entrepreneurial ecosystem in which the startup operates. This allows the model to learn both global startup patterns and region-specific relationships within a single unified architecture (Cai & Ye, 2025).

Regional Embeddings

Regional embeddings are dense numerical representations that encode geographic regions into continuous vector spaces. Rather than representing each region using one-hot encoding or arbitrary numerical labels, embedding vectors are learned automatically during model training.

Regions that exhibit similar entrepreneurial characteristics tend to develop similar embedding representations. This enables the neural network to capture relationships among different regions while reducing the dimensionality of categorical geographic data. Compared with traditional categorical encoding techniques, learned embeddings provide richer contextual information and improve the model's ability to generalize across diverse entrepreneurial ecosystems (Guo & Berkahn, 2016).

Model Interpretability in Neural Networks

Model interpretability refers to the degree to which the internal mechanisms and decision-making process of a machine learning model can be understood by a human observer. As neural network architectures grow in depth and complexity, their predictions increasingly function as outputs of a “black box,” in which the specific reasoning behind an individual prediction is not directly observable from the model's outputs alone.

Two broad families of technique address this concern. Post-hoc explainability techniques, such as SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017), approximate a trained model's behavior after training is complete, treating the model as a fixed function and estimating each input feature's contribution to a given prediction without inspecting the model's internal parameters directly. Inherently interpretable techniques, in contrast, rely on model components that have a direct, transparent mathematical role in the forward pass, such that no separate approximation step is required to explain their effect.

This study adopts the second approach. Because the SFT layer conditions the model on regional context through explicit scale (γ) and shift (β) parameters, these parameters can be read directly as literal indicators of how the network adjusts its shared representation for each region, without requiring a post-hoc attribution method such as SHAP. SHAP was evaluated as a candidate interpretability method during the design of this study and was not adopted, both because γ and β already provide a more direct evidentiary source for this specific architecture, and because scoping the project to interpretability methods with a lower-risk implementation path was a deliberate choice given the study's three-month timeline (Section 4.8).

Grad-CAM (Selvaraju et al., 2017), a widely used interpretability technique for convolutional neural networks, was likewise evaluated and not adopted. Grad-CAM produces its explanations by computing gradients with respect to a convolutional layer's spatial feature map (organized by height, width, and channel) and combining them into a pixel-level heatmap over an image. The SFTNN contains no convolutional layers and no image-like spatial feature maps; the term “spatial” in Spatial Feature Transform refers to geographic region rather than pixel-grid structure, so Grad-CAM has no principled counterpart in this architecture.

3.3 Theoretical Foundations

The proposed Spatial Feature Transform Neural Network (SFTNN) is built upon several established theories and principles in machine learning. These theoretical foundations explain how the model learns from labeled startup data, optimizes its predictions, adapts to regional differences, and provides interpretable results. Together, these concepts form the scientific basis for the architecture and training process presented in Chapter 4.

3.3.1 Supervised Learning and Empirical Risk Minimization

The proposed SFTNN follows the supervised learning paradigm, in which a model learns the relationship between input features and known output labels using labeled training data. During training, each startup in the dataset is represented by an input feature vector X and its corresponding target label Y , where the label indicates whether the startup is considered successful or unsuccessful. The objective of supervised learning is to learn a function that accurately predicts the correct label for previously unseen startups.

Mathematically, supervised learning seeks to learn a mapping function:

$$f: X \rightarrow Y$$

where X represents the input feature space, Y represents the output label space, and f is the predictive function learned by the model. Rather than memorizing individual training examples, the model aims to learn general patterns that can be applied to unseen data.

To evaluate how well the model performs during training, the SFTNN minimizes empirical risk, which represents the average prediction error over the training dataset. Because startup success prediction is formulated as a binary classification problem, this study employs the Binary Cross-Entropy (BCE) loss function, expressed as

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

where L is the loss value, y is the true class label, and $\hat{y}$ is the predicted probability produced by the model. Binary Cross-Entropy penalizes incorrect predictions by assigning larger loss values when predicted probabilities differ significantly from the true labels. As outlined in standard deep learning classification frameworks (James et al., 2023), minimizing this loss encourages the model to assign high probabilities to successful startups and low probabilities to unsuccessful ones, resulting in more accurate classification.

3.3.2 Feedforward Neural Networks and Backpropagation

A Multi-Layer Perceptron (MLP) consists of stacked layers, each computing a nonlinear transformation of its input. For layer l , the transformation is:

$$h^{(l)} = \sigma(W^{(l)} h^{(l-1)} + b^{(l)})$$

where $h^{(l)}$ is the output of the current layer, $W^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector, σ is the activation function, and $h^{(l-1)}$ is the output of the previous layer.

This study uses the Rectified Linear Unit (ReLU) activation function because of its computational efficiency and ability to reduce the vanishing gradient problem during training. The ReLU function is defined as

$$ReLU(z) = \max(0, z)$$

After each forward pass, the prediction error is propagated backward through the network using the backpropagation algorithm. Backpropagation applies the chain rule of calculus to compute the gradient of the loss function with respect to every learnable parameter in the network. These gradients indicate how each weight contributes to the prediction error, allowing the optimizer to update the parameters iteratively until the model converges toward an optimal solution.

3.3.3 Adaptive Optimization

Training deep neural networks involves repeatedly updating millions of parameters to minimize the loss function. Instead of using a fixed learning rate for every parameter, this study employs the Adaptive Moment Estimation (Adam) optimizer, one of the most widely used optimization algorithms in deep learning.

Adam combines the advantages of Momentum and Root Mean Square Propagation (RMSProp) by maintaining estimates of both the first and second moments of the gradients.

Compared to conventional gradient descent, Adam automatically adjusts the learning rate for each parameter based on previous gradient information. In their foundational work, Kingma and Ba (2015) demonstrated that this adaptive approach enables faster convergence and more stable training, particularly when dealing with noisy, sparse, or imbalanced gradients. This specialized mechanism makes the algorithm highly effective for processing complex data environments, such as startup prediction datasets.

3.3.4 Domain Adaptation

A key challenge addressed in this study is that startup data collected from different geographic regions often follow different statistical distributions. This phenomenon is commonly known as domain shift, where the relationship between input features and prediction labels varies across different environments.

Domain Adaptation is a machine learning approach that aims to improve a model's ability to generalize across domains with different data distributions while maintaining predictive performance. Instead of learning separate models for each region, domain adaptation enables a single model to adapt its internal representations according to contextual information.

Wilson and Cook (2020) explain that representation learning is one of the most effective approaches for reducing domain shift because it allows models to

learn feature representations that remain informative across multiple domains. Similarly, Ge et al. (2023) demonstrate that conditioning mechanisms can further improve adaptation by allowing contextual information to influence intermediate feature representations.

In this study, each geographic region is treated as a separate domain. Rather than training independent regional models, the proposed SFTNN incorporates regional context through Spatial Feature Modulation (SFM), enabling the model to learn both globally shared startup characteristics and region-specific relationships within a unified neural network architecture.

3.3.5 Spatial Non-Stationarity

Traditional machine learning models generally assume that the relationship between predictor variables and target outcomes remains consistent across all observations. However, this assumption is often unrealistic when dealing with geographically distributed data.

Spatial non-stationarity refers to the condition in which relationships between variables change across different geographic locations. In startup prediction, this means that identical startup characteristics may have different levels of importance depending on the entrepreneurial ecosystem in which the startup operates.

For example, investor participation, funding availability, and government support differ significantly between developed and emerging economies. Consequently, startup features that strongly indicate success in one region may not have the same predictive value elsewhere.

Boegl and Kounadi (2024) demonstrate that explicitly incorporating spatial heterogeneity into machine learning models generally leads to higher predictive performance than models that assume geographic uniformity. Instead of partitioning the dataset into separate regional models, the proposed SFTNN models spatial variation through learned feature modulation, allowing regional

context to dynamically influence the neural network's internal feature representations while preserving a shared global model.

3.3.6 Direct Parameter Interpretation and Counterfactual Ablation

Beyond producing accurate predictions, this study is also concerned with explaining how the proposed SFTNN's regional conditioning mechanism influences its predictions. Two complementary theoretical justifications underlie the interpretability approach adopted in this study: direct parameter inspection and counterfactual ablation.

Direct parameter inspection rests on the observation that not all model components require external approximation to be understood. Post-hoc attribution methods, such as Shapley-value-based approaches (Lundberg & Lee, 2017), are necessary when a model's internal computations cannot themselves be mapped transparently onto individual predictions — this is the case for the dense, entangled weight matrices of a standard feedforward layer, where no single parameter corresponds cleanly to a single interpretable effect. The scale (γ) and shift (β) parameters generated by the SFT layer are structurally different: by design, they are applied as a direct, literal element-wise transformation of the shared feature representation, $F' = \gamma \odot F + \beta$. Because this transformation is the actual mechanism, and not an approximation of it, γ and β can be read directly as a faithful record of how much scale and shift correction the network applies to a given region's feature representation, without requiring separate estimation.

Counterfactual ablation provides a second, complementary form of evidence, grounded in the general principle that a component's contribution to a model's behavior can be estimated by observing the effect of removing or neutralizing that component while holding all else constant. In this study, the effect of the regional conditioning mechanism is assessed by substituting a startup's learned region embedding with the mean embedding computed across all regions at inference time, and measuring the resulting change in classification performance. A substantial performance decline following this substitution

indicates that the model relies meaningfully on regional context to produce accurate predictions; a negligible change indicates that regional conditioning contributes little beyond what the shared trunk already captures. This ablation-based evidence does not depend on the qualitative plausibility judgments involved in reading γ/β heatmaps, and therefore serves as an independent, quantitative check on the same underlying claim.

Together, these two techniques allow this study to interpret the SFTNN's regional-adaptation behavior without adopting a post-hoc attribution framework such as SHAP, which was considered during the design of this study but not adopted, for the reasons discussed in Section 3.2.

Two further techniques are held as conditional extensions to this core interpretability approach, contingent on project time and on whether the primary evidence indicates a genuine regional signal worth decomposing further. Regional embedding clustering projects the learned region embedding table into two dimensions using Principal Component Analysis, following the entity-embedding visualization convention of Guo and Berkhahn (2016), to provide a qualitative check on whether ecosystems with similar entrepreneurial characteristics occupy nearby positions in the learned embedding space. Integrated Gradients (IG) (Sundararajan et al., 2017) is a gradient-based attribution technique that interpolates an input from a baseline (in this study, a zero-vector region embedding) to its true value across a fixed number of steps, integrating the model's gradient at each step to attribute the resulting change in prediction to individual input dimensions. Unlike the ablation check described above, IG decomposes this contribution at the level of an individual prediction and individual embedding dimension rather than as a single aggregate before/after comparison, and its attributions are guaranteed by construction to sum exactly to the difference between the baseline and true predictions — a completeness property the ablation check does not provide. This additional rigor comes at the cost of requiring gradient access at every interpolation step and depending on the interpolation path remaining within a region of the embedding

space the model has meaningfully learned from; for this reason, IG is treated in this study as a supplement to, rather than a replacement for, the ablation check, to be pursued only if the primary evidence and remaining project time both justify it.

3.4 Existing Technologies and Frameworks

This section describes the specific tools and frameworks employed in the development, training, and evaluation of the proposed Spatial Feature Transform Neural Network (SFTNN). Each technology is discussed in terms of its role within the study, its underlying design, and its known strengths and limitations, providing the practical foundation that complements the theoretical concepts presented in the previous section.

3.4.1 PyTorch

PyTorch is the primary deep learning framework used in this study. Unlike earlier libraries that require statically defined computation graphs, PyTorch constructs graphs dynamically at runtime through its Autograd engine, making it highly flexible for research prototyping and well suited to implementing conditional computation structures such as the SFT module, where the transformation applied to startup features depends on regional context (Cai & Ye, 2025). Its strength lies in this flexibility and the ease with which models can be debugged during development; its limitation is that production deployment generally requires additional optimization tooling compared to frameworks built around statically compiled graphs.

3.4.2 Kaggle and Regional Startup Databases

Kaggle and region-specific startup repositories serve as the primary data sources for this study, providing structured tabular records of venture characteristics such as funding history, organizational attributes, geographic location, and outcome labels. A known limitation of these sources is uneven regional coverage and high rates of missing values, particularly for early-stage

startups with limited reporting histories (Eranki et al., 2023). Preprocessing procedures address these limitations through imputation, normalization, and class balancing techniques consistent with best practices identified in the tabular machine learning literature (Hakami, 2024).

3.4.3 Scikit-learn

Scikit-learn is used for preprocessing pipelines, including feature normalization, missing value imputation, and evaluation metric computation. It provides standardized implementations of preprocessing transformations and cross-validation procedures that ensure reproducibility across experimental conditions. Because its components follow consistent and well-documented interface conventions, they integrate readily into the broader PyTorch-based pipeline used in this study, keeping the preprocessing and modeling stages modular and easy to verify.

3.4.4 Visualization and Interpretability Tooling

Interpretability outputs in this study are generated using Matplotlib and Seaborn, both of which provide the plotting utilities used to render the extracted γ_r and β_r vectors as region-by-hidden-dimension heatmaps. Because the interpretability approach adopted in this study depends on direct parameter extraction and counterfactual ablation rather than post-hoc attribution, no additional third-party explainability library is required for the core interpretability package: both core techniques are implemented directly within the existing PyTorch training and inference pipeline used elsewhere in this study (Section 3.4.1), by reading trained parameter tensors and by substituting embedding values at inference time, respectively. The region-embedding cluster visualization (Section 3.3.6) reuses Scikit-learn's Principal Component Analysis implementation (Section 3.4.3). Captum, PyTorch's native model-interpretability library, is held as a conditional dependency for the Integrated Gradients extension (Section 3.3.6) and is only installed if that conditional check is pursued.

This keeps the core interpretability stage of the pipeline consistent with the zero-cost, minimal-dependency constraint described in Section 4.8.3.

3.4.5 Entity Embeddings via PyTorch nn.Embedding

Geographic regions are represented as dense continuous vectors using PyTorch's nn.Embedding module, which maintains a lookup table of regional embedding vectors updated during backpropagation alongside all other model parameters. As demonstrated in the pioneering work of Guo and Berkhahn (2016), entity embeddings significantly reduce dimensionality compared to one-hot encoding and allow the model to discover latent relational structures, such that regions with similar entrepreneurial characteristics naturally develop similar embedding representations. A known limitation is that regions absent from the training set receive no learned embedding, requiring initialization strategies for unseen regions at inference time.

3.5 Problem Context

Standard neural network architectures applied to startup prediction learn a single globally optimized set of parameters across all observations, implicitly assuming that the relationship between startup features and success labels is geographically invariant. This assumption conflicts with extensive empirical evidence demonstrating that entrepreneurial ecosystems differ substantially in their institutional structures, investment norms, and developmental trajectories (Argaw & Liu, 2024; Vedula & Kim, 2019). Identical startup characteristics, such as a given funding amount or investor count, may carry fundamentally different predictive significance depending on whether a venture operates in a mature venture capital market or in an emerging ecosystem with more constrained investment availability.

Existing approaches address this limitation only partially. Feature gating mechanisms that operate globally may suppress features that are informative within specific regional contexts while appearing noisy in aggregate (Wei et al., 2025). Static categorical encodings of geographic location append regional

identifiers to the input vector without allowing regional context to influence how other features are processed internally. Graph-based and federated approaches provide richer relational representations but require extensive ecosystem connectivity information that is often unavailable in real-world startup datasets, particularly across emerging regions (Pan et al., 2026; Ma et al., 2025).

These architectural limitations are compounded by structural challenges in startup data itself, including class imbalance, high missingness rates, and uneven regional coverage. Models optimized on imbalanced global datasets risk learning predominantly from majority-class observations in dominant ecosystems, further marginalizing the patterns associated with startups in less-represented regions (Rodriguez-Belenguer et al., 2026). Together, these limitations establish the technical gap that the SFTNN addresses: the need for an architecture that jointly captures global entrepreneurial regularities and ecosystem-specific variation within a unified, scalable framework applicable to conventional tabular startup data.

3.6 Illustrations

The following figures illustrate the core architectural and conceptual components of the proposed SFTNN. All figures are original and are referenced throughout the discussion of the methodology in Chapter 4.

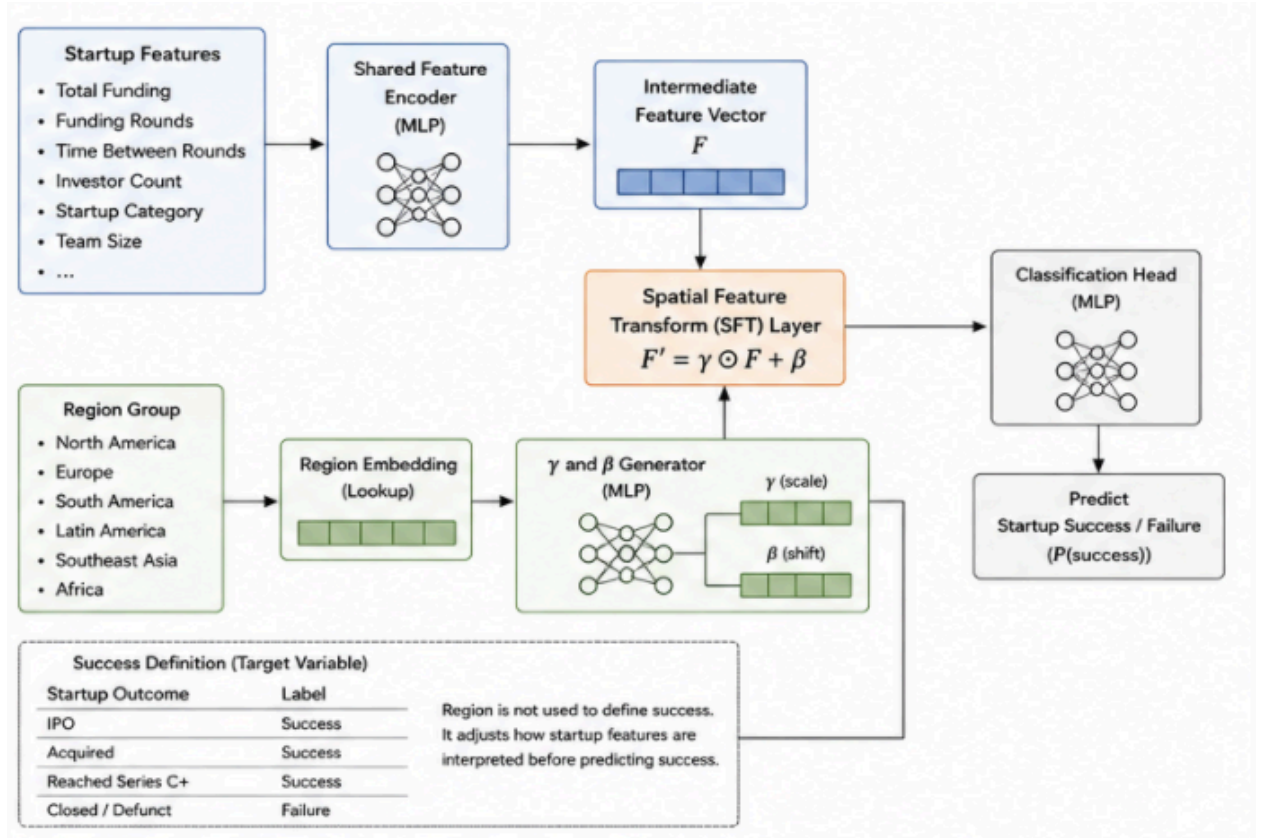


Figure 1. Architecture of the Spatial Feature Transform Neural Network (SFTNN).

Figure 1 presents the overall architecture of the SFTNN. Startup features are first processed by a shared feature encoder, producing an intermediate feature vector. In parallel, the startup's region group is mapped to a region embedding, which is passed through a γ and β generator to produce scale and shift parameters. These parameters are applied to the intermediate feature vector within the Spatial Feature Transform (SFT) layer, after which the modulated features are passed to a classification head that predicts the probability of startup success. As shown in the figure, region is not used to directly define success; rather, it adjusts how the startup's own features are interpreted before the final prediction is made.

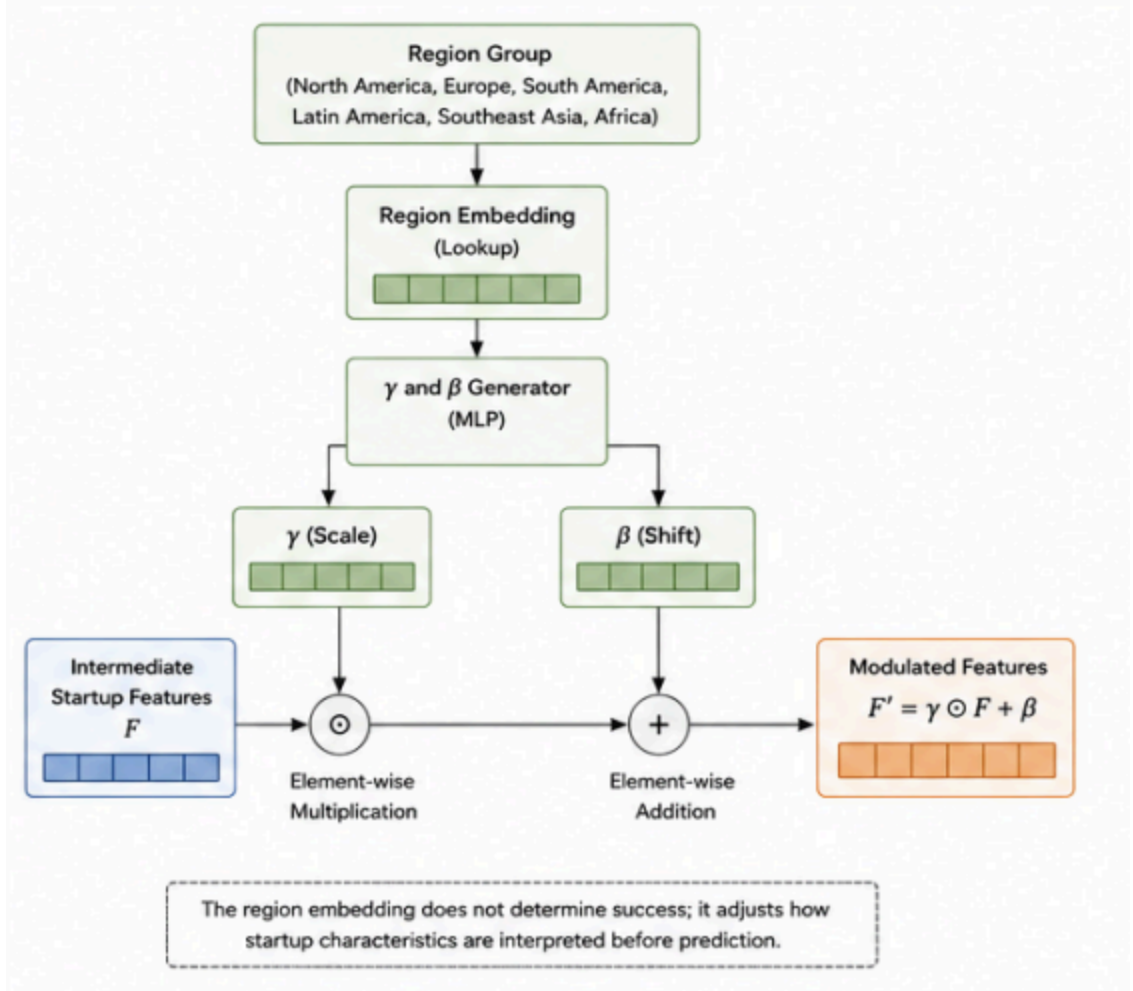


Figure 2. Feature-wise Linear Modulation (FiLM) conditioning mechanism used in the SFT module.

Figure 2 details the Feature-wise Linear Modulation (FiLM) mechanism underlying the SFT layer. The region embedding is passed through a γ and β generator to produce a scale vector (γ) and a shift vector (β), which are then applied to the intermediate startup feature vector through element-wise multiplication and addition, respectively. This process produces the modulated feature representation used for classification. The diagram reinforces a key design principle of the proposed model: the region embedding does not determine the predicted outcome on its own but instead conditions how the startup's characteristics are weighted and combined.



Figure 3. Regional variation in entrepreneurial ecosystem characteristics across global markets.

Figure 3 illustrates the regional entrepreneurial ecosystem groups used for constructing region embeddings in this study, namely North America, Europe, South Asia, East Asia, Latin America, Southeast Asia, and Africa. Each region is characterized by distinct ecosystem attributes, such as funding availability, investor activity, and market maturity, which collectively justify the need for region-aware feature modulation rather than a single globally shared set of model parameters.

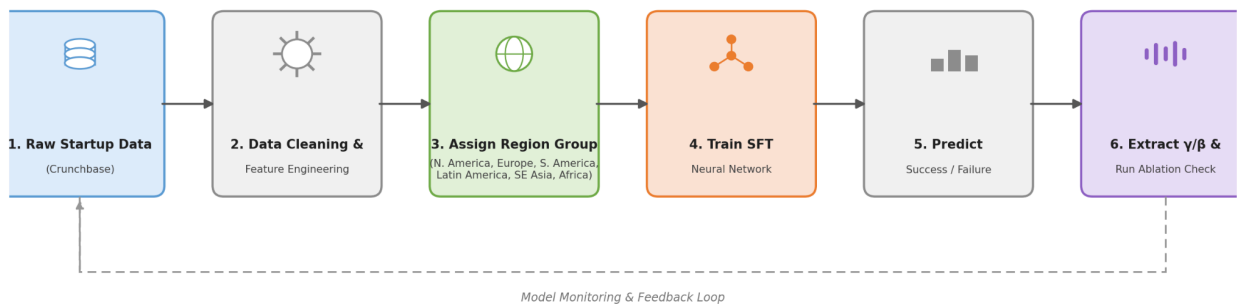


Figure 4. End-to-end data pipeline for the SFTNN, from raw data collection to interpretable prediction output.

Finally, Figure 4 presents the end-to-end data pipeline of the proposed system, from raw data collection through to interpretable prediction output. The pipeline begins with raw startup data sourced from Crunchbase, followed by data cleaning and feature engineering, assignment of region groups, training of the SFT neural network, prediction of startup success or failure, and extraction of the γ/β parameters together with the counterfactual ablation check described in Section 3.3.6 for interpretability. A feedback loop connects the interpretability stage back to the initial data stage, supporting continuous model monitoring and refinement.

Having established the foundational concepts of tabular startup prediction, the theoretical basis of feature-wise modulation and domain adaptation, the spatial non-stationarity problem motivating region-aware learning, and the tools and frameworks employed in this study, the next chapter presents the specific architecture, training procedure, dataset configuration, and experimental design of the proposed Spatial Feature Transform Neural Network.

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter will present the research environment, respondents, design, and methodology to be adopted to develop and evaluate the proposed Spatial Feature Transform Neural Network (SFTNN) architecture for region-aware startup success prediction. It will first describe the digital research environment and the dataset that will serve as the study's respondents, then will detail the complete methodological workflow: dataset collection and construction, the formal protocol for gathering the data, data cleaning and heuristic-based target label generation, and the conceptual and architectural design of the proposed SFTNN model relative to the baseline models to be used for comparison. The chapter will then present the requirements analysis and system architecture underlying the modeling pipeline, the development model and development approaches to be followed, the training strategy and hyperparameter optimization process, and the evaluation framework to be employed to assess model performance across multiple regions. Finally, it will present the project implementation timeline, budget, and the procedures to be used to ensure the methodological soundness and reproducibility of the study.

The methodology presented in this chapter will be informed by established studies on startup success prediction, data preprocessing, and interpretable machine learning. The data collection, integration, and feature engineering procedures will follow the preprocessing framework of Ferrati and Muffatto (2020), particularly in constructing time-dependent startup features such as funding-round counts and capital velocity. To prevent temporal data leakage, the preprocessing protocol will adopt the training-only fitting strategy described by Mashhadi et al. (2026), while their definition of IPO and acquisition as observed startup success will also guide the direct labeling procedure. In addressing the inherent uncertainty of startup datasets, the treatment of ambiguous operating companies and the discussion of label noise will be informed by the Positive-Unlabeled learning framework proposed by Pei et al. (2026), which

recognizes that some apparently unsuccessful startups may represent unobserved or delayed successes. These methodological foundations will collectively inform the design decisions presented throughout this chapter.

In addition, the binary definition of startup success adopted by Ningrum et al. (2024), which categorizes IPO and acquisition as successful outcomes and company closure as failure, will serve as the basis for the direct target labeling strategy to be employed in this study. Their findings will also support the inclusion of tree-based learning algorithms as competitive benchmarks for structured startup datasets, motivating the use of XGBoost as one of the baseline models against which the proposed SFTNN architecture will be evaluated.

4.1 Research Environment and Respondents

Research Environment

The research environment for this study will consist of an open-source data collection environment and a local computational environment for machine learning development. Startup datasets will be obtained primarily from publicly available Kaggle repositories and GitHub mirrors, supplemented by targeted web scraping of regional startup directories described in Section 4.2.2.

Data integration, preprocessing, model development, training, and evaluation will be implemented using Python 3.x and will be executed on two personally owned development machines described in Section 4.9.3. For computationally intensive tasks, particularly model training and hyperparameter optimization, Google Colab's free-tier GPU runtime will be used as a supplementary computing resource.

Throughout the study, all raw datasets, intermediate processing outputs, and finalized modeling-ready datasets will be maintained under version control to ensure data provenance, reproducibility, and traceability, consistent with the documentation procedures described in Section 4.2.3.

Respondents

As this study will rely strictly on secondary, publicly available, and web-scraped startup records rather than survey instruments or human participants, there will be no direct human respondents involved in the data collection process. Instead, the "respondents" of this study will be represented implicitly by the 38,090 usable startup records within the modeling-ready dataset described in Section 4.2, each representing a distinct company profile drawn from the global Crunchbase-derived baseline and the South Asian, African, Latin American, and Southeast Asian regional registries detailed in Section 4.2.2. Each record will encode a startup's funding history, industry vertical, founding information, geographic region, and observed or inferred exit outcome, collectively constituting the unit of analysis on which the proposed Spatial Feature Transform Neural Network (SFTNN) architecture will be trained and evaluated.

4.2 Research Instrument or Sources of Data

The primary research instrument of this study will be a constructed startup dataset developed through a multi-stage data collection, integration, and preprocessing pipeline. Rather than relying on surveys or physical measuring instruments, this study will utilize a consolidated research corpus specifically prepared for machine learning model development. The resulting dataset will serve as the primary dataset used throughout all model training and evaluation procedures. It will consist of three integrated data sources: (1) a global baseline of Crunchbase-derived startup records, (2) supplementary regional datasets covering South Asia, Africa, Latin America, and Southeast Asia, and (3) the processed and merged master dataset produced after data cleaning, validation, and integration. The global dataset will provide a diverse representation of startup characteristics for learning general startup behavior, while the regional datasets will enrich underrepresented ecosystems and will improve the representation of emerging markets within the proposed Spatial Feature Transform Neural Network (SFTNN) framework.

4.2.1 Primary Corpus: Global Crunchbase-Derived Baseline

The primary dataset will be constructed from historical Crunchbase-derived startup records obtained through publicly available Kaggle repositories containing company profiles, founding information, funding history, and exit outcomes. These datasets will be matched using their filenames, column structures, and data schemas to ensure compatibility prior to integration. Following the preprocessing framework of Ferrati and Muffatto (2020), duplicate records will be removed, inconsistent fields will be standardized, and schema validation will be performed before merging the datasets into a unified corpus. To provide richer startup representations, the baseline dataset will be further supplemented with transaction-level funding records containing detailed investment rounds, including Seed, Venture, Debt, and Series A through H financing events. These records will be matched using unique startup permalinks, achieving approximately 98.5% entity overlap between both datasets. Incorporating funding-round histories will enable the study to capture the progression of startup financing over time rather than relying solely on static organizational attributes.

Within this global baseline, startup records will be assigned to regional groups based on standardized country codes. North America, Europe, and East Asia will therefore be represented directly through the Crunchbase-derived corpus rather than through the targeted supplementary registries described in Section 4.2.2, which will be developed specifically to improve coverage in South Asia, Africa, Latin America, and Southeast Asia.

4.2.2 Regional Supplementary Startup Registries

To improve regional representation beyond the primary Crunchbase-derived baseline, four supplementary registries will be collected and integrated, targeting South Asia, Africa, Latin America, and Southeast Asia. The collection method for each will be described in Sections 4.2.2.1 to 4.2.2.4.

4.2.2.1 South Asia Regional Startup Registry

To improve the representation of South Asian startups, supplementary funding records from the Indian technology startup ecosystem covering the period from 2015 to 2021 will be collected and integrated into the primary dataset. These records will be programmatically extracted, normalized, and merged with the global Crunchbase-derived baseline after validating their schema compatibility. The data lineage will be further verified using a publicly available GitHub mirror of the original repository. Incorporating these records will increase the representation of South Asian startups and will provide additional regional diversity for model training.

4.2.2.2 African Regional Startup Registry

To address the limited representation of African startups within publicly available global datasets, startup funding records will be collected from Disrupt Africa's annual technology startup reports for 2022, 2023, and 2025. The extracted records will be standardized and merged into the primary dataset, with the resulting row counts matching the published figures reported by Disrupt Africa. Although no 2024 report will be available, this will not affect feature construction because temporal variables in this study will be derived relative to each startup's operational age rather than the publication year of the source dataset. Consequently, the absence of a 2024 report will not introduce missing values into the engineered feature space.

4.2.2.3 Latin American Regional Startup Registry

To expand the representation of Latin American startups, targeted web scrapers will be developed to collect historical funding information from GrowthList startup directories, with particular emphasis on startups based in Mexico and Brazil. Additional startup records will be obtained from the Y Combinator international startup directory to further improve

regional coverage. After data cleaning and schema standardization, these records will be merged into the primary dataset to strengthen the representation of Latin American startup ecosystems.

4.2.2.4 Southeast Asian Regional Startup Registry (Focal Regional Domain)

Since Southeast Asia will be the primary region of interest in this study, additional efforts will be made to address its limited representation within publicly available startup datasets. Initial data collection will determine that comprehensive row-level datasets for the region are generally accessible only through commercial platforms such as Crunchbase Pro, Tracxn, and DealStreetAsia, which will be beyond the financial scope of this research. To overcome this limitation, two supplementary datasets will be developed and integrated into the primary corpus.

First, a Playwright-based web scraper will be developed to collect startup information from Failory's publicly accessible country-specific startup directories for Singapore, Indonesia, Malaysia, Thailand, Vietnam, and the Philippines. This process will produce 359 additional startup records containing funding stage, industry sector, founding year, and related company information. Second, a Singapore-focused startup registry obtained from GrowthList will contribute an additional 84 startup records representing one of the region's most active entrepreneurial ecosystems.

When combined with the 564 Southeast Asian startups already present in the global Crunchbase-derived dataset, the augmentation process will increase the collected Southeast Asian dataset to 1,007 startups, representing a 78.5% increase in regional coverage. The resulting distribution will consist of Singapore (498), Malaysia (154), Vietnam (115), Thailand (90), Indonesia (90), the Philippines (59), and

Cambodia (1). This supplementary collection process will improve the representation of Southeast Asian startups available for model training, allowing the proposed Spatial Feature Transform Neural Network (SFTNN) model to learn regional characteristics from a substantially larger and more diverse sample. As will be described in Section 4.2.3, a portion of these collected records will not proceed to the final modeling-ready dataset due to the target-label generation criteria that will be applied uniformly across all regions.

4.2.3 Regional Composition of the Final Modeling Dataset

Table 1 summarizes the resulting regional composition of the full 38,090-row modeling-ready dataset, combining the primary global baseline (Section 4.2.1) with the four supplementary registries described in Section 4.2.2.

Region	Records	% of 38,090	Success-Labeled (%)
North America	23,976	62.95%	53.7%
Europe	7,679	20.16%	41.5%
Unmapped/Missing	2,499	6.56%	27.6%
East Asia	1,556	4.09%	45.3%
South Asia	791	2.08%	34.8%
Latin America	745	1.96%	37.9%
Southeast Asia	699	1.84%	26.6%

Africa	145	0.38%	22.8%
Total	38,090	100.00%	47.9%

Table 1. Regional Distribution of the Modeling-Ready Dataset

Note. Records with an unmapped or missing regional identifier (n = 2,499) will be excluded prior to model training (Section 4.3.2), yielding 35,591 region-assigned records for train/validation/test partitioning (24,913 / 5,339 / 5,339).

The supplementary registries described in Section 4.2.2 will substantially improve the representation of the four targeted regions—most notably a 78.5% increase in collected Southeast Asian records—without which these regions would be represented almost exclusively through whatever incidental coverage exists in the primary Crunchbase-derived baseline. The final region-labeled counts in Table 1 will be lower than the collected totals reported in Section 4.2.2, reflecting the exclusion of records with indeterminate outcomes during target-label generation (Section 4.3.2): of the 1,007 Southeast Asian startups collected, 699 (69.4%) will remain after this uniformly applied exclusion criterion. Despite this augmentation, the resulting dataset will remain geographically uneven, as Table 1 will show. Section 4.2.4 will examine this imbalance in detail and will describe how it will be managed within the study.

4.2.4 Regional Data Imbalance and Its Management

4.2.4.1 Regional Representation Imbalance

North America and Europe will jointly account for 83.1% of the full 38,090-row corpus (Table 1), while South Asia, Latin America, Africa, and Southeast Asia will collectively represent approximately 6% despite the targeted supplementary collection efforts in Section 4.2.2. This concentration will reflect the broader geographic bias toward data-rich

ecosystems identified in Section 1.1, rather than a limitation specific to this study's collection process.

4.2.4.2 Regional Success-Label Imbalance

Beyond raw row-count skew, the proportion of startups labeled Success will vary independently by region, from 22.8% in Africa to 53.7% in North America (Table 1). This will not simply restate the representation imbalance—Southeast Asia, for instance, will have more region-labeled records than Africa or Latin America yet will have the lowest Success rate of any named region (26.6%)—indicating that regional context will shape not only how much data is available but how the outcome distribution itself will behave. This pattern will be consistent with the spatial non-stationarity (Section 3.3.5) motivating region-conditioned modulation over a single globally shared classifier. Despite this per-region skew, the dataset's overall class balance will be comparatively even (47.9% Success, 52.1% Failure), an outcome of the age-conditioned funding-round momentum heuristic (Section 4.3.2) that will be applied uniformly across regions during distant-supervision label generation, rather than a property of any individual region's raw registry.

4.2.4.3 Management of Regional Data Imbalance

This imbalance will deliberately not be corrected through data-level resampling, such as Synthetic Minority Over-sampling Technique (SMOTE) or other synthetic oversampling techniques, which may risk fabricating unrealistic patterns given the limited number of authentic African, Latin American, and Southeast Asian records available. It will instead be addressed through three complementary mechanisms: (1) the targeted supplementary collection described in Section 4.2.2, which will increase the absolute volume of authentic minority-region records rather than synthesizing artificial ones; (2) the SFTNN's region-conditioned FiLM architecture (Section 4.4.1), which will allow underrepresented regions to

share the globally learned feature trunk rather than requiring independent, data-hungry per-region models; and (3) a per-region evaluation protocol (Section 4.10.2) that will report performance separately for each region, preventing strong aggregate results on North America and Europe from masking degraded performance elsewhere. Records with unmapped geographic origin ($n = 2,499$) will be excluded prior to model training (Section 4.3.2), yielding 35,591 region-assigned records.

4.2.5 Dataset Documentation and Metadata

To ensure data transparency and reproducibility, every dataset incorporated into this study will be accompanied by a documented source record describing its origin, collection method, and validation status. Following the documentation practices recommended by Ferrati and Muffatto (2020), each dataset will be categorized according to its provenance, including datasets verified against independent public mirrors, records collected through web scraping from identifiable sources, and datasets validated through schema and filename matching with companion releases. Maintaining this documentation will provide a clear audit trail of the data integration process and will enable every dataset included in the final corpus to be traced to its original source. In addition to documenting the data sources, the target-label generation process will be fully documented through a dedicated labeling workbook. The workbook will contain the configurable parameters used to generate the binary target labels, including the reference year, company-age thresholds, and funding-round decision criteria. These parameters will be maintained as editable values rather than hard-coded constants, allowing the labeling process to remain transparent, reproducible, and easily verifiable throughout the study.

4.2.6 Limitations of the Instruments

A rigorous data screening procedure will be performed before integrating candidate datasets into the final research corpus. Datasets exhibiting characteristics indicative of synthetic or unreliable data, such as unrealistically

complete attribute values combined with highly uniform geographic distributions, will be excluded. Enterprise acquisition databases whose unit of analysis focuses on acquiring corporations rather than individual startups will also be removed to preserve the consistency of the target variable. Likewise, datasets containing insufficient funding information or exhibiting extreme geographic concentration will be excluded to minimize sampling bias and improve the overall quality of the modeling dataset.

Despite the screening procedures, two limitations of the research dataset will remain. First, regional data imbalance. The dataset will exhibit substantial regional imbalance in both representation and outcome distribution, as detailed and addressed in Section 4.2.4. Second, because no comprehensive publicly available dataset exists for Southeast Asia, the regional dataset will be assembled by integrating multiple supplementary sources described in Section 4.2.2. These limitations will be acknowledged throughout the evaluation process and will be considered when interpreting the performance of the proposed model across different geographic regions.

4.3 Research Procedure

The raw consolidated dataset collected across heterogeneous international startup registries will undergo a rigorous, multi-stage preprocessing and feature engineering pipeline prior to machine learning model development. In venture capital predictive analytics, primary source data inherently will suffer from multi-source schema variance, reporting omissions, heavy-tailed financial distributions, and right-censoring across active company lifecycles. To transform these unstandardized administrative records into modeling-ready inputs, the research procedure will be organized around four objectives: (1) data quality and integrity enhancement, correcting spreadsheet corruption and standardizing inconsistent textual vocabularies; (2) predictive signal extraction, constructing domain-specific ratios such as capital velocity and stabilizing numerical variance for neural network convergence; (3) distant-supervision target labeling,

systematically inferring outcomes for right-censored operating entities while avoiding severe class imbalance; and (4) leakage control, enforcing a strict operational firewall that isolates evaluation data prior to parameter estimation. The complete procedure will be divided into two phases: (1) multi-source data gathering and integrity verification, and (2) leakage-free preprocessing, feature engineering, and target-label generation. Figure 5 will illustrate the preprocessing stage of this workflow.

4.3.1 Data Gathering

Data Extraction and Acquisition. To construct a representative global startup dataset that counters the historical Western reporting bias of commercial venture databases, primary startup records will be ingested from five categories of sources:

1. **Crunchbase Registry (Baseline Corpus):** Will be extracted via bulk historical exports, establishing the baseline international foundation for funding rounds, lead investor syndicates, board composition, and acquisition events.
2. **Disrupt Africa Directory:** Will be ingested via customized scraping modules to capture early-stage pre-seed and seed-stage tech ventures across Sub-Saharan Africa, as described in Section 4.2.2.2.
3. **GrowthList & Regional Directories:** Will be mapped to capture high-growth tech entities operating across Southeast Asia and Latin America, as described in Sections 4.2.2.3 and 4.2.2.4.
4. **Failory & Y Combinator (YC) Public Directories:** Will be ingested to harvest structured startup metadata and early-stage accelerator cohort information for the Southeast Asian focal region, as described in Section 4.2.2.4.
5. **Automated Web Harvesting (Playwright):** For entities with incomplete temporal funding records, headless browser automation

scripts will be executed to extract primary founding dates, public press releases, and registry registration timestamps directly from web portals.

Data Integrity Verification. Prior to preprocessing, the integrated multi-source dataset will undergo four integrity validation gates, applied in sequence so that later gates will operate only on records that have already passed earlier ones:

1. **Record Count Verification:** Parsed row totals will be cross-checked against raw source file headers to confirm zero record loss during ingestion, ensuring that every source record is either retained or explicitly and traceably excluded.
2. **Duplicate Verification & Permalinks:** Startup entities across independent sources will be cross-referenced using normalized permalink strings and exact website domain matching, since the same startup frequently will appear in more than one source registry under slightly different name formatting.
3. **Schema Validation:** Strict data-type constraints will be enforced across every column, including numeric formatting on financial amounts and consistent formatting on timestamps, to prevent type-coercion errors later in the pipeline.
4. **Cross-Source Conflict Resolution:** Where financial metrics will differ across sources for the same entity, values from primary legal registries or direct investment disclosures will be prioritized over aggregator estimates, since aggregator-reported figures are more prone to reporting lag and rounding.

4.3.2 Data Preprocessing

Data preprocessing will systematically transform heterogeneous raw records into standardized numerical inputs suitable for deep learning. This pipeline will address structural anomalies, extract domain-specific predictive

indicators, infer ground-truth labels, and format feature arrays without introducing empirical data leakage. Figure 5 will summarize this pipeline end-to-end.

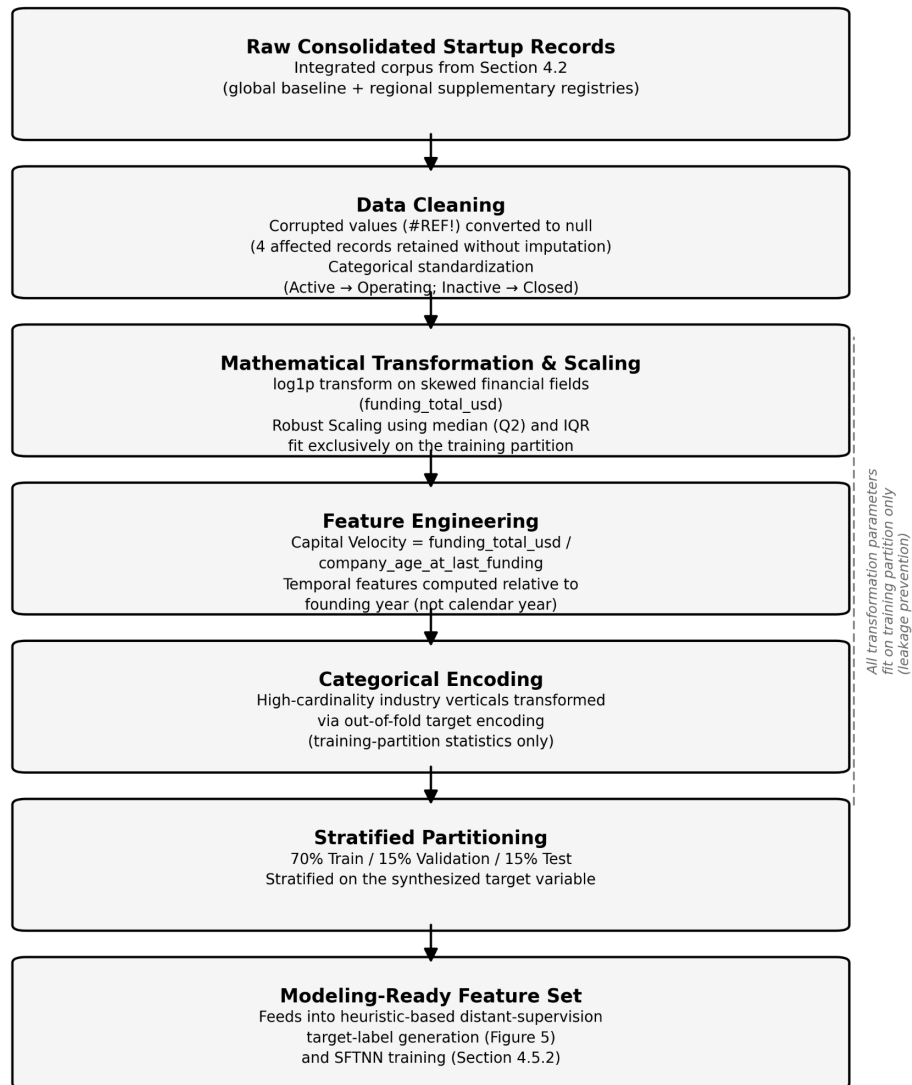


Figure 5. Data preprocessing and feature engineering pipeline, from raw consolidated records to the modeling-ready feature set.

4.3.2.1 Structural Validation and Corrupted Value Handling

Since the startup records will be collected from multiple repositories with different data formats and levels of data quality, the dataset will first be cleaned to ensure structural consistency before applying any statistical

transformations. Corrupted spreadsheet values, including Excel errors such as #REF!, will be identified and converted to null values to prevent processing errors in later stages. These values will be treated as missing data instead of being removed. Four records affected by these errors will be retained without imputation because the corrupted values will already be present in the original source datasets. Imputing these values could introduce information that will not be supported by the original records.

Categorical attributes will also be standardized to address differences in capitalization, spacing, and naming conventions across the source registries. For example, Active records will be standardized to Operating, while Inactive records will be mapped to Closed. This will ensure that a consistent operational-status vocabulary will be used during the target-label generation process. Duplicate startup records appearing across multiple sources will be consolidated using permalink and domain matching. Records with inconsistent dates, such as a last funding date occurring before the recorded founding date, will also be removed before feature engineering.

Primary Data Quality Issue	Automated Remediation Action	Rationale
Excel Formula Errors	Convert strings (#REF!, #VALUE!, #N/A) to formal null values	Four records will be retained without imputation because the corrupted values originated from the source datasets rather than the study's preprocessing pipeline.

Duplicate Entities	Merge records using permalink and domain matching	This will prevent the same startup from being counted more than once and potentially being distributed across different data partitions.
Status String Drift	Standardize Active, Operating Company → Operating; Liquidated, Dormant → Closed	This will establish a consistent operational-status taxonomy across all 38,090 usable records.
Text Casing & Spacing	Remove non-alphanumeric trailing characters and normalized text to lowercase	This will ensure that industry sub-tags are grouped consistently before target encoding.
Temporal Inconsistencies	Remove records where last_funding_date was earlier than founding_date	This will prevent invalid records from affecting company-age and capital-velocity calculations.

Table 2. Automated data cleaning operations and resolution scope

4.3.2.2 Missing Value Treatment

Missing data in startup registries may not occur randomly. Early-stage and emerging-market startups may have less complete records compared with startups from more established markets.

Therefore, applying the same imputation strategy to all missing values may introduce regional reporting patterns into the feature space.

Founding-year values that are later than the fixed 2026 reference year will be considered structurally invalid and will be treated as missing rather than being replaced through imputation. The actual founding year will not be reliably determined from the available fields. Records with an unresolvable founding year will therefore be excluded from the temporal feature calculations described in the succeeding sections. This approach will avoid generating artificial information to replace unavailable startup history.

4.3.2.3 Mathematical Transformations and Scaling

Continuous variables with highly skewed distributions, particularly `funding_total_usd`, will be transformed using the `log1p` function to reduce the effect of extreme values and improve feature stability during neural network training (Luengo et al., 2020):

$$x_log = \log1p(x) = \ln(1 + x)$$

The `log1p` transformation will be selected instead of the standard natural logarithm because a considerable portion of the startups in the dataset will have zero recorded funding. Since $\ln(x)$ is undefined when $x = 0$, using `log1p` will allow zero funding to remain a valid input and will result in a transformed value of 0 without requiring an arbitrary constant.

The log-transformed variables will then be normalized using Robust Scaling. The median and interquartile range will be computed exclusively from the training partition:

$$x_scaled = (x_log - Q2(X_train)) / (Q3(X_train) - Q1(X_train))$$

where Q2 represents the median or 50th percentile, and Q3 – Q1 represents the interquartile range (IQR). Robust Scaling will be selected because the dataset will contain highly skewed funding values, including startups with funding amounts several orders of magnitude higher than the median. Using conventional standardization based on the mean and standard deviation could allow these extreme values to strongly influence the scaling parameters (Ahsan et al., 2021). By using the median and IQR, Robust Scaling will be less affected by extreme observations.

Following the leakage-prevention framework proposed by Mashhadi et al. (2026), all preprocessing parameters, including medians, scaling bounds, and encoding statistics, will be calculated only from the training partition after data partitioning. The resulting parameters will then be applied without modification to the validation and test sets, as further discussed in Section 4.3.2.2.

4.3.2.4 Feature Engineering Mechanics

Total funding alone does not indicate how quickly a startup will accumulate its investment. For example, two startups may have the same total funding even though one will obtain the funding within its first year while the other will require several years. To capture this difference in funding momentum, a capital velocity feature will be calculated by dividing the total recorded funding by the startup's operational age at the time of its latest funding event:

$$\text{Capital Velocity}_i = \text{Total Funding USD}_i / \max(1, \text{Age at Latest Funding Event}_i)$$

The denominator will be set to a minimum value of 1 to avoid division problems for startups whose latest funding event will occur within their founding year. For example, two startups that each raise \$2,000,000 will have the same total funding value. However, if one startup will reach

this amount within its first year while the other will take eight years, their capital velocity values will differ by a factor of eight. This will allow the model to capture differences in fundraising momentum that cannot be represented by total funding alone.

All temporal features, including company age, will be calculated relative to each startup's founding year rather than a fixed calendar date. This will prevent older startups from being systematically penalized simply because more time will have passed between their founding and the date of analysis.

4.3.2.5 Categorical Encoding for High-Cardinality Fields

Industry vertical sub-tags will contain a large number of categories across the consolidated dataset. Using one-hot encoding for these fields would result in a sparse and high-dimensional feature matrix. On the other hand, simple integer encoding would introduce an artificial numerical order among industries that does not actually exist.

To address this, high-cardinality industry sub-tags will be converted using out-of-fold target encoding. Each category will be assigned a smoothed estimate of its success rate within the training partition, without using the observation's own target label when calculating its encoded value. This out-of-fold approach will help prevent data leakage that could occur with ordinary in-fold target encoding. In an in-fold approach, an observation's own label could influence its encoded feature, allowing the model to learn information specific to the training observation rather than a generalizable relationship between industry and startup success.

4.3.2.6 Heuristic-Based Distant Supervision: Target Label Generation

Publicly available startup datasets are inherently right-censored, with approximately 68% of the startups in the collected corpus remaining in an ambiguous Operating state without a recorded final outcome.

Consequently, relying solely on observed outcomes such as Acquired, IPO, and Closed would exclude most of the collected data and would result in a substantially imbalanced training dataset. This challenge will be further compounded by the one-sided label uncertainty commonly observed in venture capital datasets, where startups that appear unsuccessful may simply have unrecorded or delayed exit events. Pei et al. (2026) will address this issue through Positive-Unlabeled (PU) learning in the MIRAGE-VC framework, highlighting that incomplete outcome reporting is particularly common in emerging startup ecosystems with limited data coverage. To address this limitation, this study will adopt a heuristic-based distant supervision approach using a fixed reference year of 2026 to infer binary target labels from observable startup characteristics. Because a portion of the labels will be inferred rather than directly observed, the proportion of proxy-generated labels will be explicitly reported to ensure transparency during model evaluation.

Startups with explicitly observed outcomes will be assigned labels directly without inference. Companies that achieve an Initial Public Offering (IPO) or are Acquired will be labeled as Success ($Y = 1$), while companies whose operating status is Closed will be labeled as Failure ($Y = 0$). Before applying these rules, status values will be standardized by mapping Active to Operating and Inactive to Closed, ensuring a consistent labeling vocabulary throughout the dataset. This binary definition of startup success will follow the operationalization adopted by Ningrum et al. (2024) and will be consistent with the exit-event definition used by Mashhadi et al. (2026). Company age will be computed as $\text{company_age} = 2026 - \text{founded_year}$, with invalid founding years beyond the reference year treated as missing values rather than corrected through imputation.

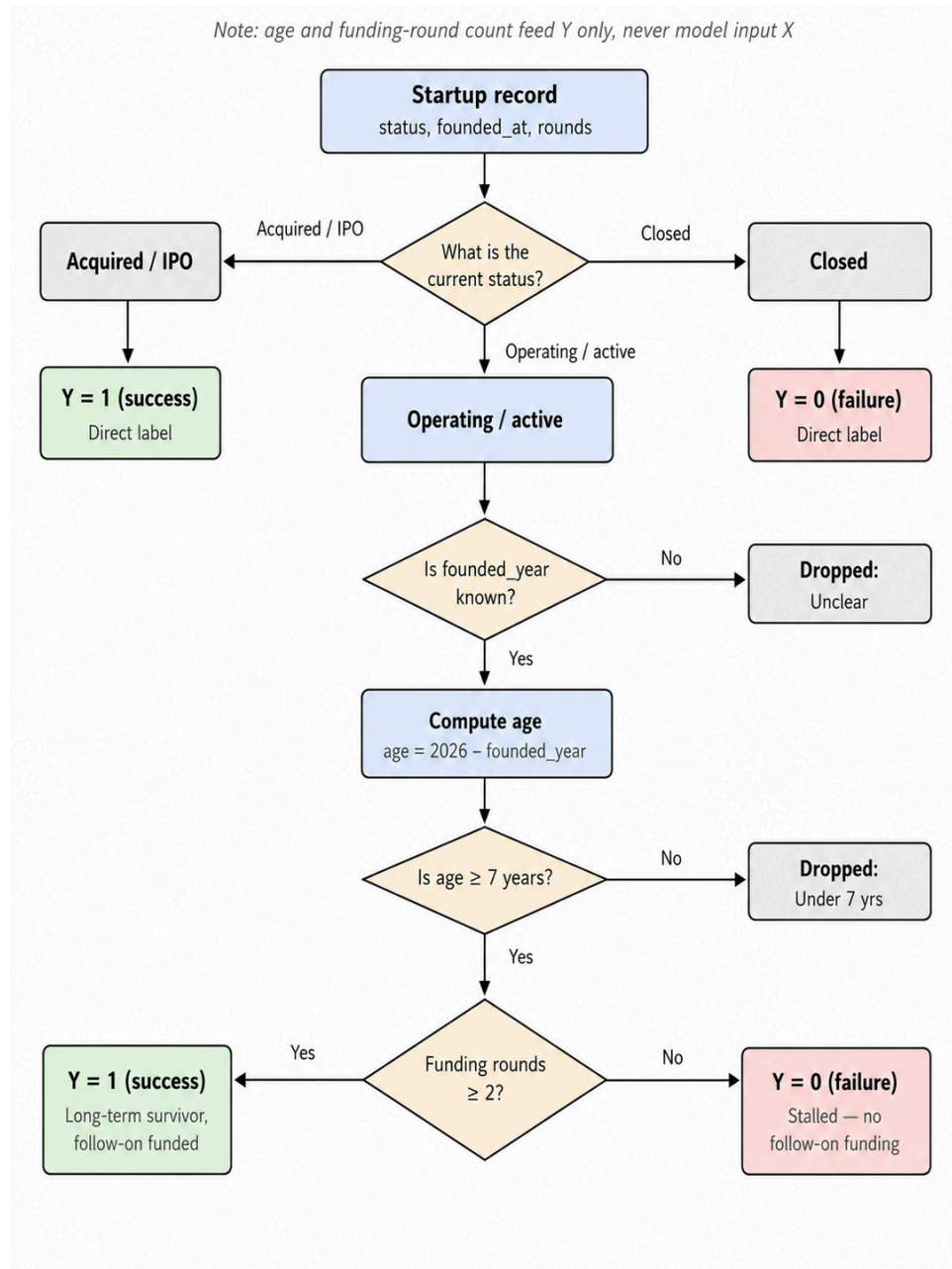


Figure 6. Decision flow for heuristic-based distant-supervision target-label generation.

For startups that will remain in the Operating state, additional heuristic rules will be applied to infer long-term outcomes. Companies with unknown founding years will be excluded because company age cannot be determined. Likewise, startups younger than three years will be

excluded since insufficient operational history will exist to evaluate their trajectory. Companies between three and six years old will also be excluded because their long-term outcomes will remain uncertain. For startups that will have operated for at least seven years, funding-round activity will serve as the deciding criterion. Companies that will have completed two or more funding rounds will be labeled as Success ($Y = 1$), representing long-term survivors with sustained investor support, whereas companies with one or no funding rounds will be labeled as Failure ($Y = 0$), representing ventures that will likely have stalled despite extended operational lifetimes.

Rejected Alternative (v1) — Calendar Funding-Gap Rule

An initial labeling approach will use a calendar-based funding-gap rule. Under this approach, operating startups with no funding events within the previous five years will be classified as failures. However, analysis will show that the historical snapshot data contain a high concentration of funding records between 2013 and 2015. When funding recency will be measured against the 2026 reference year, this approach will cause most operating startups to be classified as failures, resulting in 34,240 successes and only 3,850 failures, or approximately a 9:1 class imbalance.

The adopted age-conditioned heuristic will produce 12,673 proxy-labeled Success startups and 16,002 proxy-labeled Failure startups. When combined with the directly observed outcomes of 5,565 explicit exits and 3,850 explicit closures, the final modeling dataset will contain 18,238 Success records (47.9%) and 19,852 Failure records (52.1%) across 38,090 usable startups. This will result in a near-balanced class ratio of approximately 0.92:1. In comparison, using only directly observed outcomes will reduce the usable dataset to approximately 9,415 startups (5,565 explicit exits + 3,850 explicit closures).

Label	Count	% of 38,090	Composition
Success (Y = 1)	18,238	47.9%	5,565 explicit exits (IPO/Acquired) + 12,673 momentum-based proxy labels
Failure (Y = 0)	19,852	52.1%	3,850 explicit closures + 16,002 stagnation-based proxy labels
Total	38,090	100.0%	Class ratio $\approx$ 0.92:1

Table 3. Success-Failure Breakdown Composition

Since most of the labels will be generated through heuristic inference rather than directly observed outcomes, this uncertainty will be recognized as a limitation of the study (Section 4.2.6). To maintain label validity and prevent leakage, the variables used to generate the target labels — company age, funding-round count, and operating status — will be used only during label construction and will be excluded from the model's input features, as discussed in Section 4.3.2.3.

4.3.2.7 Data Partitioning and Leakage Prevention

After target-label generation, the modeling dataset will be divided into three stratified and non-overlapping partitions before any statistical transformation is fitted. Records with missing or unmapped regional identifiers ($n = 2,499$) will be excluded before partitioning, as described in Section 4.2.3. This will result in 35,591 region-assigned records, which will be divided into training, validation, and test sets using a 70% / 15% / 15% split:

- **Training:** $n = 24,913$
- **Validation:** $n = 5,339$
- **Test:** $n = 5,339$

The partitions will be stratified according to the target label to maintain a similar class distribution across the three sets.

Data partitioning will be performed before fitting any statistical parameters. Figure 7 will illustrate the resulting leakage-prevention process. Column medians used for imputation, Robust Scaling parameters, and target-encoding probabilities will be calculated only from the training partition. These parameters will then be applied without modification to the validation and test partitions. Re-fitting the scaler, imputer, or encoder using held-out data will not be permitted.

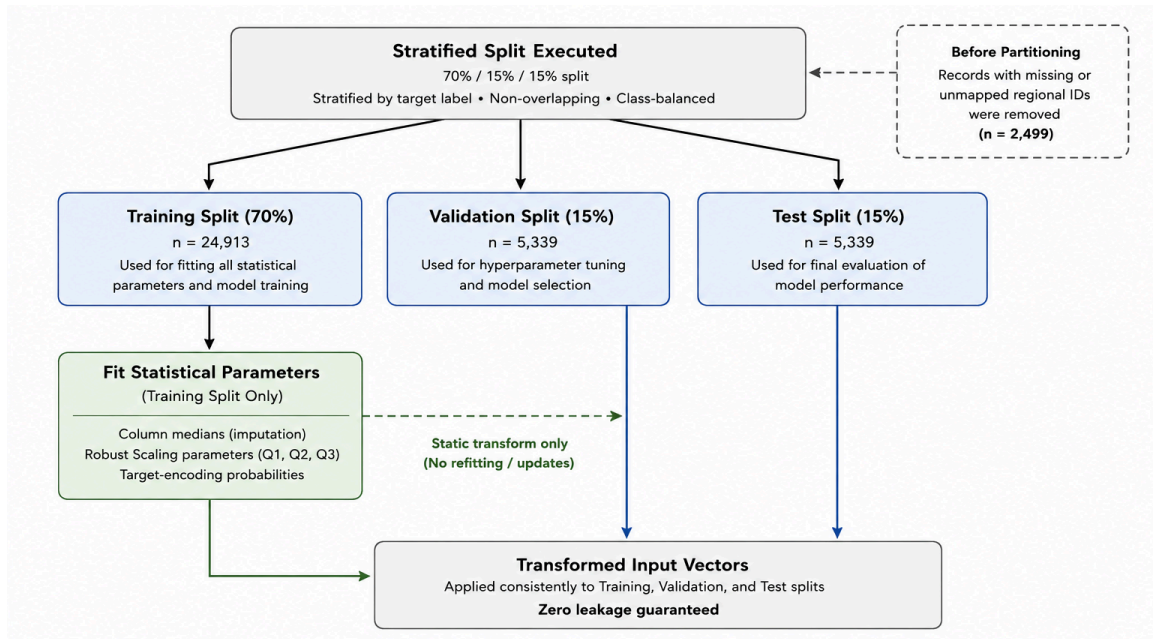


Figure 7. Leakage-prevention firewall applied during preprocessing.

4.3.2.8 Feature Separation for Model Training

To prevent target leakage, the operational status and absolute company-age variables used during target-label generation will be removed from the model's feature space. The remaining preprocessed inputs will then be separated into two tensors based on the proposed dual-track architecture.

The first will be the **universal feature tensor**, containing funding total, capital velocity, funding rounds, industry encoding, and other supporting temporal features. This tensor will be passed to the shared feature trunk. The second will be the **categorical region tensor**, which will be used as input to the Regional FiLM generator described in Section 4.4.1.

4.4 Conceptual Framework

This section will present the conceptual framework that will guide the implementation and evaluation of the proposed Spatial Feature Transform Neural Network (SFTNN) for region-aware startup success prediction. The framework will serve as the methodological blueprint of the study, illustrating the complete experimental workflow from data preparation and model development to performance evaluation and interpretability analysis. Organized as a six-stage pipeline, each stage will build on the output of the previous stage to provide a systematic, reproducible, and logically ordered process for addressing the research problem. To preserve methodological integrity, a strict operational firewall will be established at the beginning of the workflow to prevent data leakage, ensuring that all preprocessing transformations and model optimization procedures will be performed without exposing information from the validation and test datasets.

The six-stage framework will be explicitly aligned with the study's specific objectives. Stage 1 will address **Specific Objective 1** by constructing a leakage-free, modeling-ready dataset through preprocessing, feature transformation, and dataset partitioning. Stages 2 and 3 will establish optimized baseline models, namely the Standard Multi-Layer Perceptron (MLP), XGBoost, and TabNet, providing reference performance for comparison with the proposed approach. Stage 4 will address **Specific Objective 2** by implementing the proposed SFTNN, which will learn a shared representation of startup characteristics while adapting that representation to different entrepreneurial ecosystems through Feature-wise Linear Modulation (FiLM). Stage 5 will address **Specific Objective 3** by comparing the predictive performance of all models using both overall and region-specific evaluation metrics. Finally, Stage 6 will address **Specific Objective 4** by examining the learned regional modulation parameters and conducting counterfactual ablation experiments to verify the contribution of geographic conditioning to the model's predictions.

4.4.1 Six-Stage Conceptual Framework

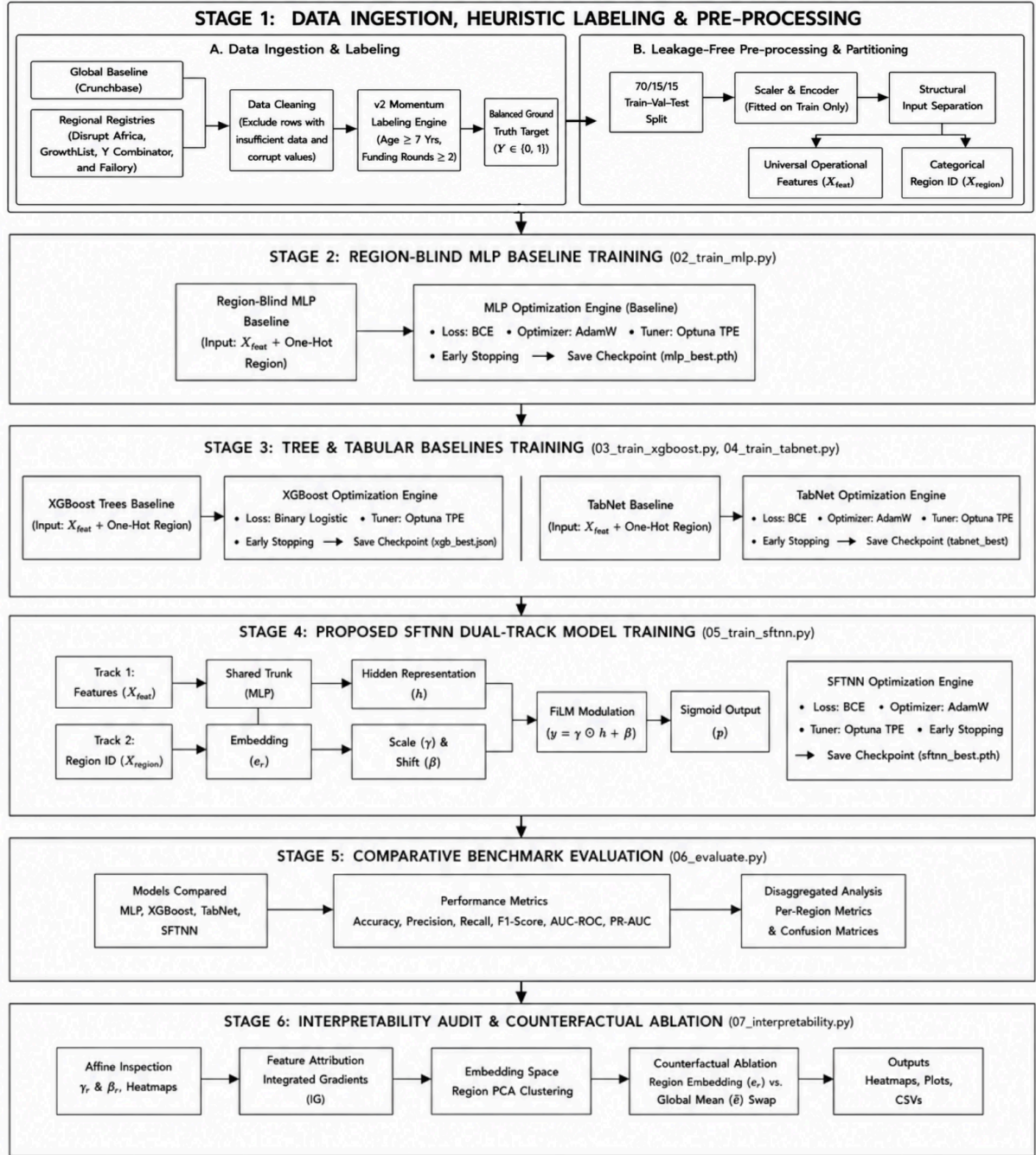


Figure 8. Conceptual and Methodological Framework of the Proposed SFTNN

As illustrated in Figure 8, the proposed methodology will be organized into six sequential stages that will collectively transform raw startup data into region-aware predictive models and interpretable evaluation results. Each stage

will perform a distinct function and will contribute directly to one or more of the study's specific objectives. The workflow will proceed as follows:

Stage 1: Data Ingestion, Heuristic Labeling & Pre-processing

Raw startup records collected from the global Crunchbase-derived dataset and four supplementary regional startup registries (Sections 4.2.1–4.2.2) will undergo systematic data preprocessing, feature preparation, and target-label generation. To preserve the integrity of model evaluation, a strict operational firewall will be established at this stage. All preprocessing parameters, including imputation values, scaling statistics, and encoding mappings, will be estimated exclusively from the training partition and subsequently applied unchanged to the validation and test sets, preventing information leakage across data partitions.

The preprocessing pipeline will include data cleaning, missing-value treatment, feature transformation, feature engineering, categorical encoding, and heuristic-based target-label generation (Sections 4.3.2.1–4.3.2.6). The processed dataset will then be partitioned into training (70%), validation (15%), and test (15%) subsets using stratified sampling based on startup outcome and geographic region (Section 4.3.2.7). The output of this stage will be a leakage-free, modeling-ready dataset that will serve as the common input for all subsequent models.

Stage 2: Region-Blind MLP Baseline Optimization

The first baseline model will be a standard Multi-Layer Perceptron (MLP), where the regional identifier will be represented as a one-hot encoded input alongside the remaining startup features. Because regional information will be treated as an ordinary categorical variable, the model will not explicitly adapt its internal feature learning to different entrepreneurial ecosystems. This baseline will represent a conventional neural-network approach and will provide a reference point for evaluating

the effectiveness of the proposed region-aware architecture. The complete model configuration and optimization procedure will be presented in Section 4.7.

Stage 3: Tree-Based and Attention-Based Baseline Optimization

To provide broader comparative benchmarks, two baseline models will be trained using the same preprocessed dataset and identical train-validation-test partitions: XGBoost, representing gradient-boosted decision trees, and TabNet, representing attention-based deep learning for tabular data. Similar to the MLP baseline, both models will receive geographic region as a standard one-hot encoded input feature without explicit spatial conditioning. Their optimization procedures will be described in Section 4.7. Together with the MLP, these models will establish strong benchmark performance against which the proposed SFTNN will be evaluated.

Stage 4: Proposed SFTNN Dual-Track Architecture Optimization

The proposed Spatial Feature Transform Neural Network (SFTNN) will introduce a dual-track architecture that will separate universal startup characteristics from regional information. Universal startup features will be processed through a shared feature extraction network to learn general patterns associated with startup success, while the regional identifier will be processed through a parallel conditioning branch that will generate learned scaling (gamma) and shifting (beta) parameters.

The two branches will converge through a Feature-wise Linear Modulation (FiLM) operation,

$$y_{\text{o}} d_{\text{u}} a_{\text{e}} d = \gamma_{\text{r}} \odot h + \beta_{\text{r}}$$

where the learned regional parameters will dynamically adjust the shared hidden representation according to the entrepreneurial ecosystem of each

startup. Instead of learning separate models for every region, this design will enable the network to preserve globally shared knowledge while adapting its internal representations to localized regional characteristics. The modulated representation will subsequently be passed to the final classification layer to predict startup success. The detailed network architecture and optimization procedure will be presented in Sections 4.5 and 4.7.

Stage 5: Comparative Benchmark Evaluation

The optimized MLP, XGBoost, TabNet, and SFTNN models will be evaluated using the reserved 15% blind test set ($n = 5,339$) under a common evaluation protocol to ensure a fair comparison. As described in Section 4.10, model performance will be measured using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC (Equations 5–10), and will be reported at two complementary levels.

At the aggregate level (Section 4.10.1, Table 7), each model will be evaluated across the entire test set to compare its overall predictive performance. At the regional level (Section 4.10.2, Table 8), the same metrics will be computed separately for each geographic region (North America, Europe, East Asia, South Asia, Latin America, Southeast Asia, and Africa). This regional evaluation will determine whether improvements observed at the aggregate level remain consistent across both established and underrepresented startup ecosystems, allowing regional performance disparities and potential geographic bias to be identified. Together, these evaluations will satisfy Specific Objective 3 by determining whether the proposed SFTNN will provide more balanced and reliable startup success prediction than the baseline models.

Stage 6: Interpretability Audit and Counterfactual Ablation

Following performance evaluation, the proposed SFTNN will undergo an interpretability analysis to examine how regional information

will influence its predictions. The procedures described in Section 4.10.2 will verify that the learned regional conditioning mechanism will contribute meaningful information rather than merely increasing model complexity.

First, the learned regional scaling (γ) and shifting (β) parameters will be extracted and visualized as heatmaps to illustrate how the model will adjust its shared feature representation for each geographic region. These visualizations will provide a direct interpretation of the region-specific adaptations learned by the FiLM conditioning mechanism.

Second, a counterfactual regional ablation experiment will be performed by replacing each startup's learned regional embedding (r) with the global mean embedding (e) during inference while keeping all other network parameters unchanged. The resulting changes in Recall, F1-Score, ROC-AUC, and PR-AUC will quantify the contribution of regional conditioning to predictive performance. A measurable decline in performance will indicate that the learned regional embeddings capture meaningful geographic information beyond the globally shared feature representation.

Finally, if the learned regional embeddings demonstrate a meaningful contribution, they will be projected into a two-dimensional space using Principal Component Analysis (PCA) to visualize relationships among regions. Although supplementary, this visualization will provide an additional qualitative assessment of whether regions with similar entrepreneurial ecosystem characteristics form coherent clusters within the learned embedding space. Collectively, these analyses will satisfy Specific Objective 4 by evaluating both the interpretability and the practical contribution of the proposed region-aware conditioning mechanism.

4.5 Analysis and Design

This section will detail the operational requirements and structural design specifications of the proposed modeling pipeline. It will bridge the gap between the conceptual framework presented in Section 4.4 and the physical implementation of the four models compared in Section 4.7.

4.5.1 Requirements Analysis

The requirements of the modeling pipeline will be divided into functional specifications, which will define the core capabilities the pipeline will provide, and non-functional specifications, which will define the quality attributes the pipeline will satisfy.

Functional Requirements

- 1. Data Integration** - The system shall ingest, validate, and integrate the multi-source, multi-regional startup corpus described in Section 4.2 into a single, schema-consistent dataset suitable for model development, training, and evaluation.
- 2. Target Label Generation** - The system shall generate binary startup success labels using the parameter-driven heuristic distant-supervision procedure described in Section 4.3.2 without relying on undocumented rules or hard-coded thresholds.
- 3. Model Training and Evaluation** - The system shall train and evaluate four predictive models—the baseline models MLP, XGBoost, TabNet, and the proposed SFTNN architecture—using an identical preprocessing pipeline, feature set, and train/validation/test partitioning to ensure fair comparison.

4. Performance Assessment - The system shall compute and report aggregate and regional evaluation metrics for all predictive models using the standardized performance measures defined in Section 4.10.

5. SFTNN Interpretability - The system shall extract and expose the learned γ (gamma) and β (beta) regional modulation parameters of the SFTNN architecture and support the counterfactual region-embedding ablation procedure described in Section 4.10.2.

6. Model Inference and Prediction Generation - The system shall accept startup feature inputs through the deployed interface, apply the serialized preprocessing pipeline, and generate startup success predictions using the trained SFTNN and XGBoost models without requiring model retraining.

7. Deployment Pipeline Integration - The system shall integrate the Streamlit user interface and FastAPI inference backend to enable real-time communication between user inputs, preprocessing operations, model execution, and prediction visualization.

8. Prediction Response Visualization - The system shall present model outputs, including prediction probabilities, model comparison results, and SFTNN regional modulation parameters (γ and β), through an interactive visualization interface.

Non-Functional Requirements

1. **Reproducibility:** All preprocessing parameters, target-label thresholds, model configurations, and experimental settings shall be externally configurable rather than hard-coded, consistent with the documentation practices described in Section 4.2.3, to enable reproducible experimentation.
2. **Leakage Prevention:** All preprocessing, feature encoding, and normalization parameters shall be fitted exclusively on the training

partition and applied unchanged to the validation and test partitions. Furthermore, no variable used in constructing the target label shall be included as an input feature during model training.

3. **Computational Feasibility:** The complete experimental pipeline, including hyperparameter optimization for all four models, shall execute within the computational resources described in Section 4.1 and remain achievable within the twelve-week implementation schedule presented in Section 4.9.1.
4. **Interpretability:** The proposed region-conditioning mechanism shall remain inherently interpretable through its learned modulation parameters (γ and β), eliminating the need for approximation-based post-hoc attribution techniques and supporting the interpretive methodology described in Section 4.10.2.
5. **Consistency** - The deployed inference pipeline shall utilize the same preprocessing configurations, feature transformations, and model parameters used during experimental evaluation to ensure consistency between research results and application predictions.
6. **Maintainability** - The system shall maintain separation between the presentation layer, inference layer, and model asset layer to allow independent modification, testing, and replacement of individual components without affecting the entire system.

4.5.2 System Architecture

4.5.2.1 Full Stack Integration Architecture

The application logic layer will be implemented as a FastAPI backend that will function as the system's inference engine. User inputs submitted through the Streamlit dashboard will be received through a data ingestion endpoint, where they will be validated, parsed, and transformed using the serialized preprocessing pipeline. This will ensure that the same feature engineering, robust scaling, target encoding, and derived feature

calculations applied during model training will be consistently reproduced during deployment. The processed feature vector will then be forwarded to the inference pipeline, where the trained SFTNN and XGBoost models will execute in parallel, enabling both models to evaluate the same startup instance under identical preprocessing conditions. The resulting prediction probabilities, together with the extracted SFTNN modulation parameters (γ and β), will be serialized into a structured JSON response and returned to the presentation layer for visualization.

The data storage and model assets layer will store the deployment artifacts required during inference, including the serialized preprocessing pipeline (*preprocessor.joblib*), the finalized SFTNN model (*sftnn_model.pth*), and the trained XGBoost baseline model. During runtime, these artifacts will be loaded directly by the backend, eliminating the need for retraining while ensuring that every prediction will be generated using the same validated preprocessing configuration and model parameters evaluated during the experimental phase. This separation of presentation, inference, and model assets will provide a modular and maintainable deployment architecture that will support consistent prediction, model comparison, and the interactive dashboard demonstration presented in Chapter 5.

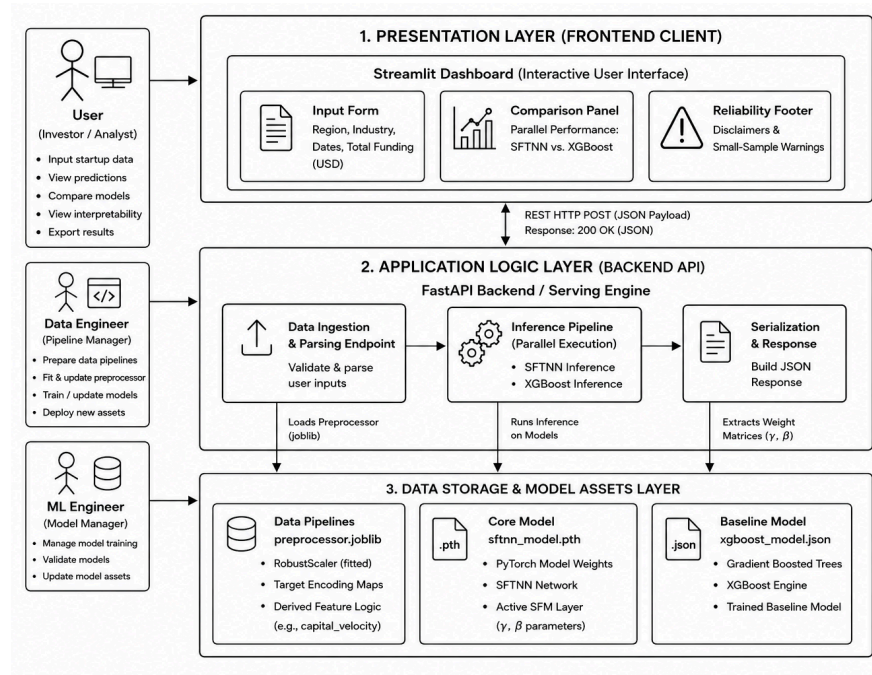


Figure 9. Three-Tier System Architecture of TechVenture-SFTNN Platform

4.5.2.2 SFTNN Modulation Architecture

Let each observation in the modeling-ready dataset be represented as a tuple (x, r, y) . The vector $x \in \mathbb{R}^n$ will denote the universal, region-agnostic corporate attributes described in Section 4.3.2 (log-scaled funding totals, capital velocity, encoded industry vertical bins). The scalar $r \in \{1, \dots, R\}$ will denote the categorical regional identifier corresponding to one of the R geographic classes represented in the dataset, spanning South Asia, Africa, Latin America, and the focal Southeast Asian markets of Indonesia, Malaysia, Singapore, Thailand, Vietnam, and the Philippines. The scalar $y \in \{0, 1\}$ will denote the binary target label synthesized via the distant-supervision procedure described in Section 4.3.2. Critically, x will contain no trace of the age, funding-round-count, or status fields used to construct y , which will serve as the structural mechanism that prevents label leakage.

The Startup Feature Trunk. The universal attribute vector x will be passed through a feed-forward trunk composed of L fully connected layers with ReLU activations, producing a region-agnostic hidden representation h .

Equation 1. Shared Trunk Forward Pass

$$h^{(l)} = \text{ReLU}(W^{(l)}h^{(l-1)} + b^{(l)}), l = 1, \dots, L$$

with $h(0) = x$ and the final hidden state $h = h(L) \in \mathbb{R}^d$ carrying region-agnostic corporate signal forward into the modulation stage.

The Region Embedding and Affine Parameter Generator. In parallel, the regional identifier r will be used to generate a dedicated pair of affine modulation parameters. The regional identifier r will first be projected into a dense trainable embedding $e_r = E[r] \in \mathbb{R}^k$ via a lookup embedding matrix $E \in \mathbb{R}^{R \times k}$; this embedding is then passed through a small dedicated feed-forward branch that outputs two vectors matching the dimensionality of h — a scaling vector γ_r and a shifting vector β_r .

Equation 2. Regional Scale and Shift Generation

$$[\gamma_r; \beta_r] = W \cdot \text{ReLU}(W_e e_r + b_e) + b$$

SFTNN. The regional scale and shift vectors will then be applied element-wise to the shared hidden representation, giving each region a learned affine correction on an otherwise universal feature pathway.

Equation 3. SFTNN Affine Transformation

$$y_{\text{modulated}} = \gamma_r \odot h + \beta_r$$

where $\odot$ will denote element-wise (Hadamard) multiplication. This operation will give the network the capacity to shift decision boundaries per region without maintaining fully separate regional sub-networks, and, critically, will leave γ_r and β_r available as directly inspectable parameters for the interpretability analysis described in Section 4.10.2.

Classifier Head. The modulated representation will be passed through a final linear layer and Sigmoid activation to produce the probability estimate described in Section 4.4.2, optimized against a Binary Cross-Entropy objective.

Equation 4. Output Probability and Loss

$$p = \sigma(w_o^T y_{\text{modulated}} + b_o)$$

$$L = -(1/N) \sum [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

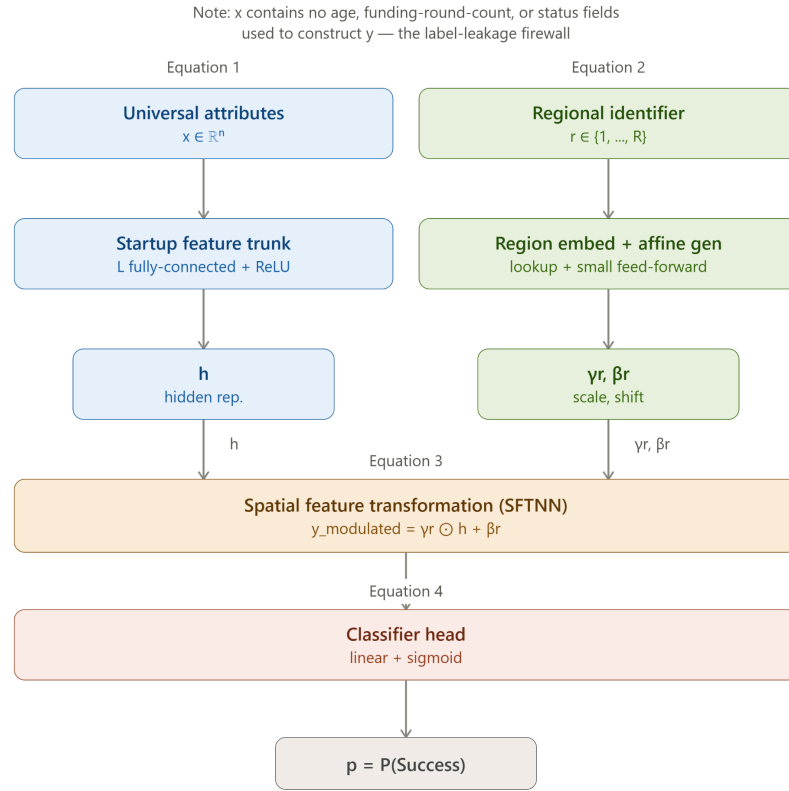


Figure 10. Full SFTNN model architecture, annotated with Equations 1–4.

Collectively, Equations 1 through 4 will define the complete forward propagation process of the proposed Spatial Feature Transformation Neural Network (SFTNN) network. The shared trunk will first learn region-independent startup representations, after which the region embedding will generate affine modulation parameters that will adapt these representations according to

regional context before producing the final probability estimate. This formulation will enable the model to learn universal startup characteristics while accounting for differences across entrepreneurial ecosystems within a single end-to-end trainable architecture.

4.5.3 Detailed Design and Process Flow

Because this study will develop both a machine learning prediction pipeline and a deployable demonstration platform, the detailed system design will be represented through complementary diagrams that will capture the system from both the functional and methodological perspectives.

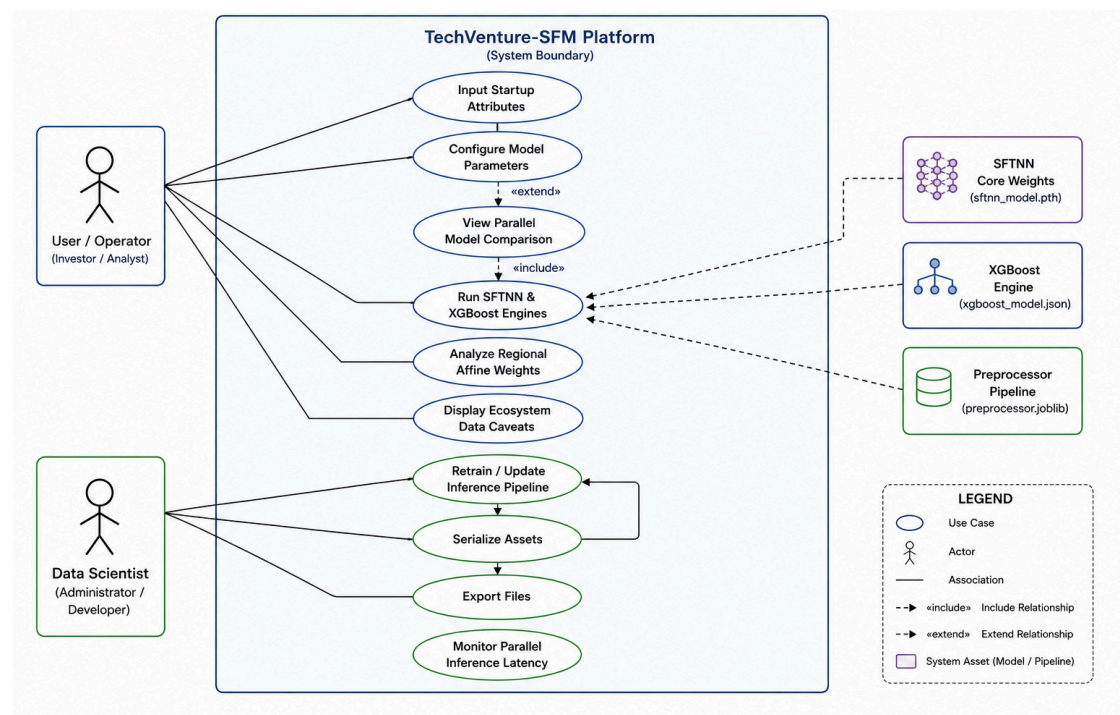


Figure 11. UML Use Case Diagram for TechVenture-SFTNN Platform

Figure 11 will describe how different users will interact with the deployed system. The User/Operator (Investor or Analyst) will perform prediction-related functions, including entering startup attributes, configuring model parameters, executing the SFTNN and XGBoost inference engines, comparing model predictions, analyzing the learned regional affine weights, and viewing

ecosystem-specific reliability warnings. The Data Scientist (Administrator or Developer) will be responsible for maintaining the deployed system by retraining the inference pipeline, serializing updated model assets, exporting deployment files, and monitoring inference performance. These use cases will interact with the deployed model assets—including the SFTNN model, XGBoost baseline, and preprocessing pipeline—which will be loaded during runtime to support real-time prediction.

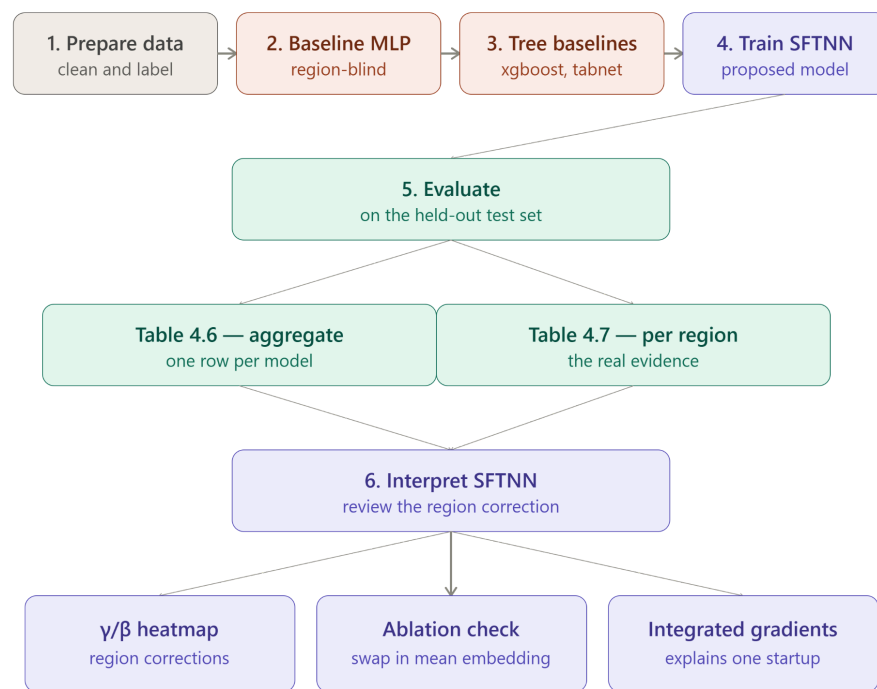


Figure 12. SFTNN Pipeline Process Diagram

Figure 12 will complement the functional view by presenting the methodological workflow that will be followed during model development and evaluation. The process will begin with dataset preparation and target-label generation, followed by the implementation of the baseline MLP, XGBoost, and TabNet models before developing the proposed SFTNN architecture under the same preprocessing pipeline and data partitions. All models will subsequently be evaluated on the held-out test set using both aggregate and per-region

performance metrics. Finally, the SFTNN will be interpreted through the analysis of the learned regional modulation parameters (γ and β), supported by γ/β heatmaps, counterfactual ablation analysis, and Integrated Gradients. Together, these diagrams will provide a comprehensive representation of both the operational behavior of the deployed platform and the end-to-end methodological workflow that will be used to develop and validate the proposed region-aware startup success prediction system.

4.6 Development Model

The study will adopt a hybrid, two-track development model to guide the implementation of both the machine learning engine and the interactive web platform. As shown in Figure 11, the development process will consist of six sequential and parallelized stages: (1) Planning and Requirements, (2) Data Preparation and Serialization, (3) Baseline Model Development, (4) Proposed Model Development, (5) Evolutionary Dashboard Prototyping, and (6) System Integration and Evaluation.

The Planning and Requirements stage will establish the study objectives, target metrics, and system boundaries, mapping these initial specifications directly to final system acceptance testing. This will be followed by Data Preparation and Serialization, where the dataset will be collected, preprocessed, and partitioned for training, before compiling the final transformations into a reusable pipeline asset (`preprocessor.joblib`). Baseline models (MLP, XGBoost, and TabNet) will then be developed to establish benchmark performance, which will be followed by Proposed Model Development, where the custom PyTorch SFTNN layers will be coded and optimized under a validation-driven V-Model structure.

Once the core model weights are serialized, the Evolutionary Dashboard Prototyping stage will be initiated, wherein a frontend Streamlit user interface will be rapidly constructed and progressively refined into a multi-panel visualization

tool. Finally, the System Integration and Evaluation stage will connect the decoupled frontend to the FastAPI serving middleware, verifying parallel inference execution and extracting the active spatial parameters (γ and β) for mathematical interpretability analysis. Each stage will produce a verified functional asset or code artifact that will serve as the primary input for the succeeding stage, ensuring a highly structured, traceable, and reproducible implementation process.

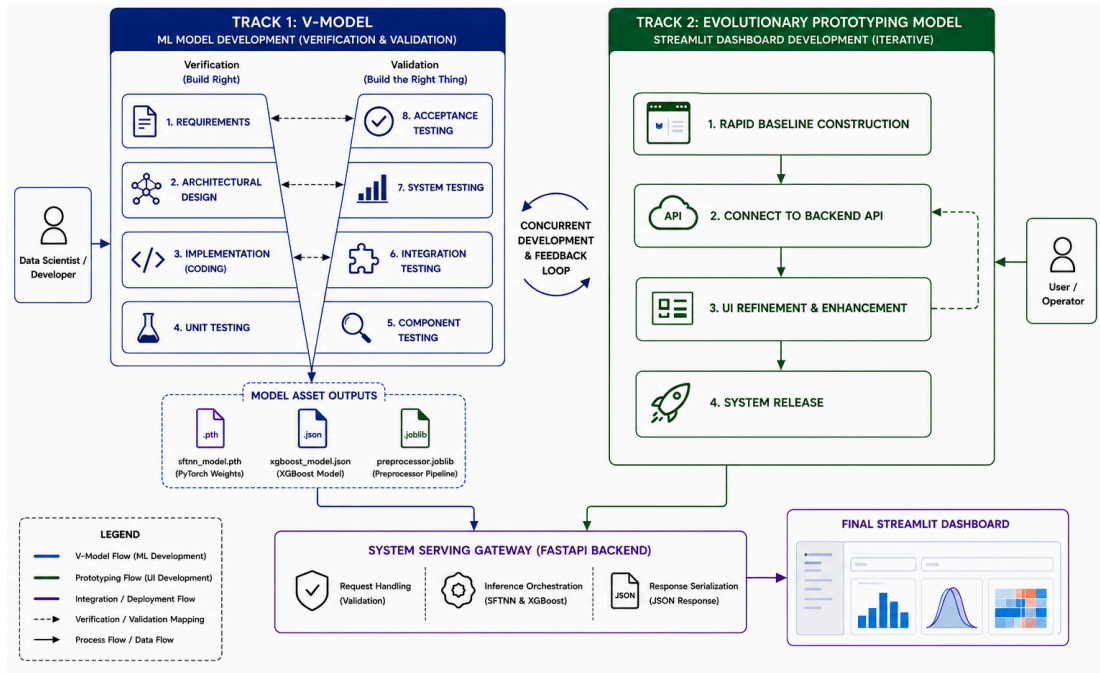


Figure 13. End-to-end SFTNN pipeline model for development

4.7 Development Approaches: Baseline Models and the Proposed SFTNN

Four models will be compared under an identical preprocessing pipeline, feature set, and train/validation/test partitioning, so that any performance difference can be attributed to model architecture rather than to differences in inputs.

Standard MLP (Conventional Baseline with One-Hot Region Encoding)

A standard Multi-Layer Perceptron (MLP) will concatenate the regional identifier (one-hot encoded) directly into the input layer alongside the universal attribute vector, with no dedicated modulation mechanism. This configuration will establish the performance floor against which the Spatial Feature Modulation architecture will be compared, isolating the specific contribution of the dual-track modulation design from the contribution of the underlying feature set.

XGBoost

A gradient-boosted decision tree ensemble (XGBoost) will be included as a non-neural baseline, consistent with prior startup-success-prediction work: Ningrum et al. (2024) benchmark classical classifiers including Random Forest, SVM, Gradient Boosting, and KNN against Crunchbase-derived startup data, finding that tree-based methods achieve strong performance on structured tabular features of this kind. The regional identifier will be supplied to XGBoost as a one-hot-encoded categorical input, mirroring the MLP baseline's treatment of region, so that the comparison will isolate architecture rather than feature representation.

TabNet

TabNet, an attention-based deep learning architecture purpose-built for tabular data, will be included as a second non-neural-network-style baseline to test whether a learned, sparse feature-selection mechanism can match or exceed the standard MLP and gradient-boosted baselines without an explicit region-conditioning mechanism. As with the MLP and XGBoost baselines, the regional identifier will be supplied as a one-hot-encoded input feature rather than through a dedicated modulation branch.

Proposed: Spatial Feature Transform Neural Network (SFTNN)

The proposed SFTNN model, described in full in Section 4.5.2, will be evaluated under the same partitioning and preprocessing pipeline as the three baselines above. Its distinguishing property relative to all three baselines will be architectural rather than representational: region will not be appended as an input feature but will instead be used to generate a dedicated affine correction on the shared hidden representation (Equations 2–3), which is the specific mechanism the per-region evaluation in Section 4.10.2 will be designed to test against the baselines.

Hyperparameter Optimization

Given the twelve-week execution window described in Section 4.9.1, hyperparameter search will be deliberately scoped to a lightweight, resource-conscious budget rather than an exhaustive sweep, using Optuna's Tree-structured Parzen Estimator (TPE) sampler for all four models. The MLP and SFTNN models will be trained using the AdamW optimizer, with a constrained search space covering learning rate (log-uniform, $1e-4$ to $1e-2$), dropout coefficient ($\{0.1, 0.2, 0.3\}$), and weight decay ($\{1e-5, 1e-4, 1e-3\}$). XGBoost's search space will cover maximum tree depth ($\{4, 6, 8\}$), learning rate (log-uniform, $1e-3$ to $3e-1$), number of estimators ($\{200, 400, 600\}$), and subsample ratio ($\{0.7, 0.85, 1.0\}$). TabNet's search space will cover the decision-step dimension n_d and attention dimension n_a (jointly, $\{8, 16, 24\}$), number of decision steps n_{steps} ($\{3, 5, 7\}$), and relaxation parameter γ ($\{1.0, 1.3, 1.5\}$). The base control MLP and the two non-neural baselines will each be allotted approximately 20 search trials; the SFTNN model will be allotted approximately 30 trials, reflecting its marginally larger parameter space from the added embedding and factor-generator branch. Early stopping on validation AUC-ROC, with a patience of 5 epochs (or, for XGBoost, 5 boosting rounds without improvement), will be applied uniformly across all four models to bound total compute time on the personally owned laptop hardware described in Section 4.9.3.

System Deployment and Full-Stack Operationalization

To demonstrate the regional adaptivity of SFTNN under production conditions, the offline mathematical weights will be compiled into the three-tier, full-stack system architecture detailed in Section 4.5.2.1. Rather than serving these models as independent CLI scripts, the runtime engine will operationalize them as serialized assets inside a decoupled, client-server pattern:

1. **The Preprocessing Pipeline Asset (`preprocessor.joblib`):** The data cleaning, robust scaling, log-transformations, and categorical target-encoding parameters will be compiled into a unified pipeline binary. This will guarantee that real-time user inputs will be mathematically scaled in exact conformity with the offline training partition boundaries.
2. **The Deep Learning Neural Weights (`sftnn_model.pth`):** The trained, context-aware SFTNN weights—including the active Spatial Feature Modulation (SFM) parameters (gamma and beta)—will be saved as a PyTorch state dictionary.
3. **The Gradient Boosting Baseline (`xgboost_model.json`):** The comparative XGBoost tree ensemble will be exported as a static JSON decision structure to serve as the live challenger baseline.

A lightweight, asynchronous FastAPI middleware backend will serve as the core application logic layer, dynamically loading these serialized assets. Upon receiving a structured JSON payload from the user-facing Streamlit client-side dashboard (Section 4.5.2.1), the API will run concurrent inference queries across both `sftnn_model.pth` and `xgboost_model.json`. This parallel execution will complete in under 50 milliseconds, returning both models' success probabilities and the active regional scaling parameters to the presentation layer interface, proving that the proposed region-aware model can be served in a responsive, real-world full-stack ecosystem.

4.8 Software Development Tools

The development of the proposed system will rely on a combination of software libraries, frameworks, and cloud-based tools that will collectively support data preprocessing, model development, evaluation, deployment, visualization, and interpretability. Each tool will be selected according to its architectural role within the machine learning pipeline and deployment framework. Table 9 will summarize the primary software development tools utilized in this study and their corresponding functions.

Software / Library / Framework	Architectural Role	Explicit Use in This Study
Python 3.x	Core Language	Implements the full end-to-end preprocessing, training, model evaluation, and deployment inference pipeline.
PyTorch	Deep Learning Framework	Builds, compiles, and trains the control Multi-Layer Perceptron (MLP) and the proposed Spatial Feature Transform Neural Network (SFTNN) .
XGBoost	Gradient-Boosted Trees	Functions as the champion, non-neural baseline benchmarked directly against the SFTNN in the parallel inference framework.

pytorch-tabnet	Tabular Deep Learning	Serves as the attention-based tabular deep learning baseline benchmarked against the SFTNN (Section 4.7).
Scikit-Learn	Machine Learning Utilities	Handles out-of-sample data partitioning, numerical feature scaling via RobustScaler , matrix transformations, and baseline metrics calculation.
Optuna	Hyperparameter Search	Executes Tree-structured Parzen Estimator (TPE) automated search optimizations for all network and tree-based architectures.
Streamlit	Presentation Layer (Frontend)	Powers the interactive, single-page web dashboard application, rendering the feature input forms and the live results panel.
FastAPI	Application Logic (Backend API)	Operates as the lightweight, high-performance middleware gateway that processes incoming client requests and serves model predictions.

Plotly	Interactive Visualization	Replaces static plotting libraries for the user interface, dynamically rendering responsive prediction gauges and baseline comparison charts.
Matplotlib / Seaborn	Static Visualization	Utilized within the experimental codebase to generate PR-AUC curves, cross-validated confusion matrices, and the post-hoc γ/β thermal heatmaps for Chapter 5.
Captum	Attribution Framework	Provides the Integrated Gradients framework used for fine-grained feature attribution and model explainability checks.
Google Colab (Free Tier)	Computing Infrastructure	Provides a containerized cloud environment with accelerated T4 GPU runtimes for model training and iterative tuning blocks.

Table 4. Software Development Tools.

4.9 Project Management

4.9.1 Schedule and Timeline

Dataset collection and preprocessing, corresponding to the data sourcing procedures in Section 4.2 and the data preparation procedures in Section 4.3, will be conducted during a dedicated pre-development phase from June 20 to July 15, 2026. This phase will include dataset collection, data cleaning and integration, feature engineering, target-label generation, dataset validation, and baseline model development, ensuring that a modeling-ready dataset and validated baseline models will be available before implementation of the proposed model.

The implementation phase will be scheduled from mid-July to the first week of October 2026. During this period, the proposed Spatial Feature Transform Neural Network (SFTNN) will be developed through the implementation of the Region Embedding Layer, Affine Parameter Generator, Spatial Feature Modulation Layer, feature fusion, and full model integration. In parallel, a web-based visualization dashboard will be developed to present prediction results, comparative model performance, and model interpretability outputs, including SHAP feature importance and affine parameter (γ/β) visualizations. Model training, hyperparameter optimization, comparative evaluation, and interpretability analyses will be completed by the first week of October, resulting in a finalized SFTNN model and fully functional visualization system.

Final manuscript preparation will be scheduled from October 4 to November 15, 2026, covering the preparation of Chapters 5–7, results interpretation, conclusions and recommendations, adviser review, revisions, formatting, and final manuscript completion. The project will conclude with camera-ready manuscript preparation, conference paper submission, and conference presentation, with the conference presentation serving as the final defense, scheduled during the last two weeks of November 2026.



Table 5. Gantt Chart of Activities ,Proposal.

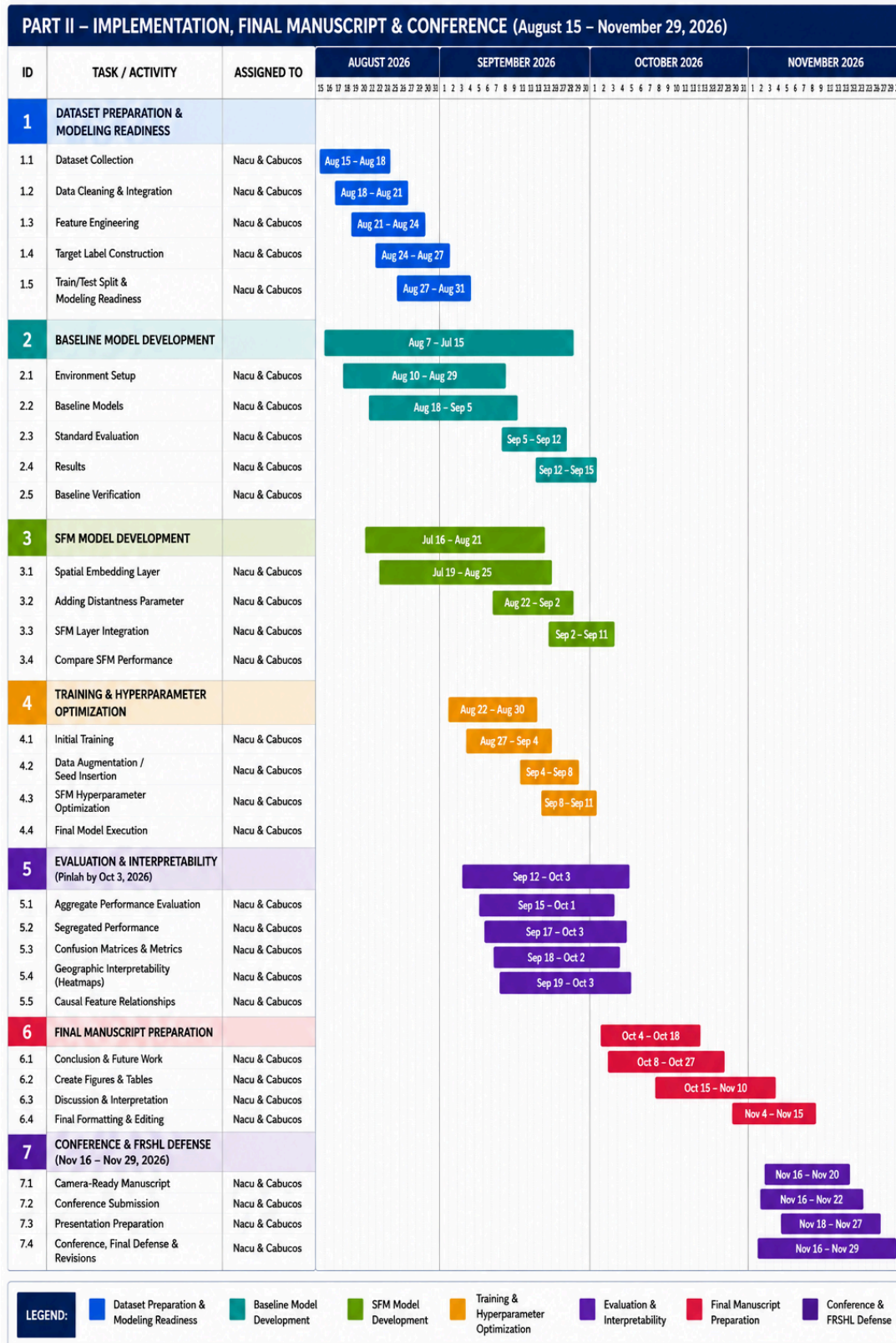


Table 6. Gantt Chart of Activities, Implementation.

4.9.2 Roles and Responsibilities

Name	Role	Responsibility
Osbev Erica Cabucos	Researcher	Design the heuristic-based distant-supervision labeling rule (age + funding-round tiebreaker, v1→v2 rationale); set the reference year and threshold logic.
	Developer	Implement the labeling workbook — editable threshold parameters, formula-driven target-label generation (derived_target).
	Researcher	Verify data provenance and lineage across the global Crunchbase-derived baseline and the South Asian, African, Latin American, and Southeast Asian regional registries.
	Developer	Implement the data-cleaning and feature-engineering pipeline (log1p + robust scaling, capital velocity, target encoding) with training-only leakage-prevention fitting.
	Researcher	Author Chapter 3 (Technical Background) and Chapter 4 Sections 4.1–4.3 (data sources, gathering, and treatment of data).
	Developer	Lead the design and development of the Chapter 5 demonstration dashboard for showcasing model predictions and evaluation results.
Drixy Reece Irish Nacu	Researcher	Lead the SFTNN architectural design — shared feature trunk, Region Embedding, Affine Parameter Generator, and Spatial Feature Modulation layer (Equations 1–4).

	Developer	Implement the standard MLP, XGBoost, and TabNet baselines and the proposed SFTNN model in PyTorch under an identical preprocessing pipeline.
	Developer	Execute the Optuna (TPE) hyperparameter search and final model retraining across all four models.
	Researcher	Generate and interpret aggregate and per-region evaluation metrics, γ/β heatmaps, the counterfactual ablation check, and the PCA embedding cluster plot.
	Researcher	Author Chapter 4 Sections 4.4–4.10 (concept, system architecture, evaluation framework) and the defense-readiness Q&A.
	Developer	Assist in the development of the Chapter 5 demonstration dashboard.

Table 7. Table of Roles and Responsibilities

4.9.3 Budget and Cost Management

This study will be conducted with minimal financial expenditure by relying on personally owned laptops and open-source software. No portion of the model implementation, hyperparameter search, or interpretability pipeline will require a paid license, API, or cloud subscription: Python, PyTorch, XGBoost, pytorch-tabnet, Scikit-Learn, Optuna, Matplotlib, and Seaborn are all freely licensed, and Google Colab's free tier will serve as an available compute fallback. Table 5 will summarize the resulting cost profile, including the two development devices and connectivity used throughout the project.

Item	Cost (₱)
Device 1 — Ryzen 7 7735U, 32GB RAM, Radeon Graphics, 477GB SSD	73,995.00
Device 2 — Intel i7-1165G7, 16GB RAM, Iris Xe, 512GB SSD	70,000.00
Software & Libraries (Python, PyTorch, XGBoost, pytorch-tabnet, Scikit-Learn, Optuna, Matplotlib, Seaborn — all open-source)	Open Source
Compute Fallback (Google Colab, free tier)	Open Source
Internet (Globe WiFi, project duration)	1,400.00
Miscellaneous Fees (printing, binding)	1,500.00
Total	146,895.00

Table 8. Budget of Expenses

4.10 Verification, Validation, and Testing

This section will describe the procedures that will be used to verify that the Spatial Feature Modulation system is built correctly, validate that it fulfills its intended purpose of predicting cross-regional venture success, and confirm through rigorous, multi-model testing that it performs reliably relative to the three baselines described in Section 4.7.

4.10.1 Reconstruction Fidelity, Baseline Comparison, and Testing Protocol

Verification will confirm that the system is built correctly according to the intended design. This phase will include systematic auditing of the logarithmic and robust-scaling transformations, the capital-velocity derivation, and the target-encoding of industry vertical bins described in Section 4.3.2, each checked against the leakage-prevention protocol in the same section. The correct isolation

of the Region Embedding and Affine Parameter Generator track from the shared trunk (Section 4.5.2) will additionally be verified through targeted unit tests confirming that gradient flow reaches both branches independently during backpropagation, and that varying the regional identifier r alone, while holding the universal attribute vector x fixed, produces a corresponding change in the modulated output.

Validation will ensure that the right model is being built to solve the specific cross-regional prediction problem without overfitting to training-set regularities. The 15% validation set, isolated per the protocol in Section 4.3.2, will be used as a passive hold-out monitor during training, triggering early stopping (Section 4.7) if validation AUC-ROC ceases to improve, and will additionally serve as the sole scoring surface for the Optuna trials across all four models, ensuring that no configuration is selected using any signal from the reserved test partition.

Testing will measure the finalized performance of all four models — the standard MLP, XGBoost, TabNet, and the proposed SFTNN architecture — against one another on the reserved 15% blind test set, simulating real-world cross-regional deployment. This four-way comparison, evaluated identically across models, will be the specific mechanism by which the added architectural complexity of SFTNN will be justified or rejected, rather than assumed. Evaluation will begin with the construction of a confusion matrix for each model, categorizing prediction outcomes against the distant-supervision ground truth established in Section 4.3.2.

	Predicted: Success	Predicted: Failure
Actual: Success	True Positive (TP)	False Negative (FN)
Actual: Failure	False Positive (FP)	True Negative (TN)

Table 9. Confusion Matrix Template (Constructed Identically for Each of the Four Models)

In the specific context of predicting cross-regional venture success, the four quadrants of this matrix will carry distinct interpretive implications, applied identically when comparing all four models. A True Positive (TP) will correctly identify a venture that goes on to achieve economic success or sustained survival. A True Negative (TN) will correctly identify a venture that fails or stalls. A False Positive (FP) will incorrectly flag a failing venture as a likely success, which in a practical deployment could misdirect investor or policy attention toward a non-viable venture. A False Negative (FN) will incorrectly flag a genuinely successful venture as a likely failure; in an ecosystem-development context this will be the more consequential error, since it will mean a viable, often underrepresented regional venture is overlooked by the very tool intended to surface it, reproducing the spatial optimization bias described in Section 4.4 rather than correcting for it. Consequently, in addition to overall accuracy, Recall on the minority-region subsets will be treated as a priority diagnostic during model selection across all four models.

Each of the four models — MLP, XGBoost, TabNet, and SFTNN — will be scored using the identical set of seven metrics defined below, computed both in aggregate and per region (Section 4.10.2).

Equation 5. Accuracy — the proportion of all predictions the model gets right

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

Equation 6. Precision — the proportion of predicted successes that are actually correct

$$Precision = TP / (TP + FP)$$

Equation 7. Recall — the proportion of actual successes the model correctly identifies

$$Recall = TP / (TP + FN)$$

Equation 8. F1-Score — the harmonic mean of precision and recall

$$F1-Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

Equation 9. AUC-ROC — separability of successful from failing ventures across all classification thresholds

$$AUC-ROC = \sum_{i=1}^n \frac{1}{2} (FPR_{i+1} - FPR_i) (TPR_i + TPR_{i+1})$$

Equation 10. PR-AUC — area under the Precision-Recall curve, reported alongside AUC-ROC given the near-balanced but still cross-regionally uneven class composition

$$PR-AUC = \int_0^1 Precision(r) dr, \text{ evaluated over Recall } r \in [0, 1]$$

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Standard MLP (baseline)	—	—	—	—	—	—
XGBoost (baseline)	—	—	—	—	—	—
TabNet (baseline)	—	—	—	—	—	—
Proposed SFTNN	—	—	—	—	—	—

Table 10. Aggregate Evaluation Reporting Template

Table 10 will serve as the fixed template to populate with measured test-set results for all four models which will be populated with empirical testing results in Chapter 5.

4.10.2 Multi-Dimensional Validation

Given the near-balanced but still cross-regionally uneven composition of the modeling-ready dataset (Section 4.3.2), classification success will be tracked along four complementary dimensions rather than a single aggregate number, for each of the four models.

The first dimension will be quantitative-aggregate: all seven metrics in Equations 5–10, computed at the aggregate level for each model, as summarized in Table 10.

The second dimension will be quantitative-regional: the same set of metrics will be computed independently per region, for each of the four models, to ensure equal tracking fidelity across minor economic domains rather than allowing performance on high-volume regions such as North America and Europe (which together account for roughly 83% of usable rows) to dominate the reported averages. This per-region view will be the specific safeguard against the regional imbalance documented in Section 4.2.4 being silently absorbed into an aggregate score that looks acceptable while masking poor performance on Southeast Asia, Africa, or Latin America, and will be the primary evidentiary basis on which the SFTNN-versus-baseline comparison will be decided: a model will be judged not only on its aggregate scores but specifically on whether it will improve Recall and F1 on the minority-region subsets relative to the standard MLP baseline.

Region	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
North America	—	—	—	—	—	—
Europe	—	—	—	—	—	—
South Asia	—	—	—	—	—	—
Africa	—	—	—	—	—	—
Latin America	—	—	—	—	—	—
Southeast Asia	—	—	—	—	—	—
East Asia	—	—	—	—	—	—

Table 11. Per-Region Evaluation Reporting Template (Repeated for Each of the Four Models)

The third validation dimension will be qualitative and will focus exclusively on the proposed SFTNN model, as only SFTNN will produce directly inspectable regional modulation parameters (γ and β). Unlike conventional models that will rely on post-hoc explainability techniques such as SHAP, SFTNN will provide inherently interpretable parameters because γ_r and β_r will participate directly in the model's forward pass (Equation 3). Consequently, the learned γ_r and β_r vectors will be extracted for each regional class and visualized as heatmaps across regions and hidden dimensions, providing a transparent representation of how the model will scale and shift its shared feature representation for different entrepreneurial ecosystems.

To assess whether these learned regional adjustments will be not only mathematically meaningful but also consistent with real-world entrepreneurial ecosystems, the generated heatmaps will be reviewed with the research adviser and, where feasible, a panel member or individual with relevant knowledge of Southeast Asian startup ecosystems. The review will examine whether the direction and relative magnitude of the learned regional corrections will align with documented characteristics of each ecosystem. For example, a historically underfunded region will be expected to exhibit modulation patterns consistent with known funding constraints rather than arbitrary or contradictory adjustments.

This qualitative assessment will complement, rather than replace, the quantitative evaluation presented in Tables 10 and 11. Together, these analyses will provide evidence that the learned regional modulation parameters will be both statistically effective and substantively meaningful within the context of cross-regional startup success prediction.

A fourth, confirmatory check will be applied specifically to the SFTNN model, alongside the γ/β heatmap analysis, to provide an independent quantitative signal that will not rely on qualitative plausibility judgment. For each region r , the trained model will be re-evaluated on the held-out test set with every startup's learned region embedding e_r replaced by the mean embedding $\bar{e}$

computed across all regions, holding the remainder of the trained network fixed. This counterfactual ablation will isolate the specific contribution of regional conditioning: if Spatial Feature Modulation meaningfully improves predictions for a given region, substituting a generic, region-uninformative embedding will measurably reduce that region's Recall and F1 relative to the true, region-specific embedding, whereas negligible change will indicate that the modulation mechanism contributes little beyond what the shared trunk already captures for that region. This check will require no additional library beyond the existing PyTorch pipeline and will be reported alongside the γ/β heatmaps as a second, independent form of evidence for the necessity of the SFTNN architecture.

Two extensions to this interpretability package will be treated as conditional or supplementary rather than core deliverables, given the study's three-month timeline. First, the learned region embedding table itself will be projected into two dimensions using Principal Component Analysis (PCA) and visualized as a scatter plot, following the entity-embedding visualization convention demonstrated by Guo and Berkahn (2016), to provide a low-cost qualitative check on whether regions with similar entrepreneurial characteristics (Section 4.2.2) will cluster together in the learned embedding space; this will require no additional library beyond Scikit-Learn (Section 4.8) and will be generated alongside the γ/β heatmaps in Week 9 (Section 4.9.1). Second, Integrated Gradients (IG) (Sundararajan et al., 2017) will be considered as a finer-grained, instance-level confirmatory method, applied conditionally in Weeks 10–11 only if the γ/β and ablation evidence will indicate a non-trivial regional signal worth decomposing further. IG will interpolate a startup's region embedding from a zero-vector baseline to its true value across a fixed number of steps, integrating the model's gradient at each step to attribute the resulting change in predicted probability to individual embedding dimensions, subject to a completeness guarantee that ablation alone will not provide. Because this guarantee will depend on the interpolation path remaining within a region of the embedding space the model will have meaningfully learned from, IG will be treated as a supplement to, rather than a replacement for, the ablation check,

and will only be pursued if project time and a genuine underlying signal both justify it. Grad-CAM will be considered and rejected as a candidate method for this study: it will be designed for convolutional architectures with image-like spatial feature maps, and will have no principled application to a tabular, region-conditioned MLP architecture in which “spatial” will denote geographic region rather than pixel-grid structure.

Taken together, these four validation dimensions will provide a comprehensive framework for evaluating both the predictive performance and methodological soundness of the proposed approach. The reliability of the heuristic-generated target labels (Section 4.3.2) will be assessed indirectly through aggregate and per-region performance across all four models rather than assumed a priori. The treatment of ambiguous Operating startups will be made transparent through the documented comparison between the rejected (v1) and adopted (v2) labeling strategies, allowing the chosen heuristic to be critically examined. Likewise, the effects of regional imbalance and the dominance of North American startups within the underlying corpus will be explicitly evaluated through per-region performance metrics instead of relying solely on aggregate results. Finally, the effectiveness of the proposed Spatial Feature Transform Neural Network (SFTNN) architecture will be established empirically by comparing its performance against the standard MLP, XGBoost, and TabNet baselines under the same evaluation framework, with additional evidence provided by the counterfactual embedding ablation analysis and, where applicable, the supplementary interpretability techniques described in this section.

BIBLIOGRAPHY

Books

- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in Python*. Springer.
<https://doi.org/10.1007/978-3-031-38747-0>
- Luengo, J., García-Vico, A. M., & Herrera, F. (2020). *Big data preprocessing: Enabling smart data*. Springer Cham.
<https://link.springer.com/book/10.1007/978-3-030-39105-8>

Journal Articles

- Abdelghani, B. (2024). Exploring the distinctions between Startups and small and medium-sized enterprises (SMEs): A comparative analysis. *International Journal of Economic Perspectives*, 18(12), 2499–2511.
<https://ijeponline.org/index.php/journal/article/view/788>
- Ahsan, Md Manjurul & MAHMUD, M. A. & Saha, Pritom & Gupta, Kishor Datta & Siddique, Zahed. (2021). Effect of Data Scaling Methods on Machine Learning Algorithms and Model Performance. *Technologies*. 9. 52. 10.3390/technologies9030052.
- Ali, Y., Petrillo, A., & De Felice, F. (2023). Macro-environmental factors and their impact on startups from the perspective of developing countries. *Cogent Business & Management*, 10(2), Article 2238754.
<https://doi.org/10.1080/19397038.2023.2238754>
- Argaw, Y. M., & Liu, Y. (2024). The pathway to startup success: A comprehensive systematic review of critical factors and the future research agenda in developed and emerging markets. *Systems*, 12(12), 541.
<https://doi.org/10.3390/systems12120541>
- Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, F., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.
<https://doi.org/10.1016/j.inffus.2019.12.012>

- Avanzi, B., Taylor, G., Wang, M., & Wong, B. (2024). Machine learning with high-cardinality categorical features in actuarial applications. *ASTIN Bulletin*, 54(2), 213–238. <https://doi.org/10.1017/asb.2024.7>
- Boegl, L., & Kounadi, O. (2024). Introducing spatial heterogeneity via regionalization methods in machine learning models for geographical prediction: A spatially conscious paradigm. *European Journal of Geography*, 15(4), 244–255. <https://doi.org/10.48088/ejg.l.boe.15.4.244.255>
- Brunsdon, C., Fotheringham, A. S., & Charlton, M. E. (1996). Geographically weighted regression: A method for exploring spatial nonstationarity. *Geographical Analysis*, 28(4), 281–298. <https://doi.org/10.1111/j.1538-4632.1996.tb00936.x>
- Dai, N., Jo, H., & Kassicieh, S. (2012). Cross-border venture capital investments in Asia: Selection and exit performance. *Journal of Business Venturing*, 27(6), 666–684. <https://doi.org/10.1016/j.jbusvent.2011.04.004>
- Gabrielli, G., Melioli, A., & Bertini, F. (2026). Corporate financial distress prediction: A machine learning approach in the era of big data. *Journal of Accounting & Organizational Change*, 22(7), 31–65. <https://doi.org/10.1108/JAOC-05-2025-0166>
- Ge, P., Ren, C.-X., Xu, X.-L., & Yan, H. (2023). Unsupervised domain adaptation via deep conditional adaptation network. *Pattern Recognition*, 134, 109088. <https://doi.org/10.1016/j.patcog.2022.109088>
- Hakami, A. (2024). Strategies for overcoming data scarcity, imbalance, and feature selection challenges in machine learning models for predictive maintenance. *Scientific Reports*, 14(1), 9645. <https://doi.org/10.1038/s41598-024-59958-9>
- Imran, G., Hossain, M. P., Hasan, M., Hasan, M. T., & Ohidujjaman. (2026). Tabular and graph-based representations for noise and missing data in robust machine learning. *Array*, 29, 100697. <https://doi.org/10.1016/j.array.2026.100697>
- Kato, A. I. (2025). Venture capital as a catalyst for innovation and economic growth in emerging economies: A systematic review and future research agenda. *Administrative Sciences*, 15(11), 405. <https://doi.org/10.3390/admsci15110405>

- Kim, J., Kim, H., & Geum, Y. (2023). How to succeed in the market? Predicting startup success using a machine learning approach. *Technological Forecasting and Social Change*, 193, Article 122614. <https://doi.org/10.1016/j.techfore.2023.122614>
- Ma, C., Zhao, H., Zhang, K., Zhang, L., & Huang, H. (2025). A federated supply chain finance risk control method based on personalized differential privacy. *Egyptian Informatics Journal*, 31, 100704. <https://doi.org/10.1016/j.eij.2025.100704>
- Mair, J., & Marti, I. (2009). Entrepreneurship in and around institutional voids: A case study from Bangladesh. *Journal of Business Venturing*, 24(5), 419–435. <https://doi.org/10.1016/j.jbusvent.2008.04.006>
- Mortada, R., & Nwachukwu, C. (2025). A review of venture capital in six global markets. *Journal of Innovations in Business and Industry*, 3(4), 309–316. <https://doi.org/10.61552/JIBI.2025.04.010>
- Ningrum, I. W. K., Ridho, F., & Wijayanto, A. W. (2024). Predicting startup success using machine learning approach. *Journal of Applied Informatics and Computing*, 8(2), 280–290. <https://doi.org/10.30871/jaic.v8i2.8338>
- Pan, C., Pan, X., & Sun, L. (2026). A multi-layer AI decision support system for startup success prediction and risk assessment using knowledge graphs and federated learning. *Scientific Reports*. Advance online publication. <https://doi.org/10.1038/s41598-026-44162-8>
- Perugini, F. (2023). Space–time analysis of entrepreneurial ecosystems. *The Journal of Technology Transfer*, 48(1), 240–291. <https://doi.org/10.1007/s10961-021-09906-w>
- Pigola, A., Fischer, B., de Moraes, G. H. S. M., do Prado, N. B., & Ribeiro, J. de A. (2025). Entrepreneurial ecosystems as spatial mosaics: Drivers of value creation across geographical spaces. *Small Business Economics*, 65(4), 2277–2300. <https://doi.org/10.1007/s11187-025-01056-9>
- Razaghzadeh Bidgoli, M., Raeesi Vanani, I., & Goodarzi, M. (2024). Predicting the success of startups using a machine learning approach. *Journal of Innovation and Entrepreneurship*, 13, Article 80. <https://doi.org/10.1186/s13731-024-00436-x>
- Ren, J., Raghupathi, W., & Raghupathi, V. (2025). Exploring the factors that impact accelerator-based startup funding. *Journal of Innovation and*

- Rodriguez-Belenguer, P., Beser-Robles, M., Marfil-Trujillo, M., Cerdá-Alberich, L., & Martí-Bonmatí, L. (2026). The challenge of data scarcity and imbalanced classes in radiomics performance. *Computer Methods and Programs in Biomedicine*, 283, 109435. <https://doi.org/10.1016/j.cmpb.2026.109435>
- Schäfer, S., Fischer, B., Rücker Schaeffer, P., & Balestrin, A. (2024). Beyond local boundaries: Unraveling the spatiality of entrepreneurial ecosystems. *Journal of Business Venturing Insights*, 22, Article e00478. <https://doi.org/10.1016/j.jbvi.2024.e00478>
- Sun, S. L., Chen, V. Z., Sunny, S. A., & Chen, J. (2019). Venture capital as an innovation ecosystem engineer in an emerging market. *International Business Review*, 28(5), 101485. <https://doi.org/10.1016/j.ibusrev.2018.02.012>
- Taghiyarrenani, Z., Nowaczyk, S., Pashami, S., & Bouguelia, M.-R. (2023). Multi-domain adaptation for regression under conditional distribution shift. *Expert Systems with Applications*, 224, Article 119907. <https://doi.org/10.1016/j.eswa.2023.119907>
- Te, Y.-F., Wieland, M., Frey, M., Pyatigorskaya, A., Schiffer, P., & Grabner, H. (2023). Making it into a successful Series A funding: An analysis of Crunchbase and LinkedIn data. *The Journal of Finance and Data Science*, 9, 100099. <https://doi.org/10.1016/j.jfds.2023.100099>
- Unger, M., & Tuzhilin, A. (2022). Hierarchical latent context representation for context-aware recommendations. *IEEE Transactions on Knowledge and Data Engineering*, 34(7), 3322–3334. <https://doi.org/10.1109/tkde.2020.3022102>
- Vasquez, E., Santisteban, J., & Mauricio, D. (2023). Predicting the success of a startup in information technology through machine learning. *International Journal of Information Technology and Web Engineering*, 18(1), 1–17. <https://doi.org/10.4018/IJITWE.323657>
- Vedula, S., & Kim, P. H. (2019). Gimme shelter or fade away: The impact of regional entrepreneurial ecosystem quality on venture survival. *Industrial and Corporate Change*, 28(4), 827–854. <https://doi.org/10.1093/icc/dtz032>

- Vedula, S., York, J. G., Conger, M., & Embry, E. (2022). Green to gone? Regional institutional logics and firm survival in moral markets. *Organization Science*, 33(6), 2274–2299. <https://doi.org/10.1287/orsc.2021.1533>
- Vicentin, D. C., Pigola, A., Fischer, B., & Moraes, G. H. S. M. de. (2026). Mapping the spatial evolution of entrepreneurial ecosystems: A clustering perspective on global investment flows. *International Journal of Entrepreneurial Behavior & Research*, 32(1), 21–42. <https://doi.org/10.1108/IJEBr-04-2025-0440>
- Wei, C.-P., Fang, E. S.-H., Yang, C.-S., & Liu, P.-J. (2025). To shine or not to shine: Startup success prediction by exploiting technological and venture-capital-related features. *Information & Management*, 62(6), 104152. <https://doi.org/10.1016/j.im.2025.104152>
- Wilson, G., & Cook, D. J. (2020). A survey of unsupervised deep domain adaptation. *ACM Computing Surveys*, 53(2), 1–36. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8323662/>
- Yildirim, A., Clarysse, B., & Wright, M. (2022). The impact of institutional voids and ecosystem logics in the spread of ecosystems in emerging economies. *Industry and Innovation*, 29(5), 649–671. <https://doi.org/10.1080/13662716.2021.2007760>
- Zhou, N. (2024). Emerging market multinationals' liability of outsidership. *Journal of World Business*, 59(3), 101505. <https://doi.org/10.1016/j.jwb.2023.101505>

Conference Proceedings Articles

- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *arXiv preprint arXiv:1603.02754*. <https://arxiv.org/abs/1603.02754>
- Dai, T., & Cai, B. (2026). Altman Z-score's predictive accuracy for financial distress in Chinese enterprises: Evidence from 32 listed manufacturing firms. In *Proceedings of the Association for Computing Machinery. Association for Computing Machinery*. <https://doi.org/10.1145/3800000.3800107>

- Eggert, M., Chwallek, C., & Wolff, F. (2022). The role of environmental factors for the success of digital start-ups. *Proceedings of the 30th European Conference on Information Systems*. https://aisel.aisnet.org/ecis2022_rp/9/
- Ferrati, F., & Muffatto, M. (2020). Setting Crunchbase for data science: Preprocessing, data integration, and feature engineering. *Proceedings of CARMA 2020: 3rd International Conference on Advanced Research Methods and Analytics*. <https://doi.org/10.4995/CARMA2020.2020.11633>
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *Proceedings of the 3rd International Conference on Learning Representations (ICLR 2015)*. <https://arxiv.org>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- Perez, E., Strub, F., De Vries, H., Dumoulin, V., & Courville, A. (2018). FiLM: Visual reasoning with a general conditioning layer. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 3942–3951. <https://doi.org/10.48550/arXiv.1709.07871>
- Pilc, L., & Taušl Procházková, P. (2025). Factors for startup success: Insights from recent research. In *Proceedings of the XB-CON International Conference 2025* (pp. 169–186). University of West Bohemia in Pilsen. <https://doi.org/10.24132/ZCU.XB-CON.2025.169-186>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144. <https://doi.org/10.1145/2939672.2939778>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision*, 618–626. <https://doi.org/10.1109/ICCV.2017.74>
- Stojanov, P., Li, Z., Gong, M., Cai, R., Carbonell, J. G., & Zhang, K. (2021). Domain adaptation with invariant representation learning: What transformations to learn? *Advances in Neural Information Processing*

Systems (NeurIPS). Curran Associates, Inc.
<https://doi.org/10.5555/3540261.3542160>

Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic attribution for deep networks. *Proceedings of the 34th International Conference on Machine Learning*, 70, 3319–3328.
<https://proceedings.mlr.press/v70/sundararajan17a.html>

Wang, X., Yu, K., Dong, C., & Loy, C. C. (2018). Recovering realistic texture in image super-resolution by deep spatial feature transform. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 606–615. <https://doi.org/10.1109/CVPR.2018.00070>

Web Articles / Encyclopedia Entries / Preprints

Arik, S. Ö., & Pfister, T. (2019). *TabNet: Attentive interpretable tabular learning*. arXiv preprint. <https://arxiv.org/abs/1908.07442>

Cai, H.-R., & Ye, H.-J. (2025). Feature-aware modulation for learning from temporal tabular data (arXiv:2512.03678). *arXiv*. <https://doi.org/10.48550/arXiv.2512.03678>

Eranki, G., Pillai, R., George, L., et al. (2023). *Challenges in data driven decision making in the early stages of a startup* (Version 1) [Preprint]. Research Square. <https://doi.org/10.21203/rs.3.rs-2341650/v1>

Feng, X., Chen, C., Li, D., Zhao, M., Hao, J., & Wang, J. (2021). CMML: Contextual modulation meta learning for cold-start recommendation (arXiv:2108.10511). *arXiv*. <https://doi.org/10.48550/arXiv.2108.10511>

Guo, C., & Berkhahn, F. (2016). Entity embeddings of categorical variables (arXiv:1604.06737). *arXiv*. <https://doi.org/10.48550/arXiv.1604.06737>

Harrison, R. T. (2025). Entrepreneurial ecosystems. In D. J. Cumming & B. Hammer (Eds.), *The Palgrave encyclopedia of private equity*. Palgrave Macmillan. https://doi.org/10.1007/978-3-031-81653-6_198

Mashhadi, S., Saghezchi, A., & Kashani, V. (2025). *Interpretable machine learning for predicting startup funding, patenting, and exits* (arXiv:2510.09465). *arXiv*. <https://doi.org/10.48550/arXiv.2510.09465>

- Pei, H., Liu, Z., Xiao, X., Du, X., Hong, S., Zhang, K., & Zhang, H. (2026). *Analyze like a venture capitalist: Information-gain and knowledge enhanced graph reasoning for startup success prediction* (arXiv:2512.23489). arXiv. <https://doi.org/10.48550/arXiv.2512.23489>
- Reinders, E. (2024). *Predicting startup success by using logistic regression, random forest and Naive Bayes classifiers* [Master's thesis, Tilburg University]. Tilburg University Repository. <http://arno.uvt.nl/show.cgi?fid=181759>

APPENDIX A: OFFICIAL SIGNED AGREEMENT



UNIVERSITY
of SAN CARLOS
SCIENTIA • VIRTUS • DEVOTIO



DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

THESIS ADVISER–ADVISEE AGREEMENT

School Year 2026–2027

This Agreement is entered into by and between:

Thesis Adviser: Christian Maderazo

and

Student Group / Advisees:

#	Full Name	Student ID No.	Program / Year Level
1	Nacu, Drixy I Reece Irish G.	24102070	BSCS - 2
2	Cabucs, Osbev Erica D.	24104013	BSCS - 2
3			
4			

Thesis Project Title (Working Title): TechVenture: Spatial Feature
Transform Neural Network for Region-Aware Startup Success
Prediction

Deliverables

DELIVERABLES	DUE DATE		
	Month	Date	Day
Signed Schedule of Deliverables	Jun	26	Fri
Chapter 2, Chapter 3 and Bibliography	Jul	3	Fri
Research Objectives / Statement of the Problem	Jul	9	Thur
Chapter 4 Design and Methodology	Jul	10	Fri
Appendices (Transmittal Letter, Interview Guide or Questionnaire, Software Requirement Specifications) and CV	Jul	16	Thur
Chapter 1 and Abstract	Jul	17	Fri
Proposal Document Draft Chapter 1-4 with References, Appendices, CV	Jul	23	Thur
Submission of Agreement Letter (with Parent's signature)	July	24	Fri
Submission of the 3 copies of Thesis Proposal Document & Sworn and Recommendation Letter	Jul	27	Mon
Thesis Proposal Defense	Aug	3-5	M-W
Submission of Rubric 1, 2.1 & 2.2	Aug	10	Mon
Submission of Revised Thesis Document and Compliance Form	Aug	14	Fri
Regular Consultations			

Agreement

The undersigned acknowledge that an initial meeting has been conducted to discuss the proposed Thesis Project, evaluate its feasibility, and establish expectations for the adviser–advisee relationship.

Both parties agree to uphold their respective roles and responsibilities as outlined in the Thesis Project Guidelines issued by the Department of Computer, Information Sciences, and Mathematics, University of San Carlos. These Guidelines shall serve as the primary reference for all matters concerning the

conduct, supervision, and evaluation of the Thesis Project.

This Agreement shall remain in force for School Year 2026–2027. Any modifications to the adviser–advisee arrangement shall be made in accordance with the procedures specified in the Thesis Project Guidelines and coordinated through the Thesis Coordinator.

By signing below, both parties affirm their commitment to the successful completion of the Thesis Project and to maintaining a professional and collaborative working relationship.

Signatures

Thesis

Adviser

Name: Christian Maderazo

Signature: _____

Date: June 26, 2026

Individual Signatures of All Advisees:

#	Full Name	Signature	Date
1	Drixyl Reece Irish G. Nacu		06/25/2026
2	Osbev Erica D. Cabucos		06/25/2026
3			
4			

APPENDIX B: SOFTWARE REQUIREMENTS SPECIFICATION

IEEE 830 / ISO/IEC/IEEE 29148 Format

Spatial Feature Transform Neural Network for Region-Aware Startup Success

Prediction

Version: 1.1

July 15, 2026

By

Cabucos, Osbev Erica

Nacu, DrixyI Reece Irish

Faculty Adviser

Mr. Christian V. Maderazo, M.Eng

Revision History

Date	Version	Description	Author
July 02, 2026	0.1	Initial requirements compilation and baseline schema drafting	O. E. Cabucos
July 10, 2026	1.0	Baseline release incorporating multi-model requirements and FiLM variables	D. R. I. Nacu
July 15, 2026	1.1	Corrected distant-supervision requirement scope, evaluation metric set, output activation, asset filename, and reference list for consistency with Thesis Chapter 4 methodology	O. E. Cabucos & D. R. I. Nacu

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the detailed functional and non-functional requirements for the TechVenture-SFTNN predictive modeling pipeline and deployment framework version 1.1. This system serves as the core software engineering implementation of the underlying undergraduate research. This document is intended for use by the thesis defense panel, faculty advisers, and future researchers evaluating or replicating the Spatial Feature Transform Neural Network (SFTNN) architecture.

The software builds an end-to-end analytical pipeline and interface to forecast startup success rates utilizing localized geographic conditioning parameters. It does not provide automated micro-investment executions or financial portfolio transactions.

1.2 Document Conventions

Requirements within this document are strictly prioritized as High, Medium, or Low based on architectural necessity. Each discrete requirement is uniquely cataloged using the standard format: REQ-<Section>-<Number>.

Mandatory systemic behavior and constraints are explicitly stated using the term “shall,” while non-mandatory or optional behavior is denoted by “should” or “may.” All mathematical symbols (γ , β , h , x) adhere to standard neural network notation.

1.3 Intended Audience and Reading Suggestions

This document is organized sequentially to align with standard software engineering practices.

- Thesis Panelists and Faculty Advisers should read Section 1 (Introduction) and Section 2 (Overall Description) to verify system bounds.
- Machine Learning Researchers and Data Engineers should focus directly on Section 4 (System Features) and Section 5.3 (Security/Leakage Prevention Requirements).
- Software Developers implementing or modifying the API and UI should focus on Section 3 (External Interfaces) and Section 4.3 (Model Inference Feature).

1.4 Product Scope

The TechVenture-SFTNN platform bridges regional data constraints in machine learning by applying specialized spatial modulation to incoming global startup data. The objective of the software is to accurately classify early-stage venture survival rates across underrepresented emerging

ecosystems (such as Southeast Asia and Africa) relative to mature ecosystems.

By executing a modular 3-tier setup, the system solves the operational problem of region-blind target evaluations, directly supporting the thesis goal of providing an interpretable, localized framework for venture capitalists and data analysts.

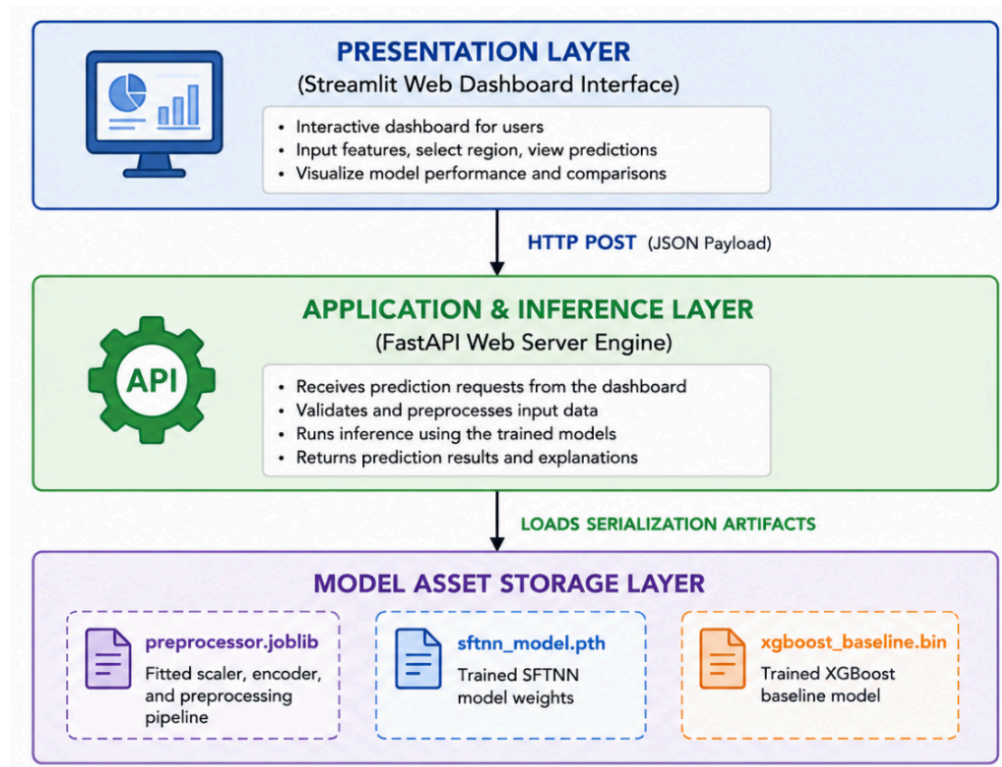
1.5 References

Ref#	Document Title	Version/Date	Source
[1]	FiLM: Visual Reasoning with a General Conditioning Layer	Perez et al., 2018	arXiv:1709.07871
[2]	Interpretable Machine Learning for Predicting Startup Funding, Patenting, and Exits	Mashhadi et al., 2026	arXiv:2510.09465
[3]	Analyze Like a Venture Capitalist: Information-Gain and Knowledge Enhanced Graph Reasoning for Startup Success Prediction	Pei et al., 2026	Findings of ACL 2026
[4]	Setting Crunchbase for Data Science: Preprocessing, Data Integration and Feature Engineering	Ferrati & Muffatto, 2020	Proceedings of CARMA 2020
[5]	Predicting Startup Success Using Machine Learning Approach	Ningrum et al., 2024	Journal of Applied Informatics and Computing, 8(2)

2. Overall Description

2.1 Product Perspective

The TechVenture-SFTNN system is an independent, newly developed software application comprising a localized data preparation track and an interactive presentation system. It acts as an autonomous predictive engine that consumes disparate multi-source CSV/tabular directories and outputs serialized mathematical weights. The application is completely decoupled into independent processing tiers, as illustrated below:



2.2 Product Functions

- **Multi-Source Data Ingestion & Alignment**: Normalizes disjoint tabular formats from Kaggle, GitHub, Failory, and Disrupt Africa into a schema-consistent layout.

- Distant-Supervision Heuristic Assignment: Automatically derives success-failure targets for ongoing operating ventures without human annotation.
- Dual-Track Neural Feature Transform: Maps input vectors through parallel conditioning paths using Feature-wise Linear Modulation (FiLM).
- Parallel Baseline Inference: Compares SFTNN outputs synchronously against baseline predictors (MLP, XGBoost, TabNet).
- Dynamic Parameter Visualizer: Extracts and renders raw inner affine weights (γ , β) to ensure native model auditability.

2.3 User Classes and Characteristics

User Class	Characteristics	Privileges/Access
Data Scientist / Researcher	Highly technical; proficient in Python, PyTorch, and data engineering pipelines.	Full administrative access via terminal to invoke raw training and serialization modules.
Venture Analyst / End User	Non-technical or domain expert; requires clean visual outputs to interpret risk metrics.	Standard access restricted to the web browser graphical user interface (GUI).

2.4 Operating Environment

The system software shall operate under the following parameters:

- Hardware Development Platform: Minimum Core i5 or equivalent processor, 16GB RAM, optionally connected to standard CUDA-capable GPU environments for deep learning iteration.

- Server Operating System: Linux Ubuntu 22.04 LTS or Windows 10/11 environments running Python 3.10+.
- Client Web Browser Compatibility: Chrome, Edge, Safari, and Firefox (latest two stable releases).

2.5 Design and Implementation Constraints

- Strict Programming Boundaries: The core predictive layers must be programmed using PyTorch, and tree architectures must utilize the native C-optimized XGBoost framework.
- No Database Storage Persistence: To comply with project boundaries, data transitions must remain in-memory or load from read-only serialized files (.joblib, .pth) rather than an active external database instance.
- Data Exclusion Matrix: Software filters must strictly exclude operating ventures with unknown founding year, operating ventures younger than 3 years, and operating ventures between 3 and 6 years of age (inclusive), to eliminate target boundary noise, consistent with the distant-supervision decision flow defined in Section 4.1.

2.6 Assumptions and Dependencies

Assumption: The system assumes that regional macroeconomic designations provided during user inference align directly with the structural categorization used during model stratification.

Dependency: The pipeline depends on the ongoing structural schema stability of third-party dependencies, including PyTorch-TabNet and Scikit-Learn scaling classes.

3. External Interface Requirements

3.1 User Interfaces

The web-based user interface is built on Streamlit as a single-page dashboard. The screen layout is divided into input mechanisms on the left and analytical prediction outputs on the right. The structural and functional components of the interface are defined in Table 3.1 below.

Interface Panel	Component Type	Functional Description
Navigation Sidebar	Navigation Control	Houses view toggles, system settings, and diagnostic readouts.
Scenario Presets	Action Cards	Instantly populates the input form with pre-configured ecosystem profiles.
Input Form	Form Controls	Collects categorical features via dropdowns, numerical financials via text fields, and stepped parameters via sliders.
Prediction Results	Comparative Visuals	Displays real-time, side-by-side success probabilities of the SFTNN and XGBoost models.
Spatial Modulation	XAI Metrics	Renders progress bars tracking the active region's conditioning parameters (gamma and beta).

3.2 Hardware Interfaces

Not applicable. This is a pure-software machine learning system that does not interface with embedded physical components, peripheral barcode readers, or biometric sensory devices.

3.3 Software Interfaces

Interface	Purpose	Data Exchanged
PyTorch Core Engine	Executes SFTNN forward tensor passes	Floating-point Torch Tensors
FastAPI Backend Server	Links UI client to serialized pipelines	JSON strings matching validated Pydantic models
Joblib Serializer	Loads RobustScaler and TargetEncoder states	Binary state dictionaries and arrays

3.4 Communications Interfaces

All interaction between the Streamlit user dashboard and the FastAPI application layer shall take place over local or networked HTTP connections using standard RESTful architecture patterns. Payload data must be transmitted exclusively as serialized JSON blocks.

4. System Features

4.1 Feature: Data Processing & Target Engineering

4.1.1 Description and Priority

The system processes unstructured, multi-source raw records to compile a uniform, noise-filtered modeling dataset and sets ground-truth labels. Priority: High.

4.1.2 Stimulus/Response Sequences

Stimulus: User/Researcher triggers the execution of the data preprocessing script via terminal.

Response: The system parses the raw input source files, removes indeterminate records, maps categorical vectors via a training-derived target encoder, runs numerical stabilization via RobustScaler, and caches a unified training matrix.

4.1.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.1-1	The system shall assign ground-truth labels directly based on explicit exit states: IPO or Acquired must evaluate to Success (Y=1), and Closed must evaluate to Failure (Y=0).	High
REQ-4.1-2	The system shall calculate proxy success targets for ongoing Operating ventures using an age-conditioned, funding-round momentum heuristic: records with unknown founding year shall be excluded; records with company age < 3 years shall be excluded; records with company age between 3 and 6 years (inclusive) shall be excluded; for records with company age ≥ 7 years, Y=1 if Funding Rounds ≥ 2, and Y=0 if Funding Rounds < 2.	High
REQ-4.1-3	The system shall programmatically drop the columns company_age, funding_round_count, and Operating status right after label extraction to ensure no tracking features spill into the training cycle.	High

4.2 Feature: Multi-Model Unified Training & Evaluation

4.2.1 Description and Priority

The training engine executes backpropagation and gradient updates synchronously across all defined architectures using uniform partitions to produce comparative evaluations. Priority: Medium.

4.2.2 Stimulus/Response Sequences

Stimulus: Data Scientist initiates the main training tracking function.

Response: The application isolates a stratified 70% partition for parameter optimization, fits weights across the four architectures, monitors validation loss against a 15% subset, and records final comparative matrices on the remaining 15% testing split.

4.2.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.2-1	The training engine shall bifurcate the SFTNN processing path into: a primary network tracking global attribute representations (h), and a conditioning network extracting spatial scaling coefficients (γ , β) based on regional metadata codes (r).	High
REQ-4.2-2	The modeling system shall calculate performance benchmarks simultaneously across four specific frameworks: Multi-Layer Perceptron (MLP), XGBoost, TabNet, and the custom SFTNN structure.	Medium

REQ-4.2-3	The reporting tool shall segment and output final evaluation performance metrics (Accuracy, Precision, Recall, F1-Score, AUC-ROC, PR-AUC) at a global level and as individual breakdowns for each recorded geographic region.	High
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4.3 Feature: Real-Time Concurrent Model Inference

4.3.1 Description and Priority

Allows end-users to pass raw inputs to the online system and receive parallel target classification evaluations without manual re-compilation. Priority: High.

4.3.2 Stimulus/Response Sequences

Stimulus: Investor configures feature parameters on the UI panel and clicks the “Predict Success Probability” action item.

Response: Streamlit transmits a JSON package to FastAPI; the backend transforms the array via `preprocessor.joblib`, runs concurrent operations across `sftnn_model.pth` and `xgboost_model.json`, and returns probability statistics.

4.3.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.3-1	The inference framework shall compute real-time probabilities using the exact feature scaling criteria generated during the original training session without running any live <code>.fit()</code> or adjustment loops.	High

REQ-4.3-2	The backend service shall return extracted spatial scaling outputs (γ_r , β_r) matching the specific region configuration index passed in the client payload.	Medium
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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- REQ-5.1-1 (API Latency Bounds): The FastAPI backend inference engine shall respond to individual incoming evaluation POST requests within a maximum time threshold of 50ms.
- REQ-5.1-2 (UI Refresh Rate): The Streamlit analytical user environment shall render comparison visuals, metrics panels, and coordinate graphs within 1.5s of receiving the backend's JSON payload.

5.2 Safety Requirements

- REQ-5.2-1 (System Outlier Alerts): If a user requests real-time inference using a geographic target region code that contains fewer than $N=10$ records within the original training archive, the system shall show a visual caveat alert message notifying the user of elevated metric variance risks.

5.3 Security Requirements

- REQ-5.3-1 (Strict Target Leakage Prevention): Under no circumstances shall target encoding transforms or normalization bounds compute mathematical attributes over the validation or testing subsets.

- REQ-5.3-2 (Read-Only Pipeline Constraints): During live inference execution, the data transformation layers must operate exclusively under fixed, read-only constraints via Scikit-Learn `.transform()` implementations.

5.4 Software Quality Attributes

- REQ-5.4-1 (Reproducibility Baseline): All dataset random seed components, architecture hyperparameters, and parameter learning schedules shall be controlled from an external `config.yaml` file to ensure independent third-party execution produces identical numerical weights.
- REQ-5.4-2 (Modular Maintainability): The application software architecture shall separate codebases cleanly so that the dashboard frontend design, API endpoints, and raw tensor model classes can be modified independently without breaking system operation.

5.5 Business Rules

- REQ-5.5-1 (Universal Macroeconomic Baseline): To maintain systematic target consistency across disparate geographic regions, the distant-supervision exclusion and threshold rules (unknown-age exclusion; age < 3 exclusion; age 3–6 exclusion; age ≥ 7 AND Funding Rounds ≥ 2 → Success) shall apply uniformly to all operating startups across the dataset to establish an un-drifted optimization benchmark.

6. Other Requirements

- REQ-6.1-1 (Environmental Virtualization Framework): The absolute system configuration framework must be isolated and managed via a

locked virtual environment using a strict requirements.txt file specifying absolute package version tags for PyTorch and Scikit-Learn.

Appendix A: Glossary

Term	Definition
FiLM (Feature-wise Linear Modulation)	An architectural conditioning mechanism that adaptively influences a network's internal representations by applying affine transformations (scaling and shifting) to intermediate feature maps based on an external conditioning variable.
SFTNN (Spatial Feature Transform Neural Network)	The specific deep learning architecture proposed in this study, which uses regional metadata to dynamically condition structural startup feature vectors via specialized affine transform layers.
Distant Supervision	A semi-supervised learning paradigm where external programmatic heuristics, proxy rules, or patterns are leveraged to automatically synthesize ground-truth training labels (Y) in the absence of manual human annotations.
Target Encoding	A categorical feature representation technique where high-cardinality nominal variables (e.g., industry sectors) are replaced with the expected scalar value or probability distribution of the target class computed over the training partition.
Data Leakage	A critical methodological flaw where information from outside the training dataset (such as validation or test set statistical distributions) is inadvertently introduced

	into the model during training, leading to overly optimistic but invalid performance metrics.
Stratified Partitioning	A sampling design method that splits a dataset into training, validation, and testing subsets while strictly maintaining the original ratio of class labels across all partitions to prevent distribution imbalances.
Hadamard Product ($\odot$)	A mathematical operation that takes two matrices or vectors of the same dimensions and produces another matrix or vector where each element (i,j) is the product of the corresponding elements of the original inputs.
Robust Scaling	A statistical normalization technique that scales numerical features by subtracting the median and dividing by the Interquartile Range (IQR), making the feature space highly resilient to extreme dataset outliers.
AUC-ROC	Area Under the Receiver Operating Characteristic curve; a performance measurement for classification tasks at various threshold settings that quantifies a model's ability to distinguish between positive and negative classes.
PR-AUC (Precision-Recall Area Under Curve)	The area under the Precision-Recall curve across all classification thresholds, reported alongside AUC-ROC to account for near-balanced but cross-regionally uneven class composition.
G-Mean (Geometric Mean)	An evaluation metric calculated as the square root of the product of Sensitivity (True Positive Rate) and Specificity (True Negative Rate). It measures the balance between classification performance on both

	majority and minority classes, which is critical for imbalanced data.
TabNet	A high-performance deep learning architecture optimized for tabular data that uses sequential attention mechanisms to select salient features at each decision step, mimicking the behavior of tree-based models.
XGBoost (Extreme Gradient Boosting)	An optimized, highly scalable distributed gradient-boosted decision tree library implemented under a supervised learning framework to optimize tree split structures using a parallelized gradient descent method.
Pydantic Schema	A data validation library utilized within Python web frameworks (like FastAPI) to enforce strict type hinting, deep structural validation, and request-response object contract conformity at runtime.
Latent Variable	An underlying, hidden variable that is not directly observed in the raw input data but is instead inferred mathematically through the collective non-linear interaction of observed features.
Inherent Interpretability	A characteristic of an algorithm where its inner decision-making mechanics, weights, and mathematical structures can be directly traced and understood without needing to build secondary approximation models like SHAP or LIME.
Target Drift	A phenomenon where the fundamental semantic meaning or real-world distribution of the target label changes over time or across different partitions (e.g.,

	across regions), creating conflicts in mathematical optimization objectives.
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Appendix B: System Predictor Interface Mockup

Interactive Predictor Dashboard



Figure C.1 illustrates the functional UI mockup for the TechVenture-SFTNN predictor dashboard. The dashboard is fully responsive and supports both high-contrast dark-mode and light-mode interfaces to accommodate different operating environments.

APPLICANT/CANDIDATE RESUME

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PERSONAL INFORMATION

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Computer PMS and Software skills:

- Languages & Frameworks: C#, JavaScript, Python, FAST API, SQL, ASP.NET Core, Blazor, Node.js, React.js, Supabase, Entity Framework
- AI & Automation: Prompt and Context Engineering, Agentic Workflows, Tool Calling Systems, LLM Integration (OpenAI, Claude, Gemini, Grok, OpenRouter), Cursor, Claude Code, AI-Assisted Software Development, AI Evaluation & Benchmarking, Data Enrichment Pipelines, Workflow Automation
- Software, Product & DevOps Tools: Systems & API Design, Modular Architecture, Database Schema Design, Git/GitHub, CI/CD, JWT Authentication, REST APIs, Azure DevOps, Jira, Trello
- Methodologies: Agile (Scrum/Kanban), Sprint Planning, Issue Tracking, Workflow Management, Cross-Functional Team Collaboration

EDUCATIONAL BACKGROUND

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Tertiary*

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Certifications & Accreditations

- C# HackerRank Certificate (2025)
- Consistent Dean's Lister, University of San Carlos (2024 – Present)
- 4th Placer – UP Interhigh Programming Competition (2024)
- Top 3 in Overall Campus – University of Cebu (2024)
- Top 1 in Overall Strand – University of Cebu (2024)
- Maintained Scholarship, University of Cebu (2022 – 2024)

STRENGTHS/TRAITS & SKILLS

- High degree of initiative in AI automation, backend architecture, and system design
- Hands-on experience building production-ready AI pipelines, full-stack applications, and agentic workflows
- Strong presentation skills, feature ideation, and code review capabilities
- Good interpersonal skills and active cross-functional collaboration within Agile teams
- Able to work within tight schedules, sprint planning cycles, and CI/CD development deadlines

CAREER OBJECTIVES

Further develop professional career path within AI engineering and full-stack software development, leveraging expertise in scalable backend design, tool-calling automation, and LLM systems within high-impact technology environments.

References

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Computer PMS and Software skills:

- Languages & Frameworks: Java, JavaScript, Python, PHP, SQL, C, React.js, Node.js, Express.js, HTML5, CSS3, Bootstrap, Android XML, Streamlit
- AI & Machine Learning: Prompt Engineering, LLM Integration (OpenAI API, OpenRouter), AI Chatbots, AI-Assisted Software Development, Web Scraping, Machine Learning Fundamentals
- Software & Development Tools: Git/GitHub, REST APIs, Postman, Android Studio, Visual Studio Code, Docker (Basic), XAMPP, Figma, WinSCP, Postman
- Engineering Practices: Full-Stack Web & Mobile Development, Database Design, System Integration, SDLC, Agile (Scrum)

EDUCATIONAL BACKGROUND

Educational Level

Aug 2024 – Present

BS Computer Science, University of San Carlos – Talamban

Tertiary

Jun 2024

Senior High School (STEM Track), University of Cebu – Banilad

Secondary

Certifications & Accreditations

- Dean's Lister, University of San Carlos (2024 – 2025)
- High Honors Awardee, University of Cebu (2022 – 2023)

STRENGTHS/TRAITS & SKILLS

- High degree of initiative in AI-assisted development and software engineering
- Hands-on experience in full-stack web and mobile application development
- Strong presentation skills and technical documentation capabilities
- Good interpersonal skills and team collaboration within Agile/Scrum environments
- Able to work within tight schedules, release cycles, and sprint deadlines

CAREER OBJECTIVES

Further develop professional career path within software engineering and artificial intelligence, leveraging expertise in full-stack web development, machine learning integration, and product engineering within innovative technology environments.

References

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Date: August 9, 2026

APPROVAL SHEET

This thesis proposal entitled: **TECH-VENTURE: SPATIAL FEATURE TRANSFORM NEURAL NETWORK FOR REGION-AWARE STARTUP SUCCESS PREDICTION**

submitted by: **OSBEV ERICA D. CABUCOS, DRIXYL REECE IRISH G. NACU,**

for the degree of: **BACHELOR OF SCIENCE IN COMPUTER SCIENCE** has been approved by the Committee assigned to the students.

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Adviser

KATRINA BELLEZA FUENTES, Ph.D.

Chair

CHRISTINE F. PEÑA, MMath

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Thesis Proposal Coordinator

THESIS COMMITTEE RECOMMENDATIONS

COMPLIANCE FORM

(Proposal Hearing)

SUGGESTIONS	NOT APPLICABLE	COMPLIED WITH	
		YES Page No.	NO Reason/s
1. Revise the conceptual framework (Fig. 6) to a more comprehensive framework including the step-by-step process and ensuring alignment with the objectives of the problem statement.		pp. 78–87 Figure 8, p. 79	
2. Improve the writeup on the discussion relating to data pre-processing; make it comprehensive.		pp. 64–77 Section 4.3	
3. Discuss what a startup is (include in the abstract and also in the introductory chapter) and differentiate this from other types of companies. Also include the definition of what a successful startup is.		p. ii; p. 1 Abstract and Section 1.1	

4. Generally, improve Chapter 4: Section 4.2.2, recommending to illustrate the data per region through a table and discuss data imbalance and how this is managed.		pp. 60–62 Table 1 and Regional Data Imbalance and Its Management	
5. In comparing the results of the models, recommend representing them through a table.		pp. 109–110 Tables 10 and 11	
6. Revise the introductory chapter writeup to shorten and include only what is relevant and necessary to the study.		pp. 1–3 Section 1.1	
7. Check the verb tenses in the manuscript, in which some are using past tense.		Throughout the manuscript, particularly Chapter 4	
8. Create a conceptual framework that is aligned with the specific objectives of the problem.		pp. 78–87 Figure 8, p. 79	
9. Make the timeline readable (put in 2 pages if necessary).		pp. 102–103 Tables 5 and 6	
10. Indicate in the limitation the characteristics of a start-up that are not included in the study.		p. 11 Section 1.6.3	

Submitted by:



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